

AGILEX™ 5 FPGA E-SERIES 065B MODULAR DEVELOPMENT KIT - SOM

PB : N19944-003

PBA : N19360-002

TA : N19359-002

BOM : 99CA38
MMID

Project Drawing Number	
Raw PCB	100-0330723-B1
Gerber Files	110-0330723-B1
PCB Design Files	120-0330723-B1
Assembly Drawing	130-0330723-B1
Fab Drawing	140-0330723-B1
Schematic Drawing	150-0330723-B1
PCB Film	160-0330723-B1
Bill of Materials	170-0330723-B1
Schematic Design Files	180-0330723-B1
Functional Specification	210-0330723-B1
PCB Layout Guidelines	220-0330723-B1
Assembly Rework	320-0330723-B1

Mon Nov 4 13:06:00 2024

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER		REV
		C+	34649	150-0330723-B1		2.1
		SCALE:		DO NOT SCALE DRAWING		SHEET 1 OF 62

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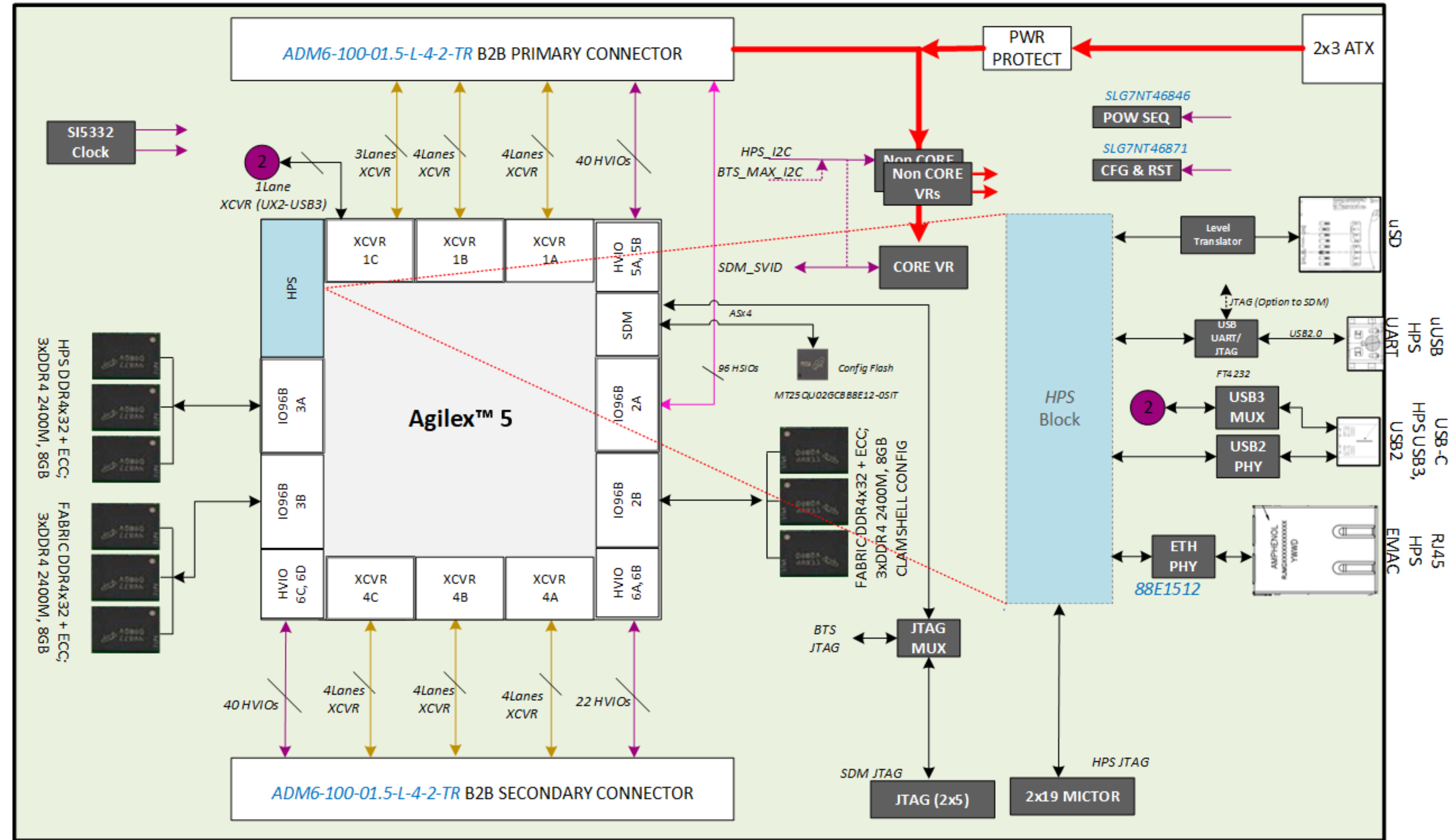
SHEET NO.	SHEET NAME
1	AGILEX_5E_SOM
2	TOC
3	BLOCK DIAGRAM
4	POWER TREE
5	POWER SEQUENCE
6	CLOCK TREE
7	I2C TREE
8	JTAG TREE
9	RESET TREE
10	TELEMETRY TREE
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12	AG5E BANK - HPS
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36	CLOCK_S15332
37	CLOCK_S1549 & GPIO_EXPANDER
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40	USB2.0PHY
41	USB3.0 MUX
42	USB3.0 CONN
43	SD CARD
44	USB - JTAG/UART CONVERTER
45	TEMP_SENSOR & EEPROM
46	LED AND SWITCH
47	I2C ISOLATOR
48	B2B_PRIMARY_CONN
49	B2B_SECONDARY_CONN
50	12V_POWER INPUT

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[illegible]

B

A



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BLOCK DIAGRAM

DEPARTMENT

altera

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2

1



POWER TREE

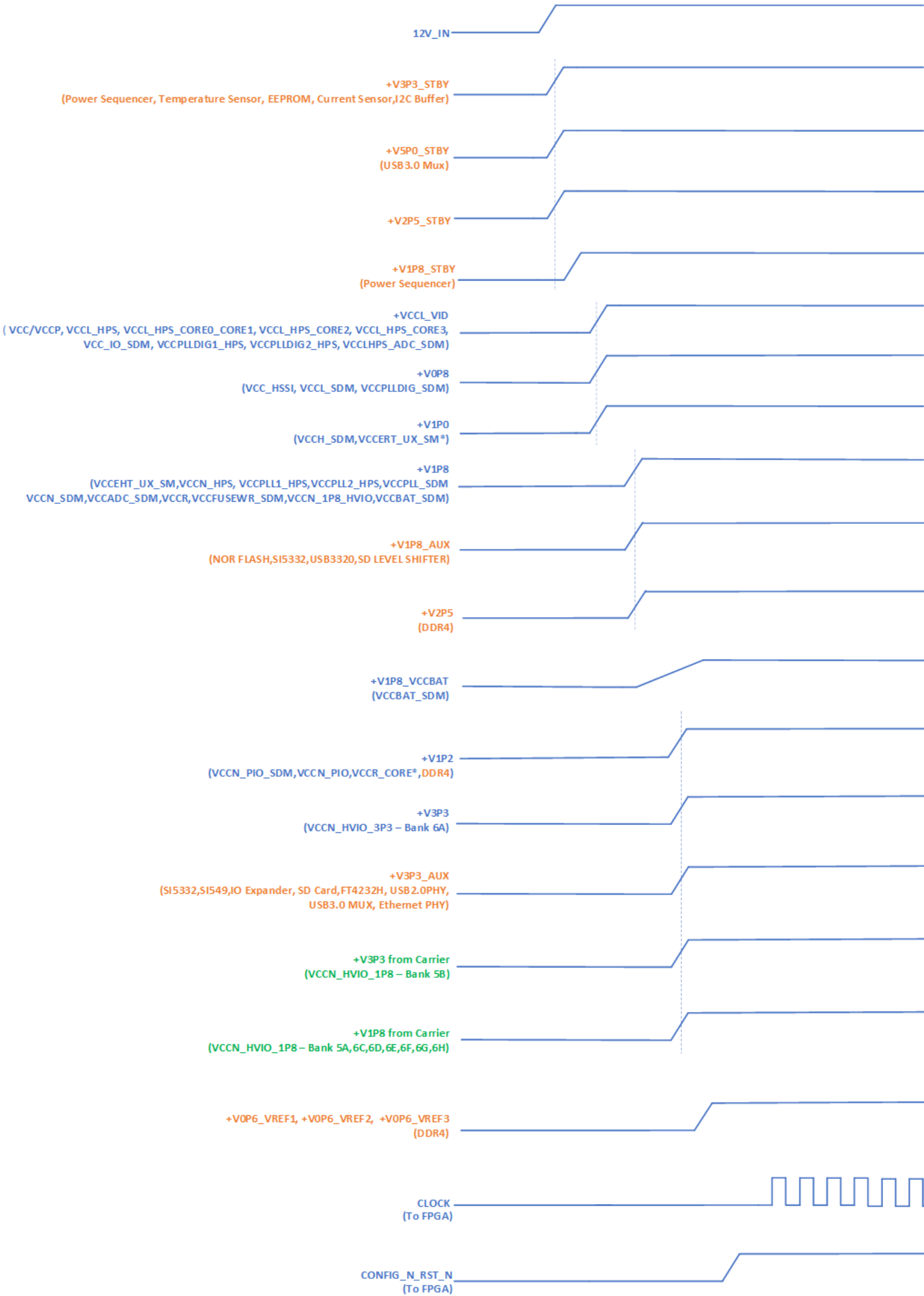
B

B

A

A

Agilex™ 5 SOM Board Power Sequence



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POWER SEQUENCE

DEPARTMENT



SIZE

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REV

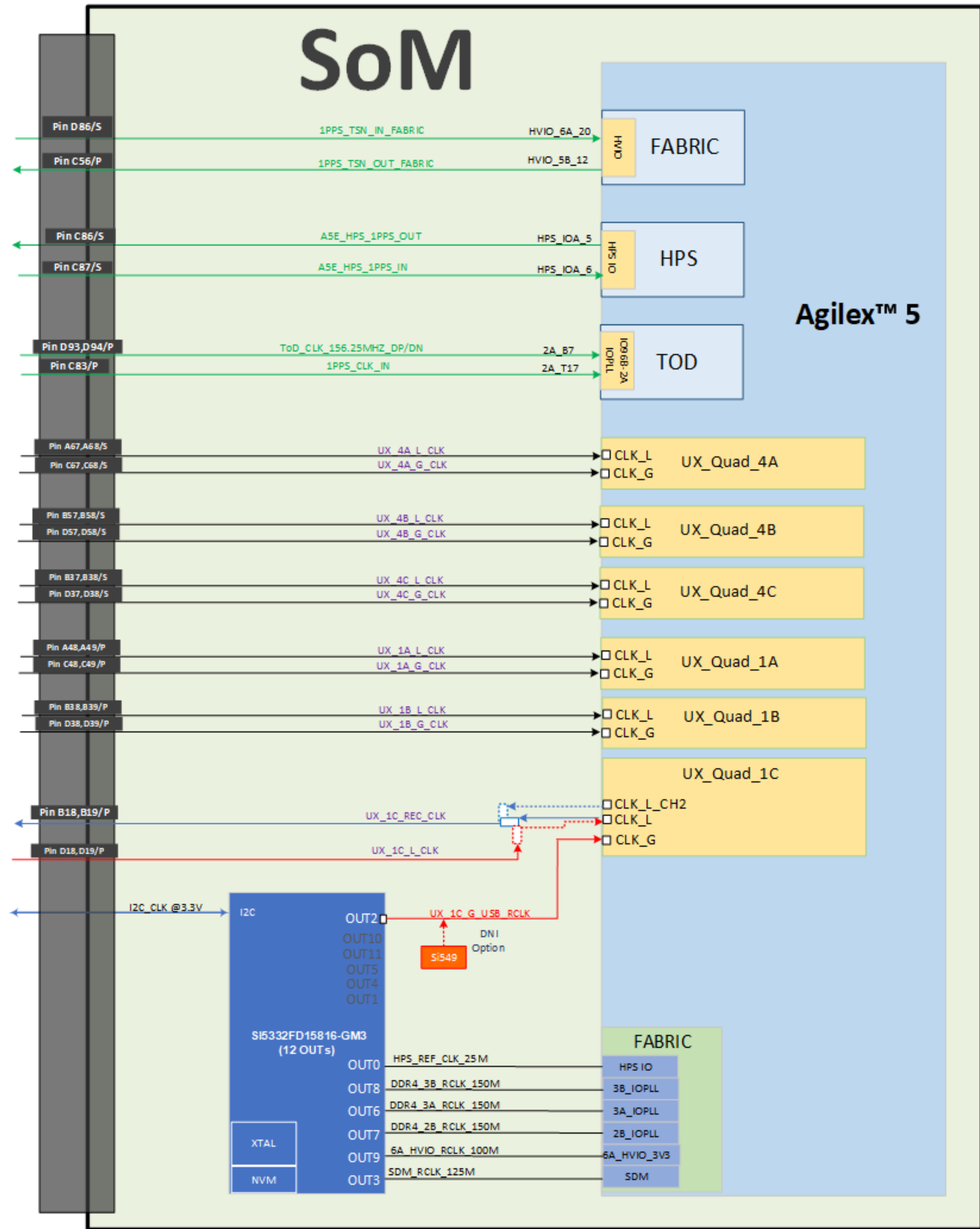
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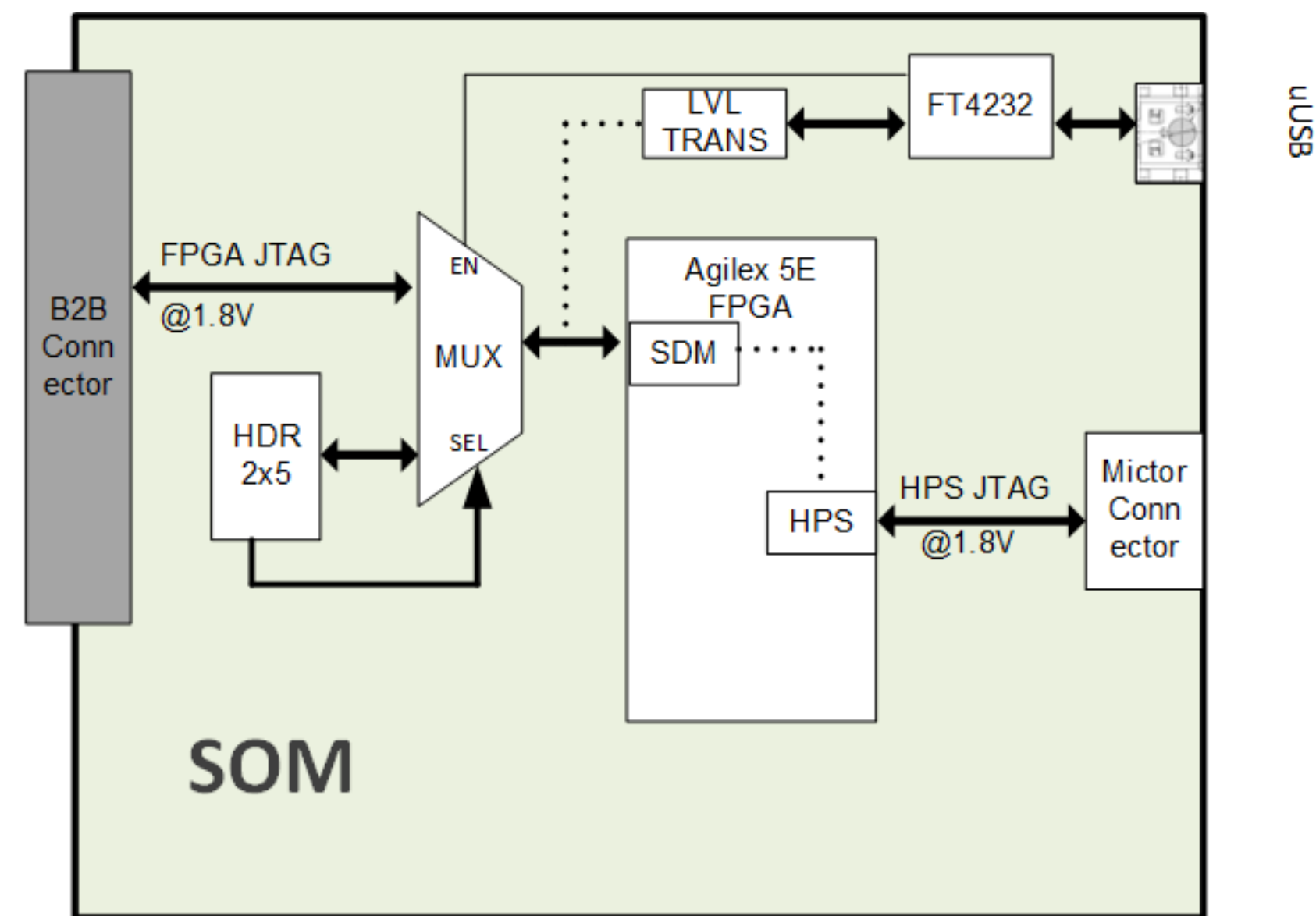
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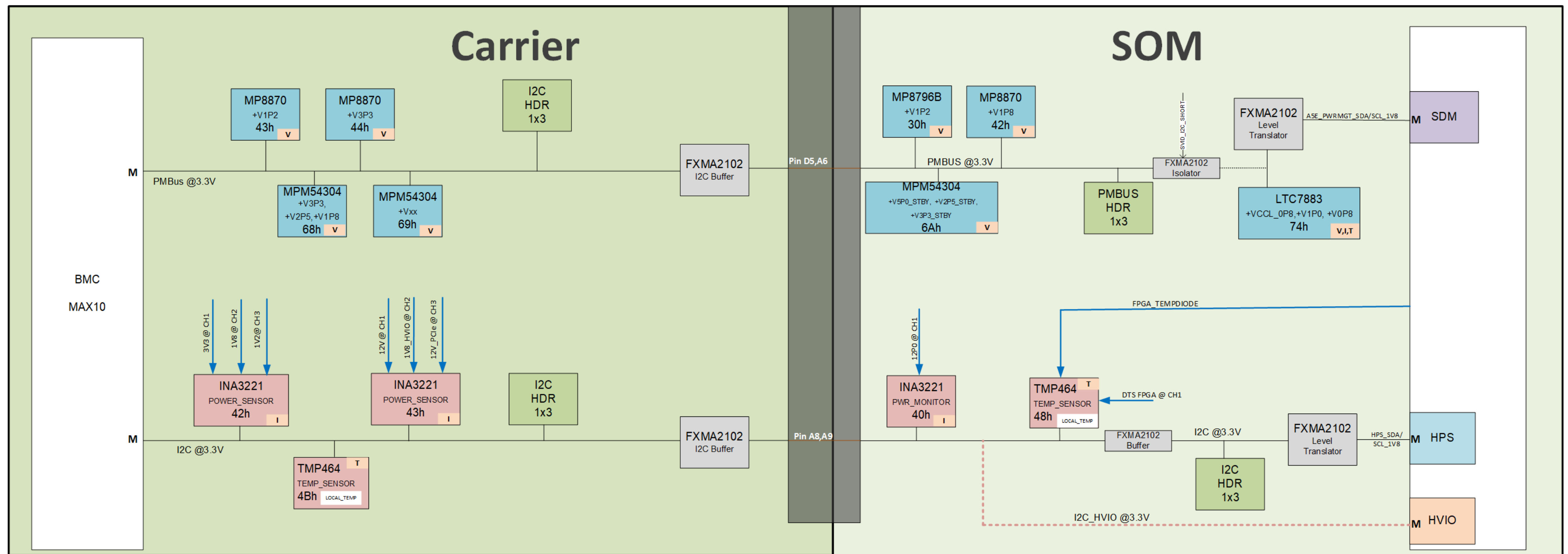


CLOCK TREE

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TELEMETRY TREE

DEPARTMENT



SIZE

C+

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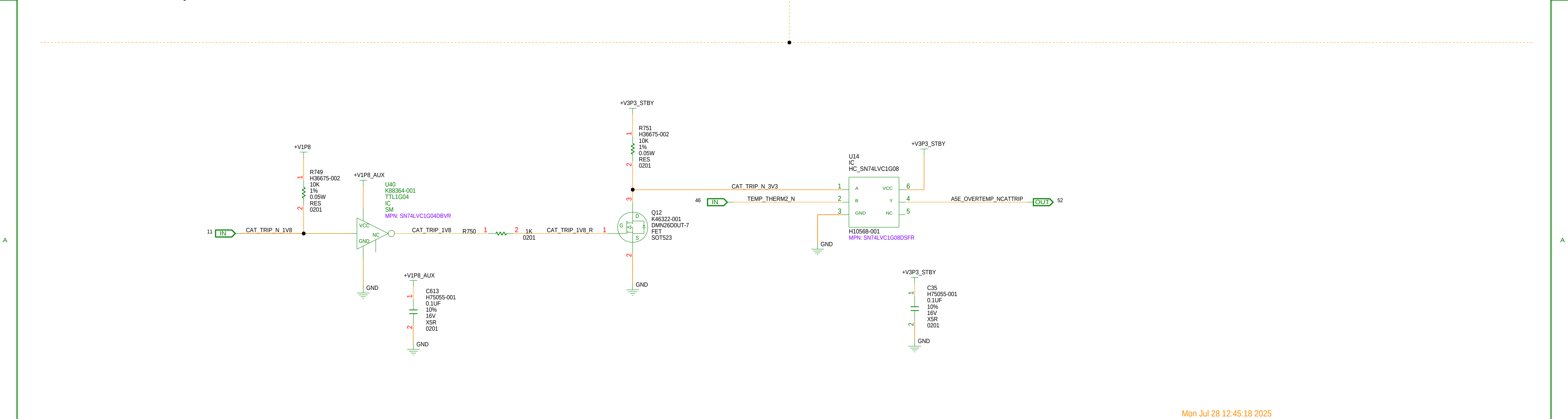
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AG5E BANK - SDM

DEPARTMENT



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4	3	2	1
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HPS IO MAP

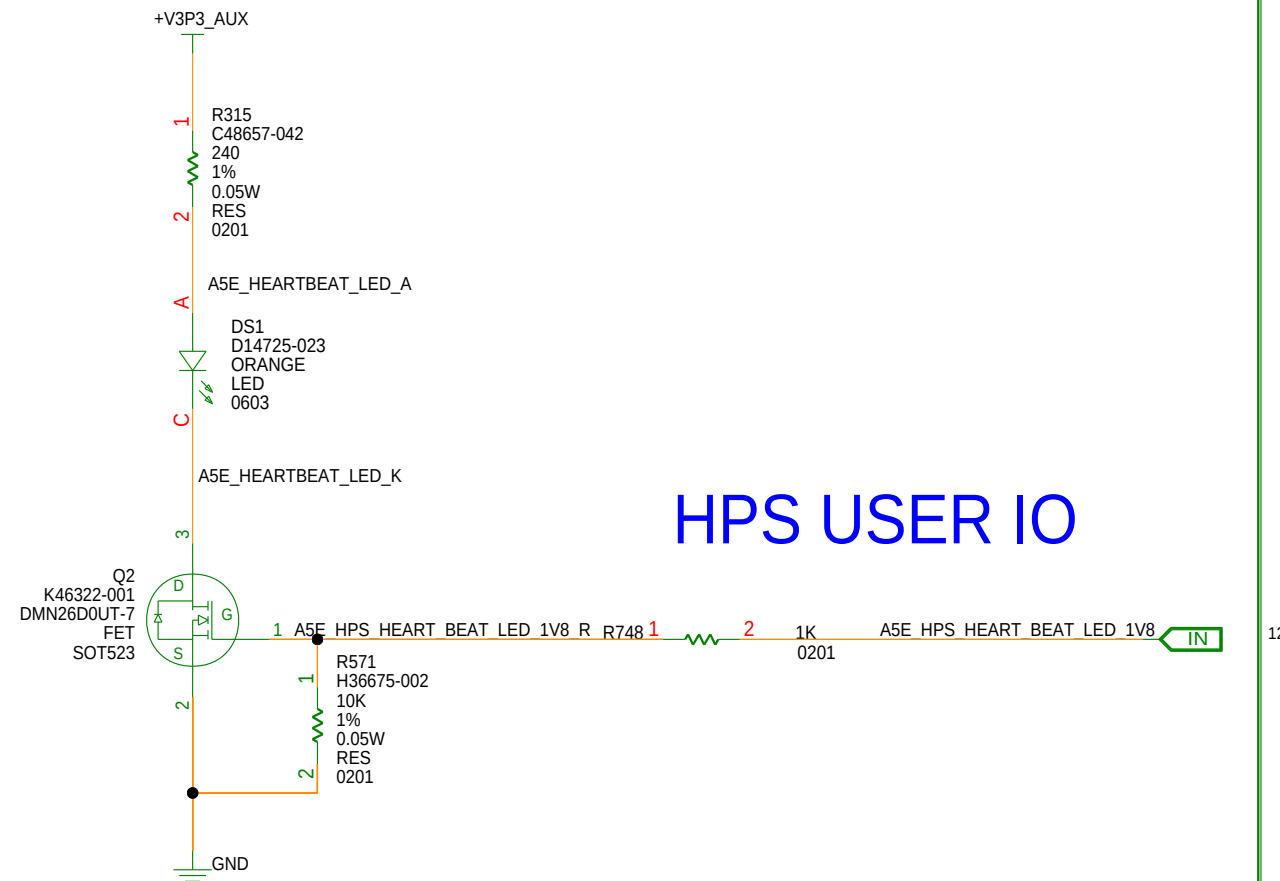
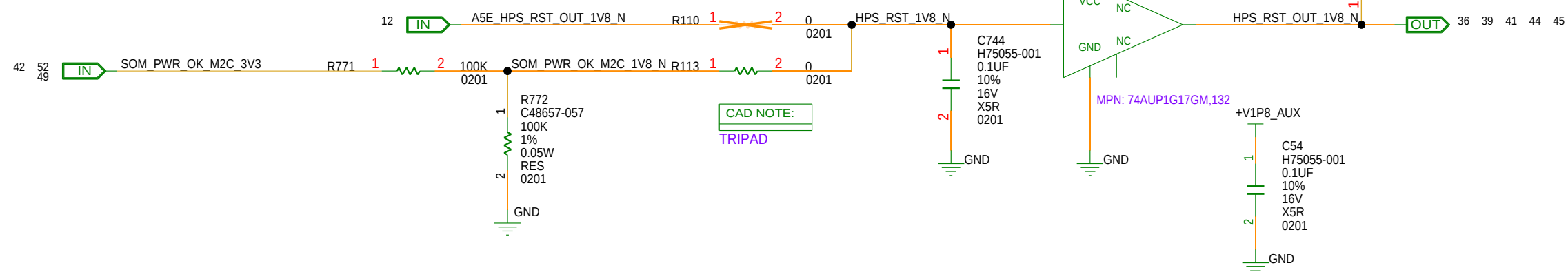
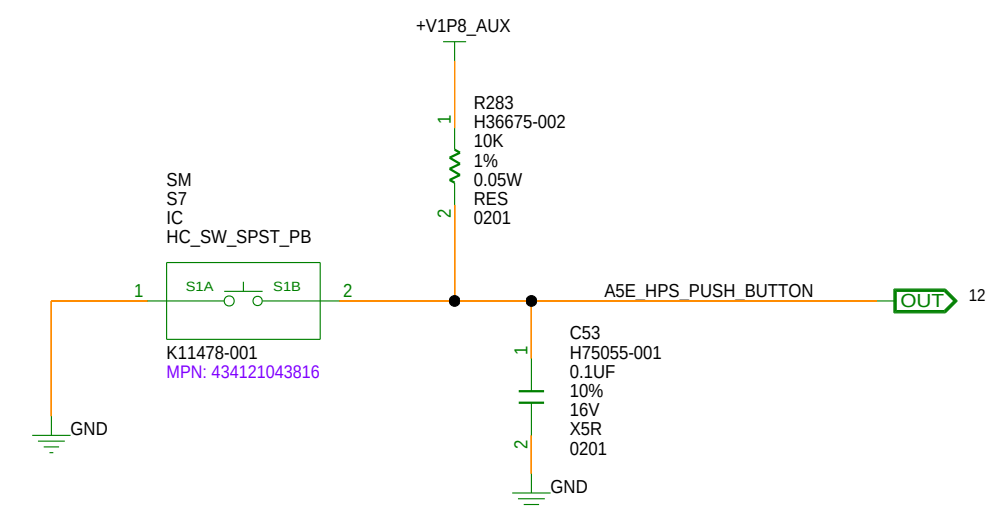
PIN NUMBER	PIN NAME	SIGNAL NAME
W135	HPS_IOA_1	ASE_HPS_HEART_BEAT_LED_1V8
U135	HPS_IOA_2	ETH_INT_N_1V8
W134	HPS_IOA_3	ASE_UART_FT4232_TXD
AK115	HPS_IOA_4	ASE_UART_FT4232_RXD
U134	HPS_IOA_5	ASE_HPS_1PPS_OUT
AL120	HPS_IOA_6	ASE_HPS_1PPS_IN
R134	HPS_IOA_7	ASE_PHY_1V8_MDIO
AG115	HPS_IOA_8	ASE_PHY_1V8_MDC
N135	HPS_IOA_9	ASE_HPS_I2C_SDA_1V8
AK120	HPS_IOA_10	ASE_HPS_I2C_SCL_1V8
N134	HPS_IOA_11	ASE_HPS_RST_OUT_1V8_N
T132	HPS_IOA_12	CLK_ASE_HPS_25M_1V8
P132	HPS_IOA_13	CLK_ULPI_60M
L135	HPS_IOA_14	USB1_ULPI_STP
J135	HPS_IOA_15	USB1_ULPI_DIR
AD135	HPS_IOA_16	USB1_ULPI_DATA0
M132	HPS_IOA_17	USB1_ULPI_DATA1
AD134	HPS_IOA_18	USB1_ULPI_NXT
K132	HPS_IOA_19	USB1_ULPI_DATA2
AG129	HPS_IOA_20	USB1_ULPI_DATA3
J134	HPS_IOA_21	USB1_ULPI_DATA4
AG120	HPS_IOA_22	USB1_ULPI_DATA5
G134	HPS_IOA_23	USB1_ULPI_DATA6
G135	HPS_IOA_24	USB1_ULPI_DATA7
E135	HPS_IOB_1	M_SD_DATA0_1V8
F132	HPS_IOB_2	M_SD_DATA1_1V8
D132	HPS_IOB_3	M_SD_CLK_1V8
AG123	HPS_IOB_4	ASE_INTN_C2M_1V8
B134	HPS_IOB_5	ASE_HPS_PUSH_BUTTON
AA135	HPS_IOB_6	M_SD_DATA2_1V8
V127	HPS_IOB_7	M_SD_DATA3_1V8
AB132	HPS_IOB_8	M_SD_CMD_1V8
T127	HPS_IOB_9	ASE_JTAG_HPS_TCK
Y132	HPS_IOB_10	ASE_JTAG_HPS_TMS
T124	HPS_IOB_11	ASE_JTAG_HPS_TDO
P124	HPS_IOB_12	ASE_JTAG_HPS_TDI
M127	HPS_IOB_13	RGMII_TX_CLK
K127	HPS_IOB_14	RGMII_TX_CTRL
M124	HPS_IOB_15	RGMII_RX_CLK
AB127	HPS_IOB_16	RGMII_RX_CTRL
K124	HPS_IOB_17	RGMII_TXD_0
Y127	HPS_IOB_18	RGMII_TXD_1
H127	HPS_IOB_19	RGMII_RXD_0
AB124	HPS_IOB_20	RGMII_RXD_1
F127	HPS_IOB_21	RGMII_TXD_2
Y124	HPS_IOB_22	RGMII_TXD_3
F124	HPS_IOB_23	RGMII_RXD_2
D124	HPS_IOB_24	RGMII_RXD_3

HPS PUSH BUTTON SWITCH

HPS RESET OUT

HEART BEAT LED

HPS USER IO



AG5E BANK - HPS

DEPARTMENT

altera

SIZE

CODE

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REV

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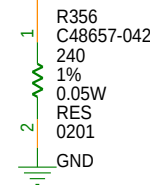
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SWIZZLING ON BANK 2B

BYTE_SWIZZLE_CH0=1,X,X,X,0,2,3,ECC;

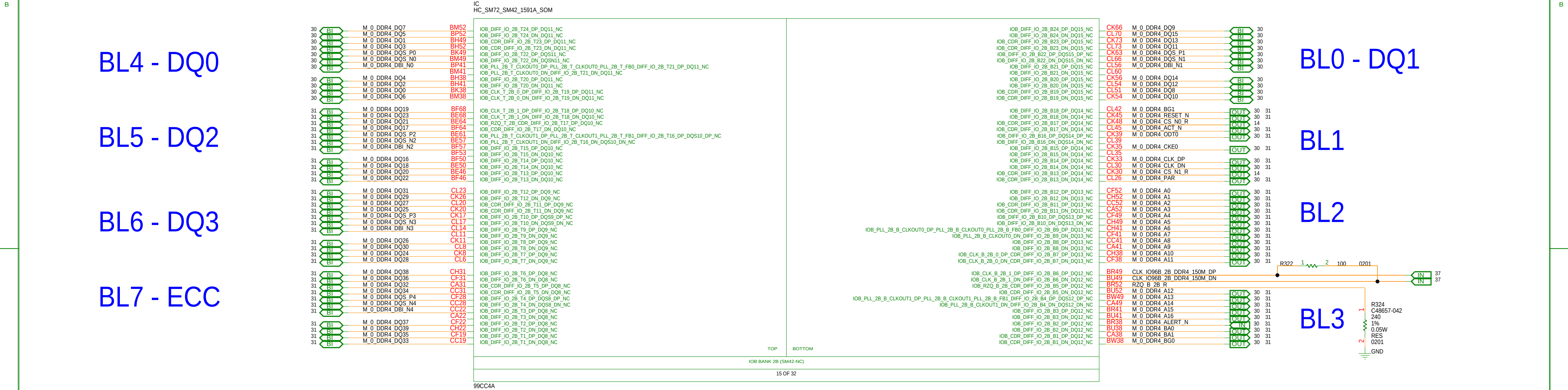
PIN_SWIZZLE_CH0_DQS0=7,5,1,3,4,2,0,6;

PIN_SWIZZLE_CH0_DQS1=9,15,13,11,14,12,8,10;

PIN_SWIZZLE_CH0_DQS2=19,23,21,17,16,18,20,22;

PIN_SWIZZLE_CH0_DQS3=31,29,27,25,26,30,24,28;

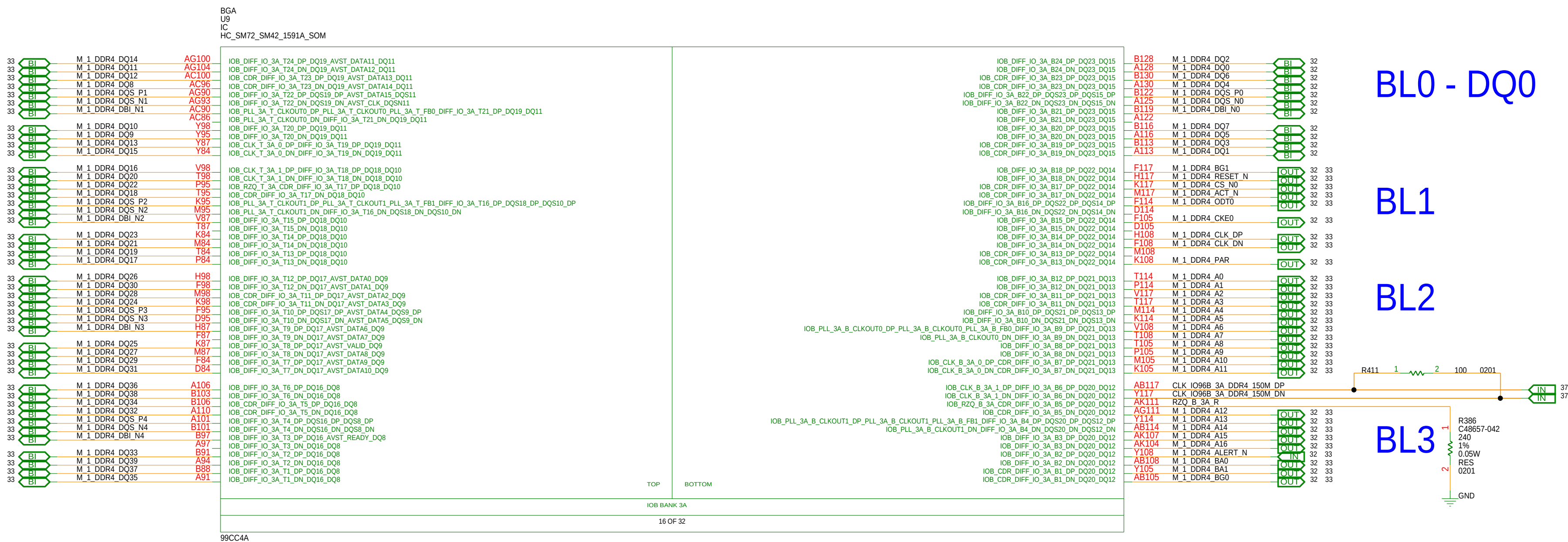
PIN_SWIZZLE_CH0_ECC=6,4,0,2,5,7,3,1;



```

BYTE_SWIZZLE_CH0=0,X,X,X,1,2,3,ECC;
PIN_SWIZZLE_CH0_DQS0=2,0,6,4,7,5,3,1;
PIN_SWIZZLE_CH0_DQS1=14,11,12,8,10,9,13,15;
PIN_SWIZZLE_CH0_DQS2=16,20,22,18,23,21,19,17;
PIN_SWIZZLE_CH0_DQS3=26,30,28,24,25,27,29,31;
PIN_SWIZZLE_CH0_ECC=4,6,2,0,1,7,5,3;

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SWIZZLING ON BANK 3B

BYTE_SWIZZLE_CH0=0,X,X,X,1,2,3,ECC;
PIN_SWIZZLE_CH0_DQS0=2,0,6,4,7,3,5,1;
PIN_SWIZZLE_CH0_DQS1=12,14,13,10,8,11,15,9;
PIN_SWIZZLE_CH0_DQS2=16,17,18,19,20,21,22,23;
PIN_SWIZZLE_CH0_DQS3=24,25,26,27,28,29,30,31;
PIN_SWIZZLE_CH0_ECC=0,2,6,4,1,3,7,5;

BL4 - DQ1

BL5 - DQ2

BL6 - DQ3

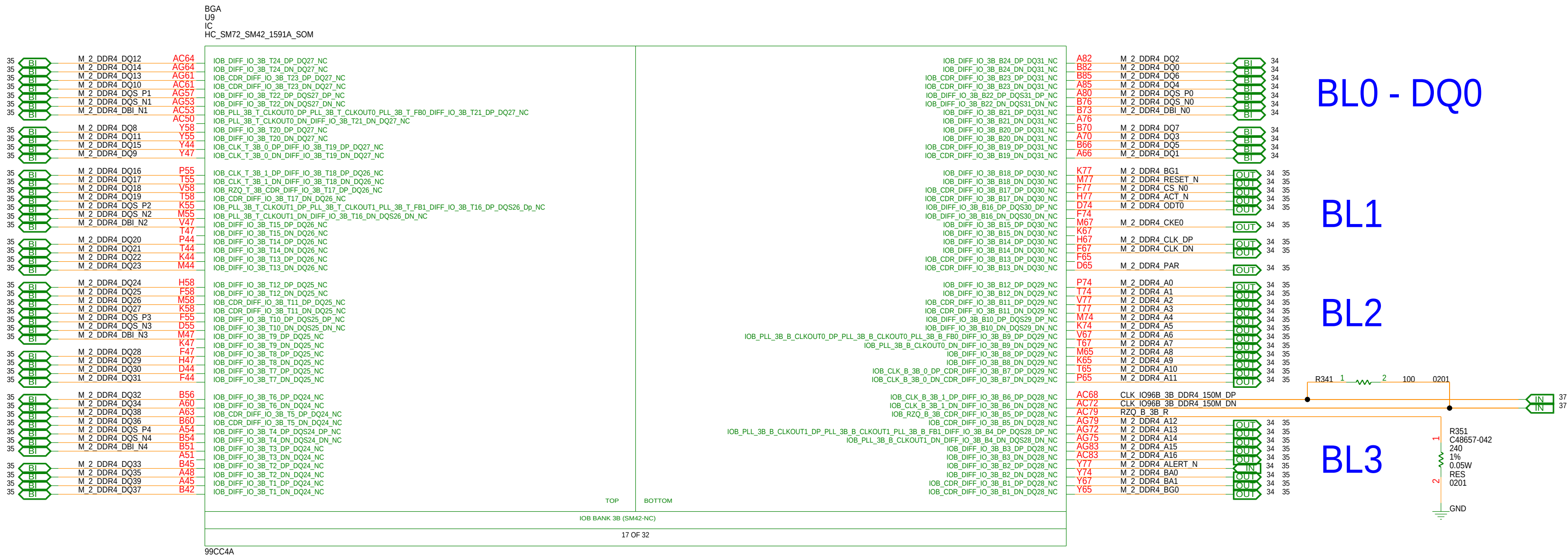
BL7 - ECC

BL0 - DQ0

BL1

BL2

BL3



AG5E BANK- IO96B-3B

DEPARTMENT



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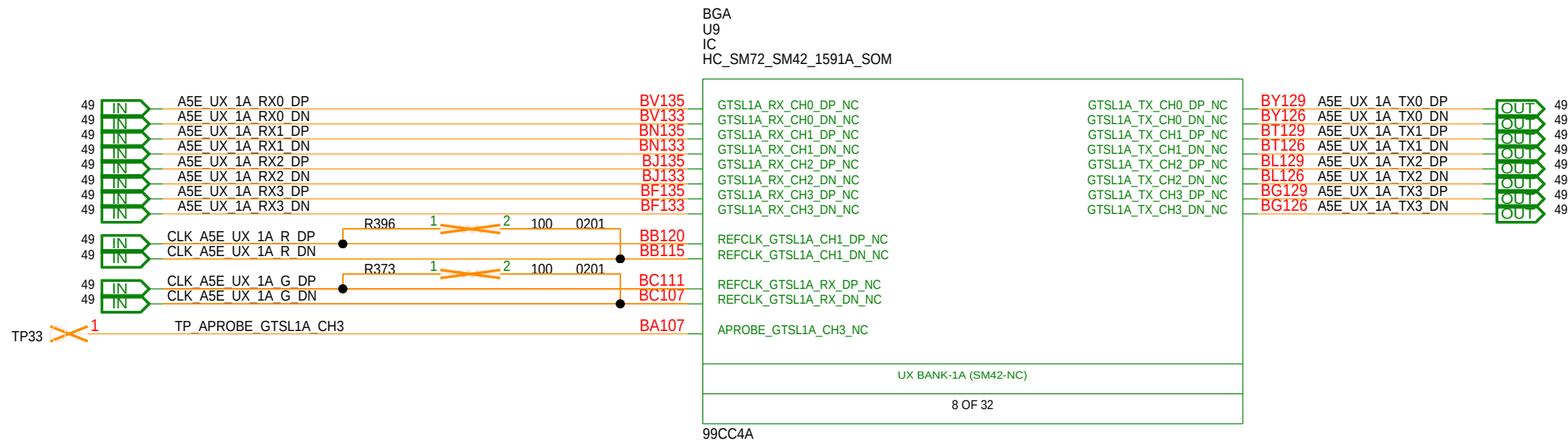
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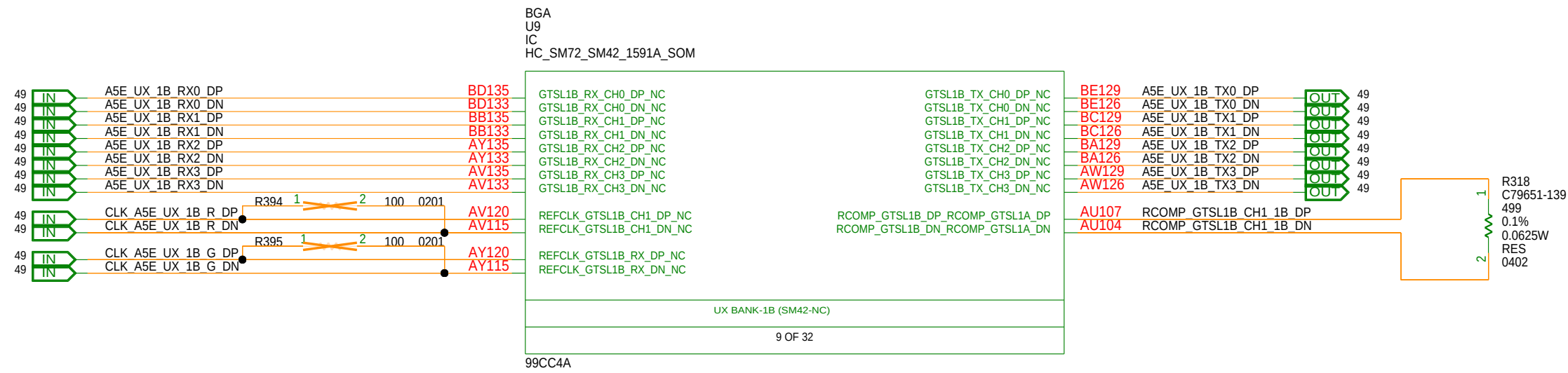
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B

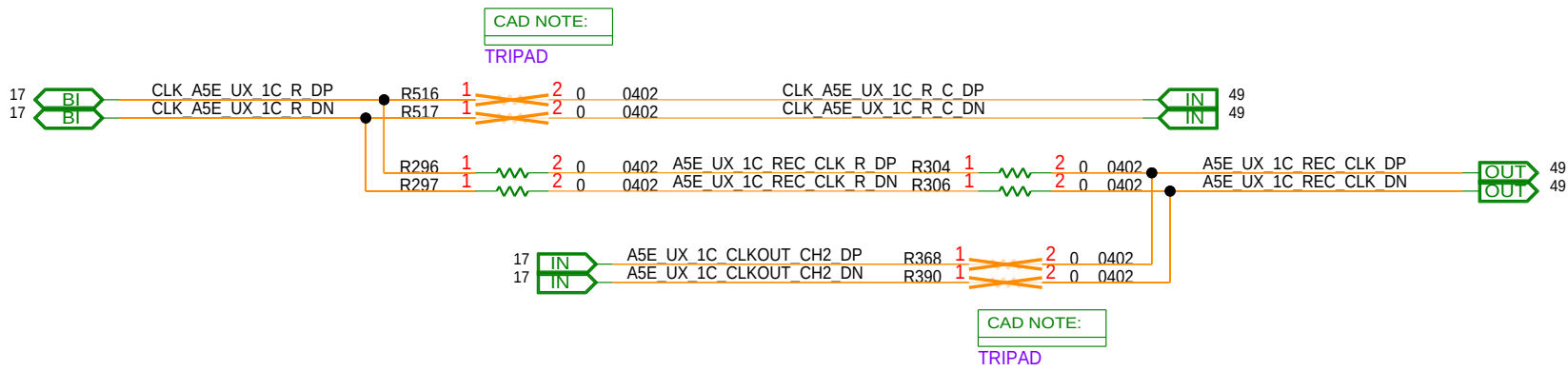
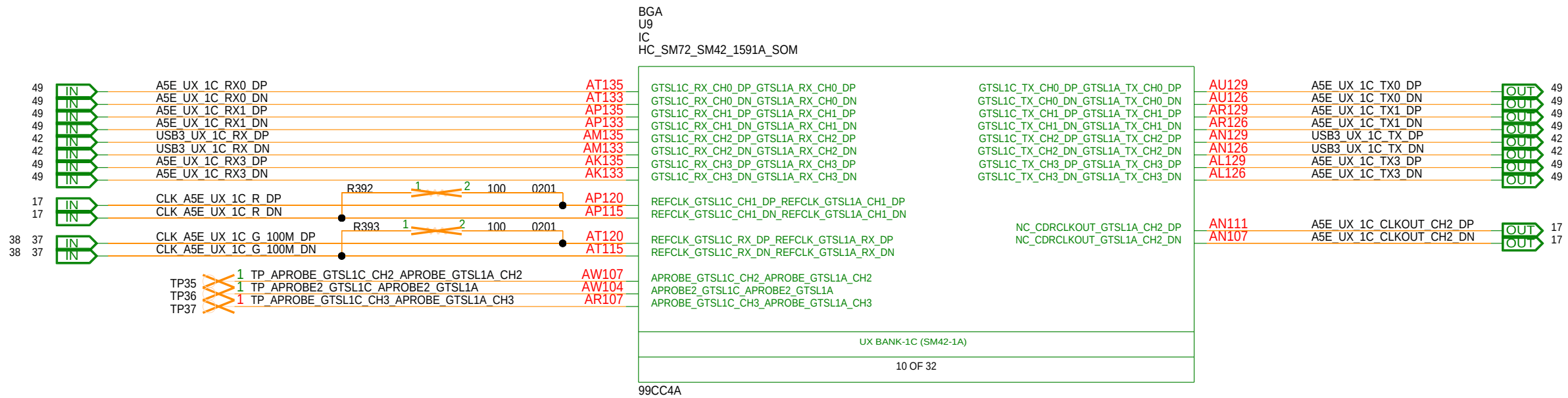


A

A



1C REC CLOCK



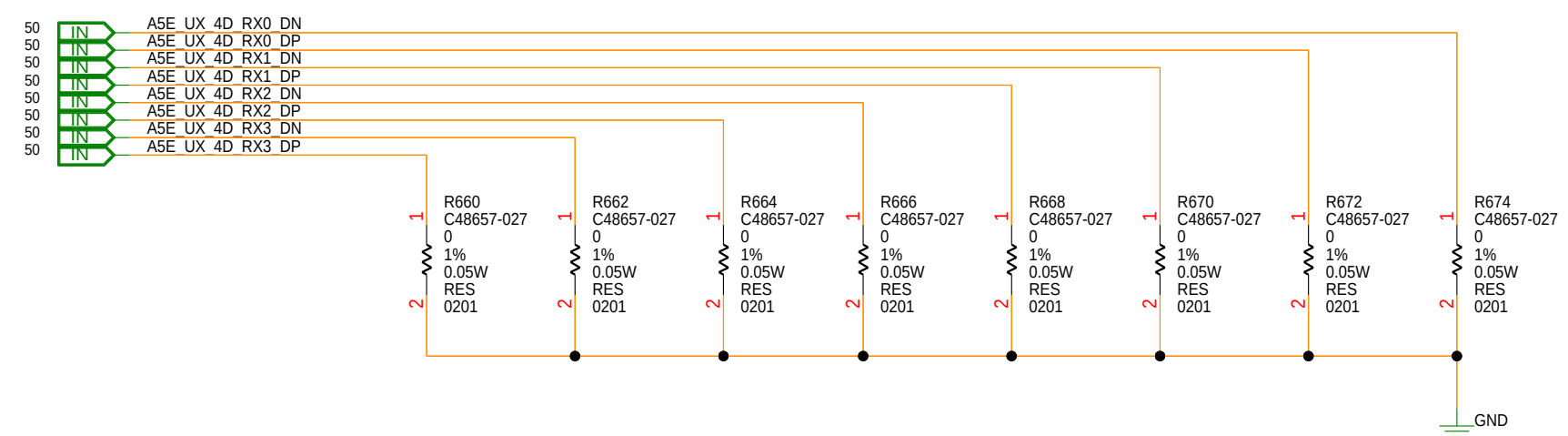
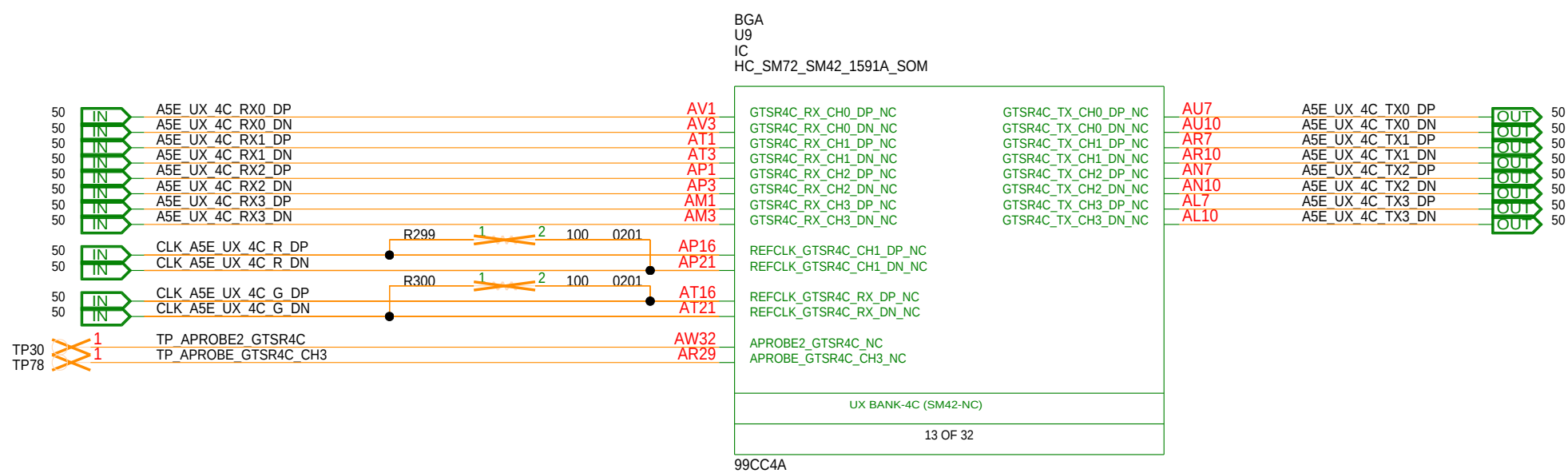
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AG5E BANK- UX -1A ,1B,1C

DEPARTMENT



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B



A



AG5E BANK - HVIO - 5A,5B

4

3

2

1

BANK 6A

VCCIO_HVIO= +V3P3

CAD NOTE:
PLACE THEM AS CLOSE AS POSSIBLE TO AVOID LONGER STUB

51 48 46 49
IN BI I2C_SCL1 C2M_3V3 R54 1 2 0 0201
51 49 48 46
BI I2C_SDA1 C2M_3V3 R79 1 2 0 0201

49 42 38 37
BI I2C_SDA2 C2M_3V3 R123 1 2 0 0201
42 38 49 37
IN IN I2C_SCL2 C2M_3V3 R119 1 2 0 0201

41 42
IN USB_ID_3V3 BR22
42 IN USB3_MUX_INT_N BK28
42 IN USB3_FAULT_N BP22
37 IN CLK_ASE_HVIO_6A_100M BK31

42 OUT USB3_MUX_EN_N BM31
50 BI ASE_I2C_PERSTN HVIO_6A_7_3V3 BW28
42 OUT ASE_USB_VBUS_EN_3V3 BU28
42 IN ASE_VBUS_DET_3V3 BU31

52 IN I2C_SDA2_R BR31
52 IN I2C_SCL2_R BR28
52 IN ASE_WARM_FPGA_RESET_3V3 BP31
52 IN ASE_FPGA_RESET_N_3V3 BU28

BGA U9
IC HC_SM72_SM42_1591A_SOM

HVIO_6A_12_SOURCE_SYNC_CLK2_TXCLK12_RXCLK4_DATA_CTRL12
HVIO_6A_11_SOURCE_SYNC_CLK1_TXCLK11_RXCLK3_DATA_CTRL11
HVIO_6A_10_PLLREFCLK2_TXCLK10_RXCLK2_DATA_CTRL10
HVIO_6A_9_PLLREFCLK1_TXCLK9_RXCLK1_DATA_CTRL9

HVIO_6A_8_TXCLK8_DATA_CTRL8
HVIO_6A_7_PIN_PERST_N_R4C_0_TXCLK7_DATA_CTRL7
HVIO_6A_6_PIN_PERST_N_R4B_0_TXCLK6_DATA_CTRL6
HVIO_6A_5_PIN_PERST_N_R4A_0_TXCLK5_DATA_CTRL5

HVIO_6A_4_SYSPLLREFCLK_R4B_1_TXCLK4_DATA_CTRL4
HVIO_6A_3_SYSPLLREFCLK_R4B_0_TXCLK3_DATA_CTRL3
HVIO_6A_2_SYSPLLREFCLK_R4A_1_TXCLK2_DATA_CTRL2
HVIO_6A_1_SYSPLLREFCLK_R4A_0_TXCLK1_DATA_CTRL1

LEFT RIGHT

HVIO BANK 6A

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99CC4A

CF9 ASE_HVIO_6A_20_3V3 BI 50
BK19 REMOTE_EN_C2M_3V3 IN 49
CF12 ASE_GPIO_LED OUT 47
BM22 REMOTE_DIP_FPGA_3V3 IN 52

BH28 REMOTE_PB_FPGA_3V3 IN 52
BW19 ASE_HVIO_PUSH_BUTTON IN 47
BU22 ASE_GPIO_DIP_SWITCH0 IN 47
CH12 ASE_GPIO_DIP_SWITCH1 IN 47

[illegible]

AG5E BANK - HVIO - 6A,6B		DEPARTMENT	altera™	SIZE C+	CODE 34649	DOCUMENT NUMBER 150-0330723-B1		REV 2.1
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COD
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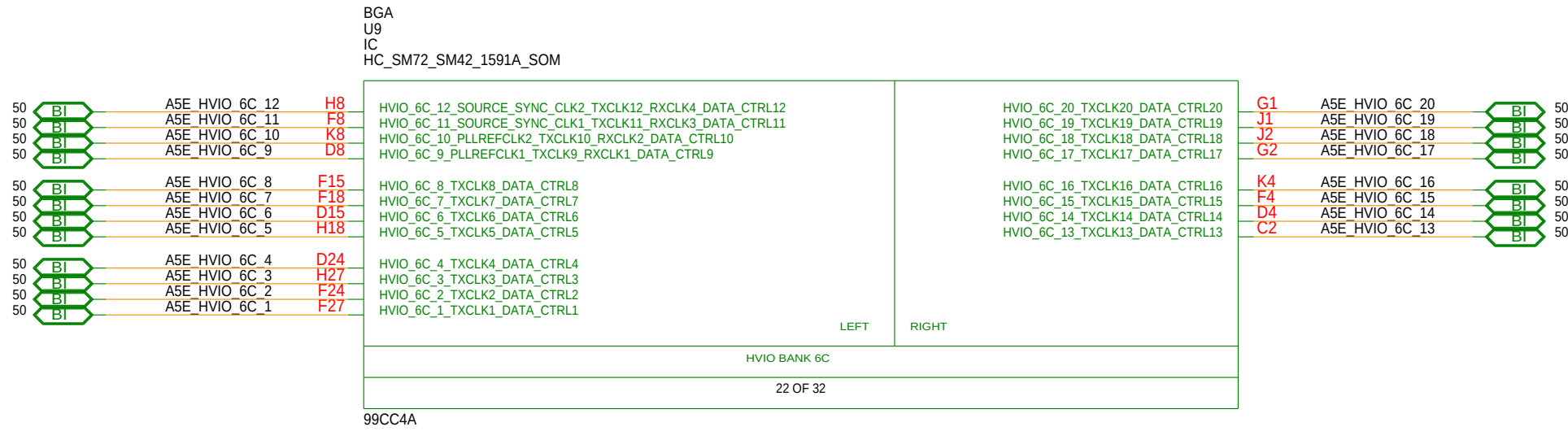
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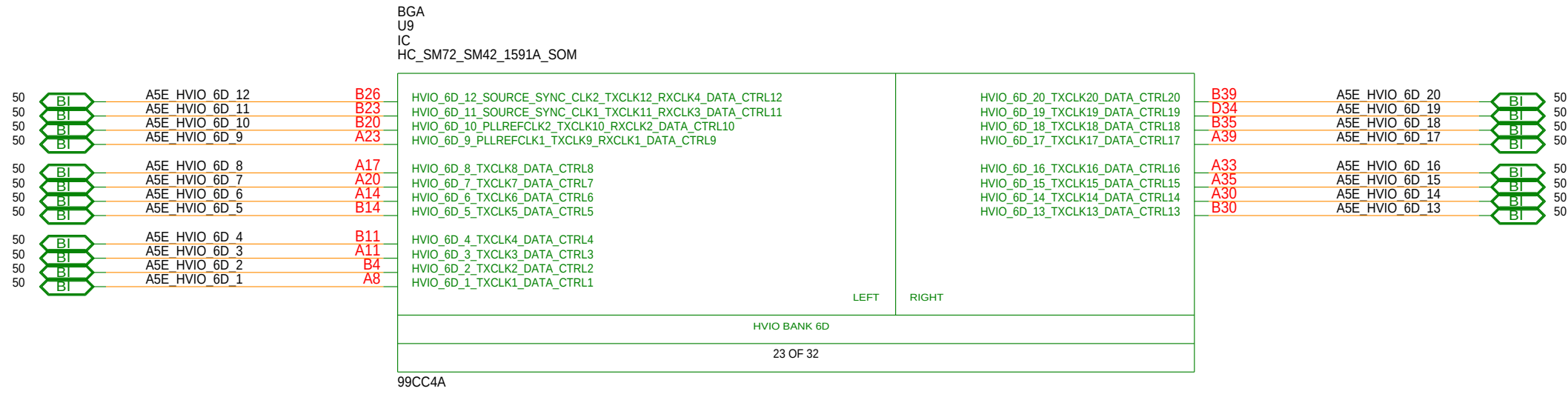
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BANK 6C



BANK 6D



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AG5E BANK - HVIO - 6C,6D

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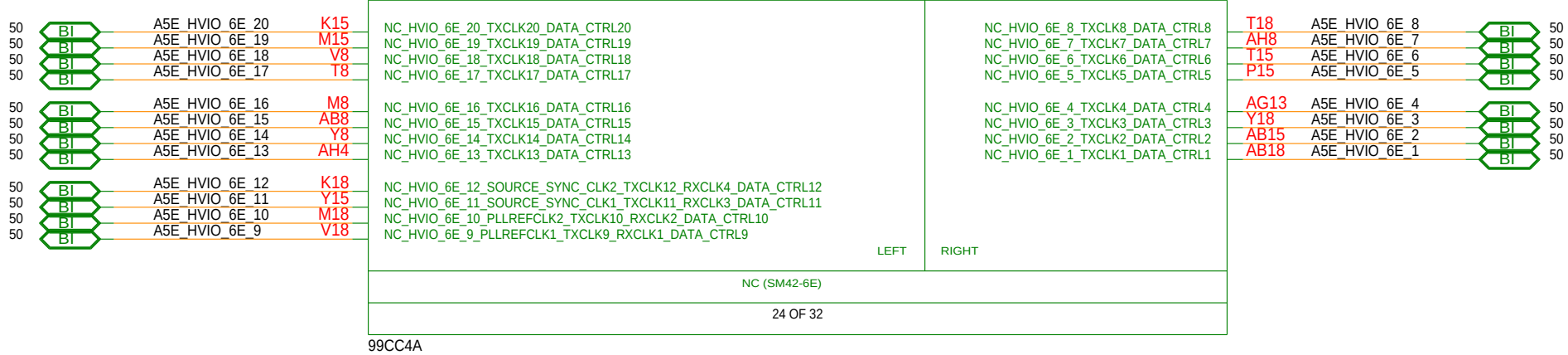
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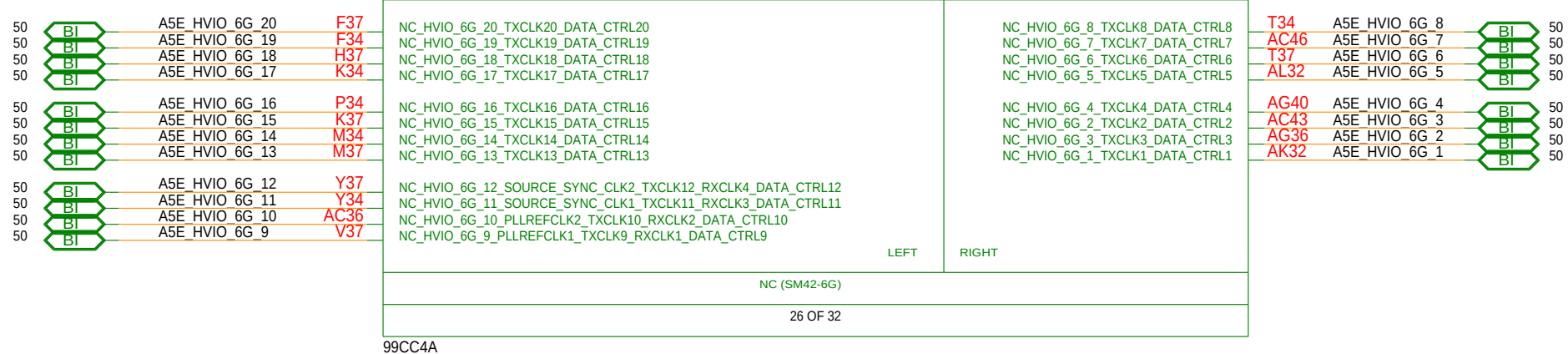
BANK 6E

BANK 6G

BGA
U9
IC
HC_SM72_SM42_1591A_SOM



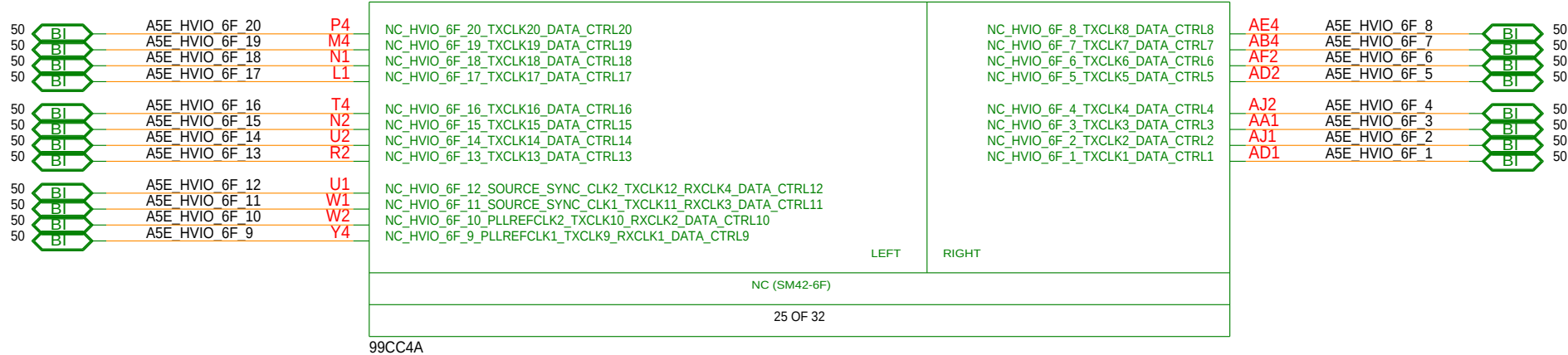
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IC
HC_SM72_SM42_1591A_SOM



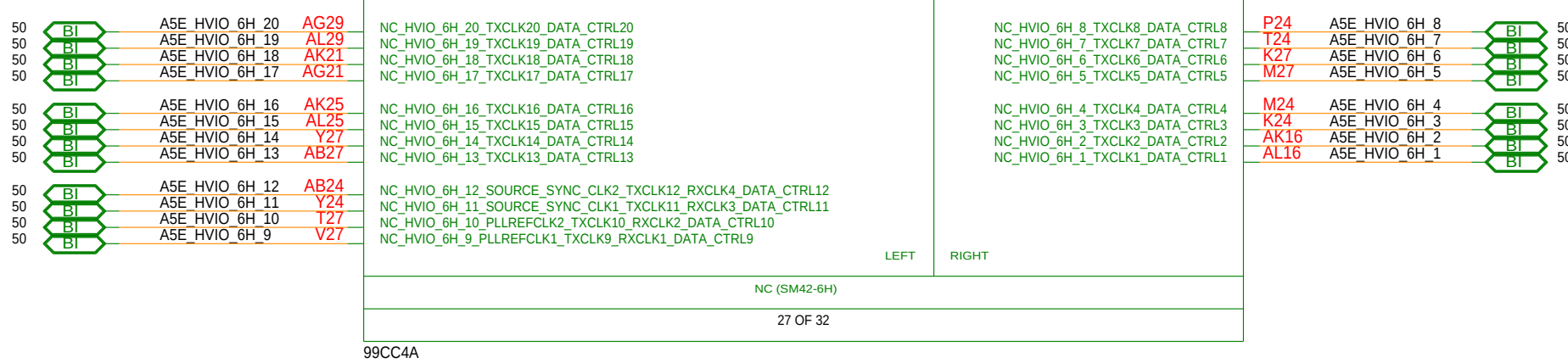
BANK 6F

BANK 6H

BGA
U9
IC
HC_SM72_SM42_1591A_SOM



BGA
U9
IC
HC_SM72_SM42_1591A_SOM



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AG5E BANK - HVIO - 6E,6F,6G,6H

DEPARTMENT



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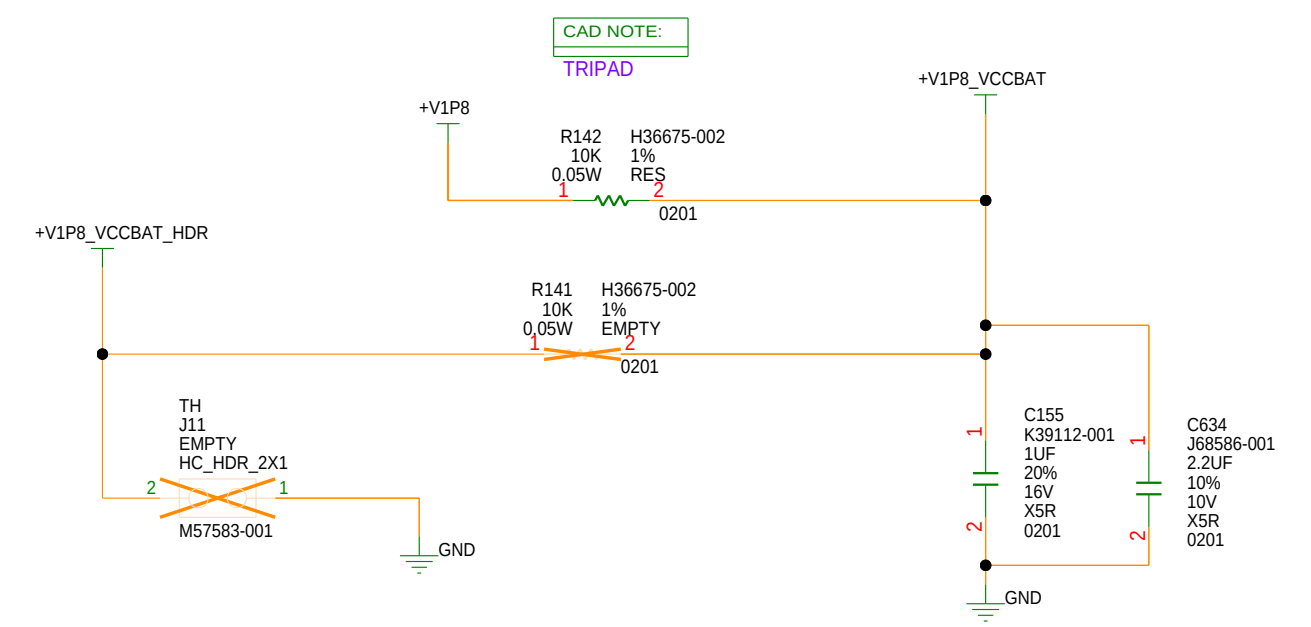
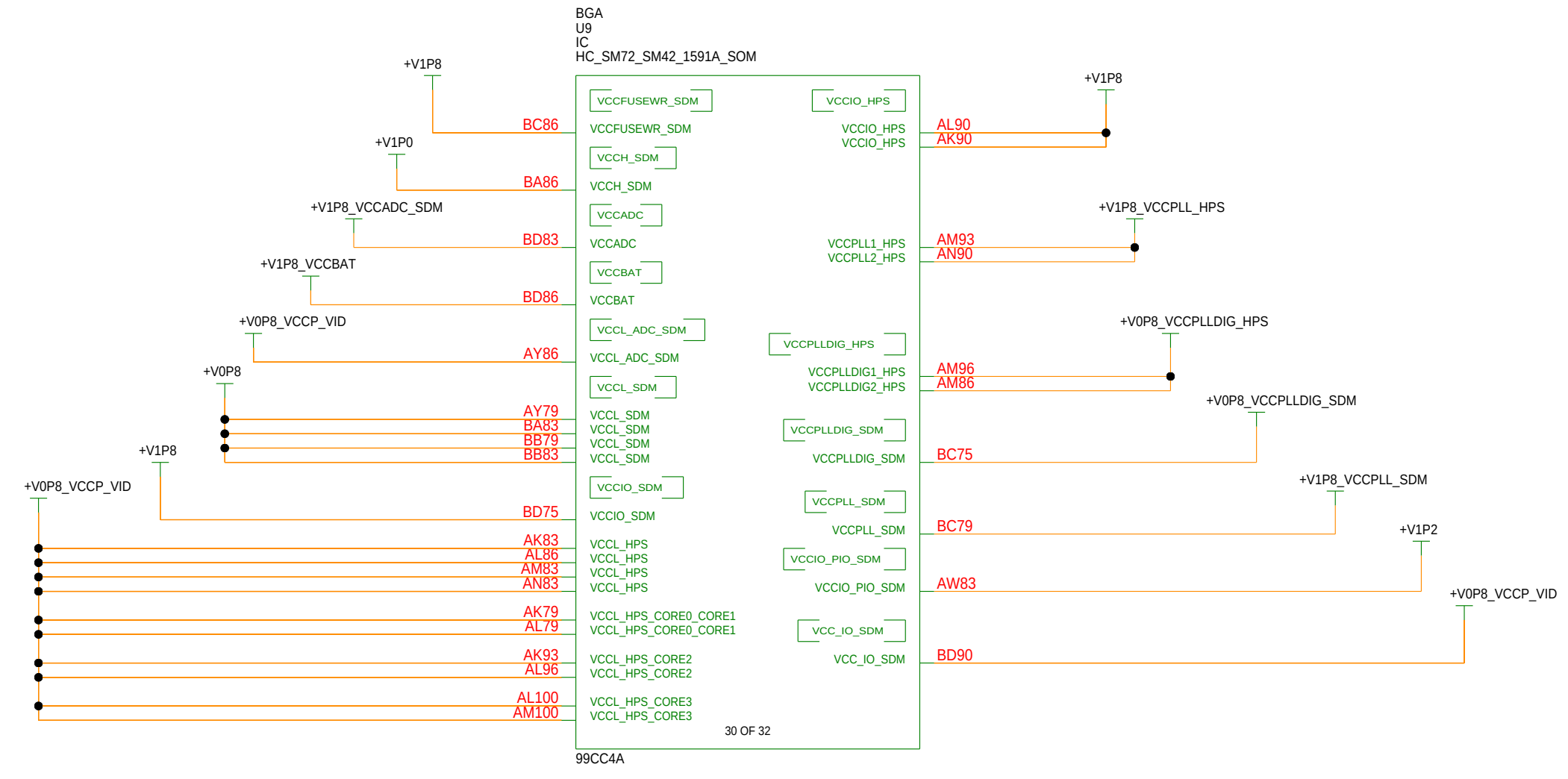
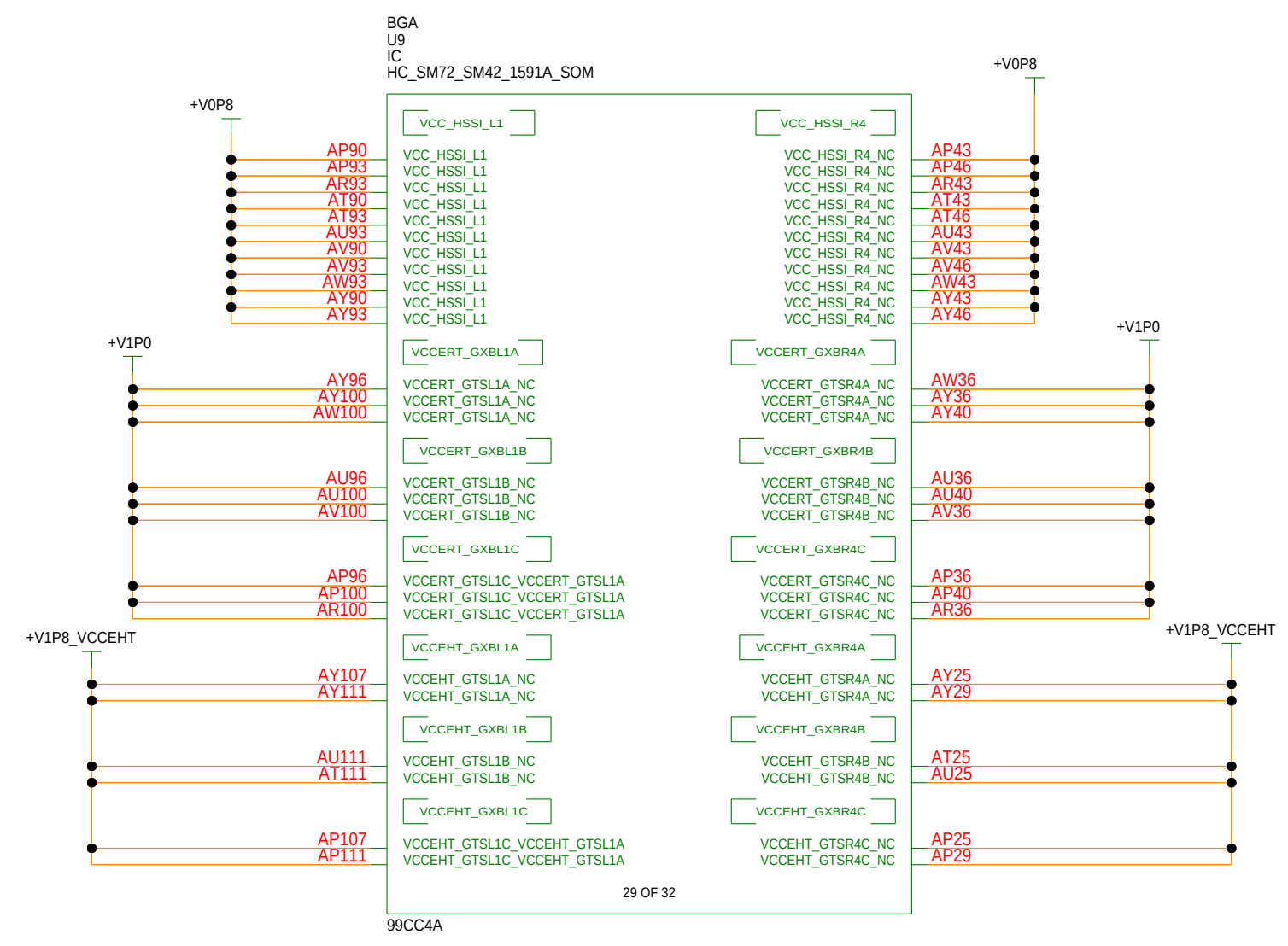
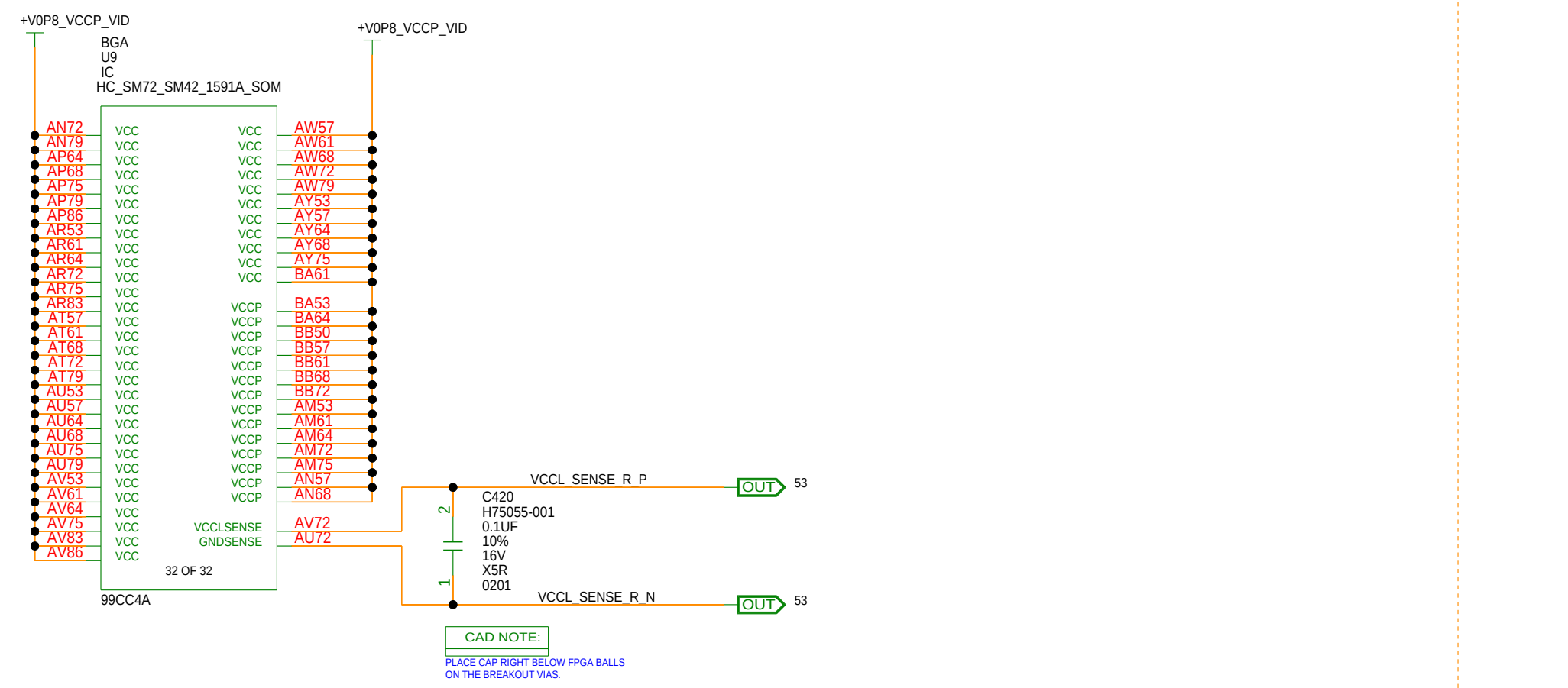
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AG5E - POWER

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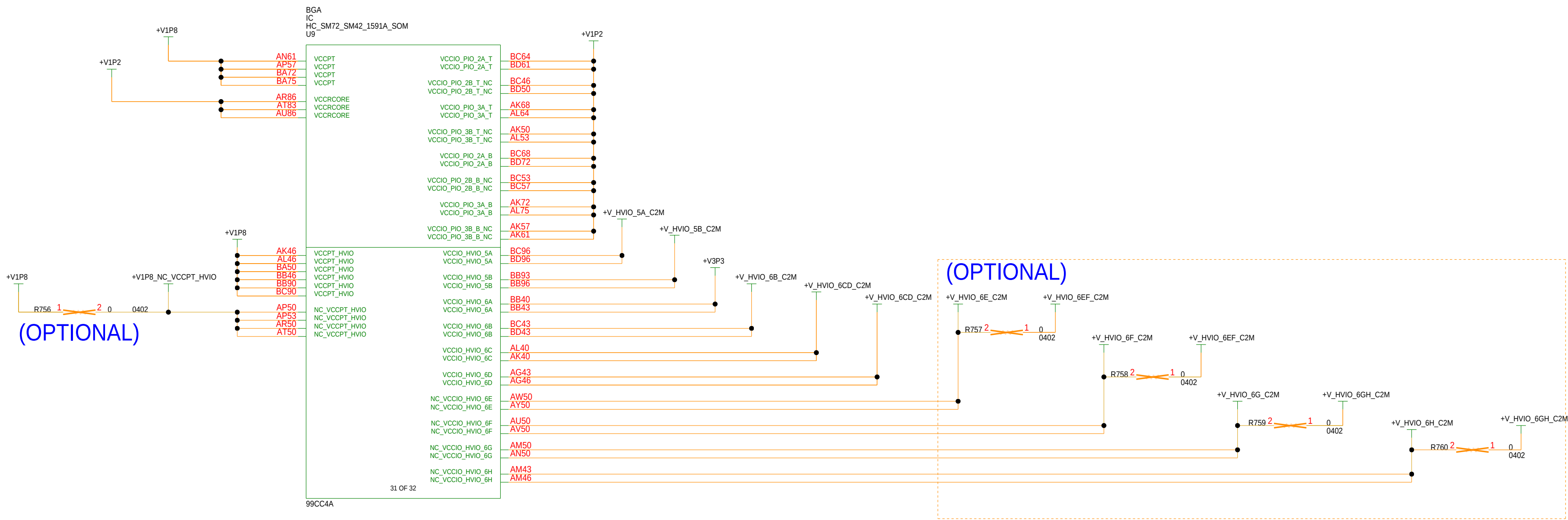
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B

B

A

A



AG5E - POWER

DEPARTMENT



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C+

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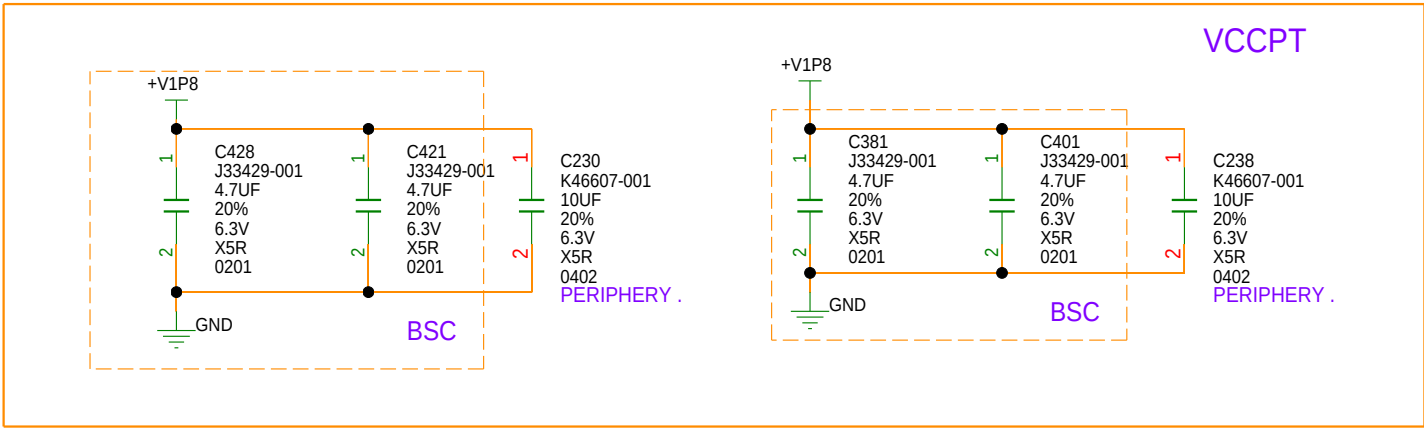
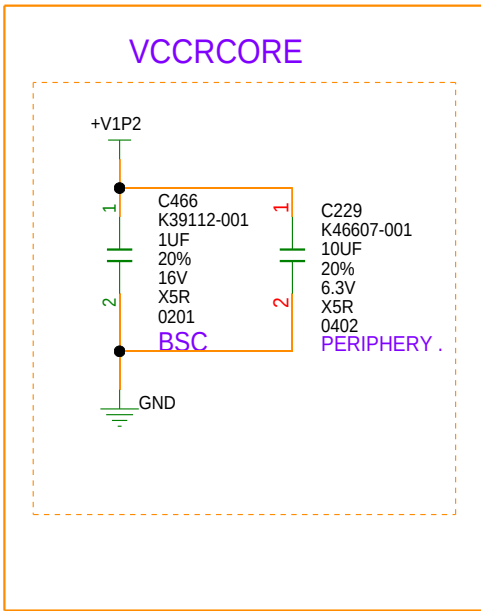
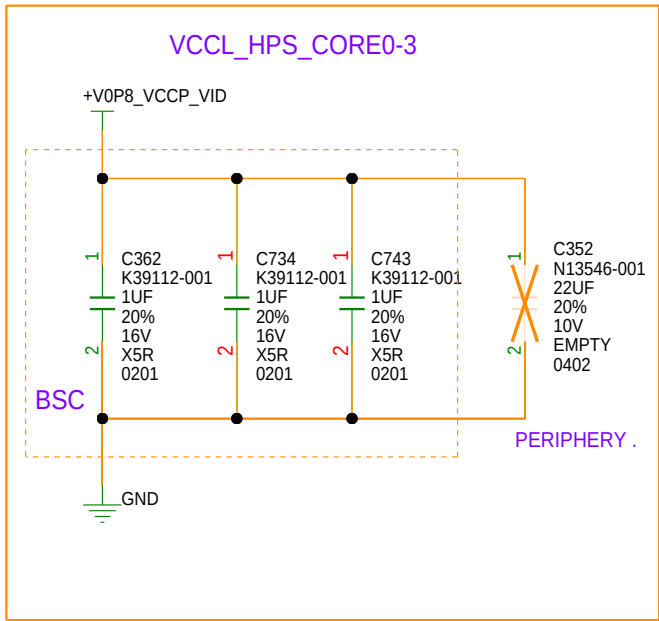
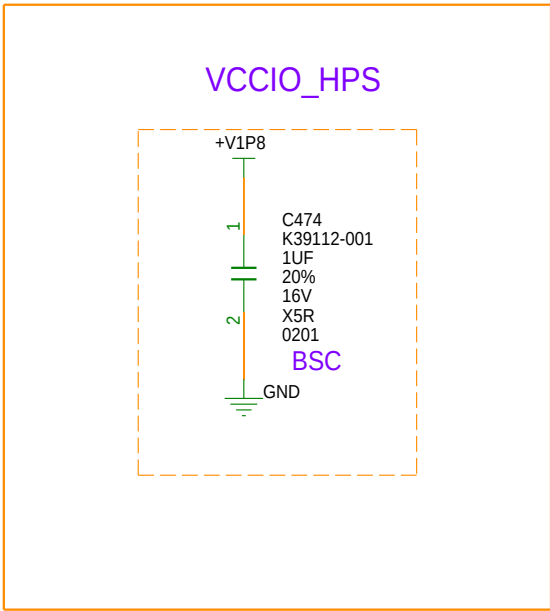
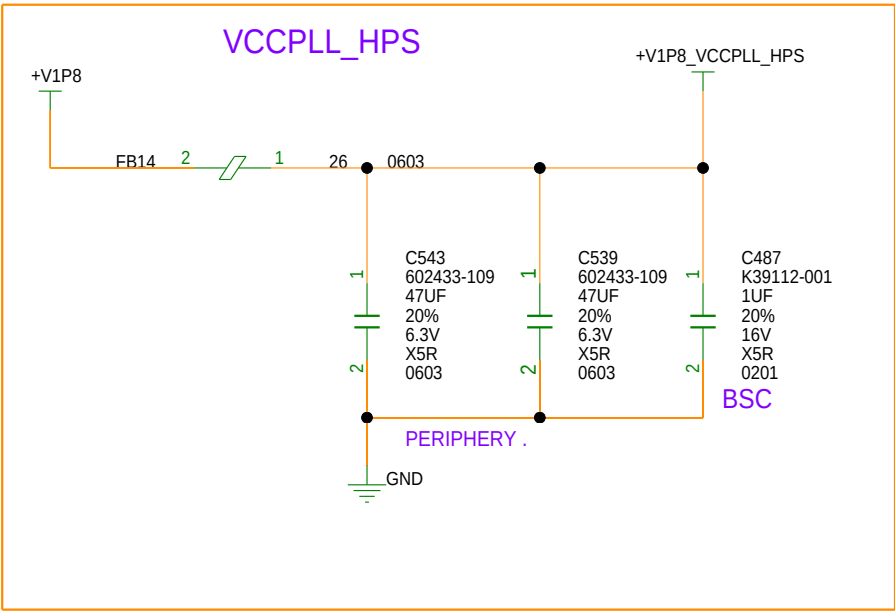
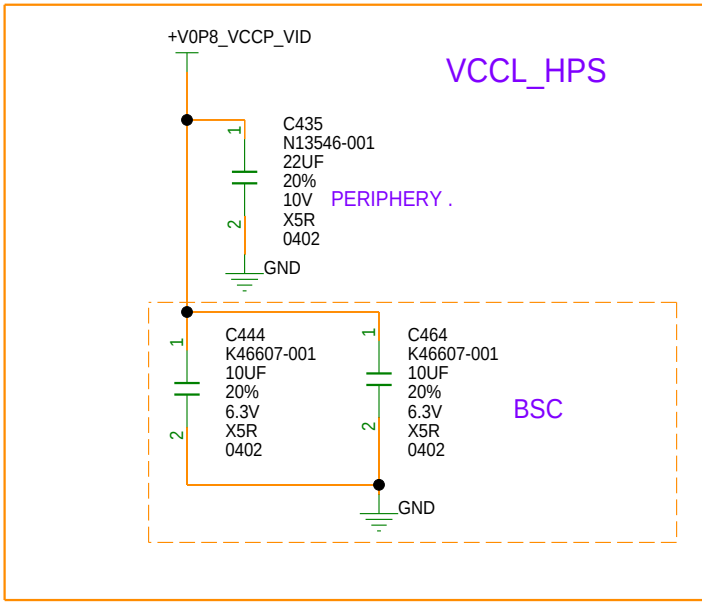
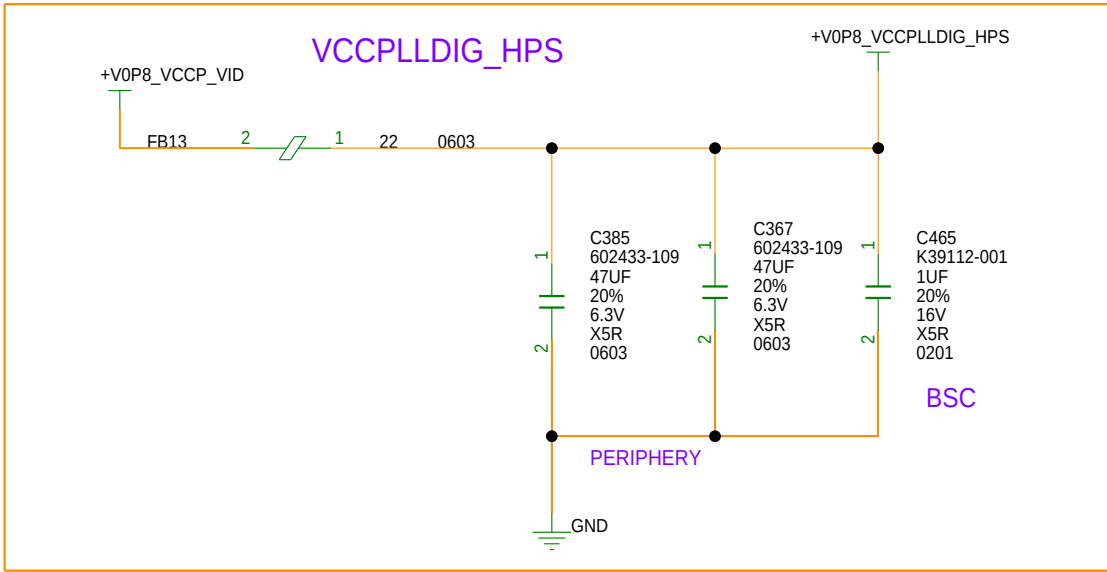
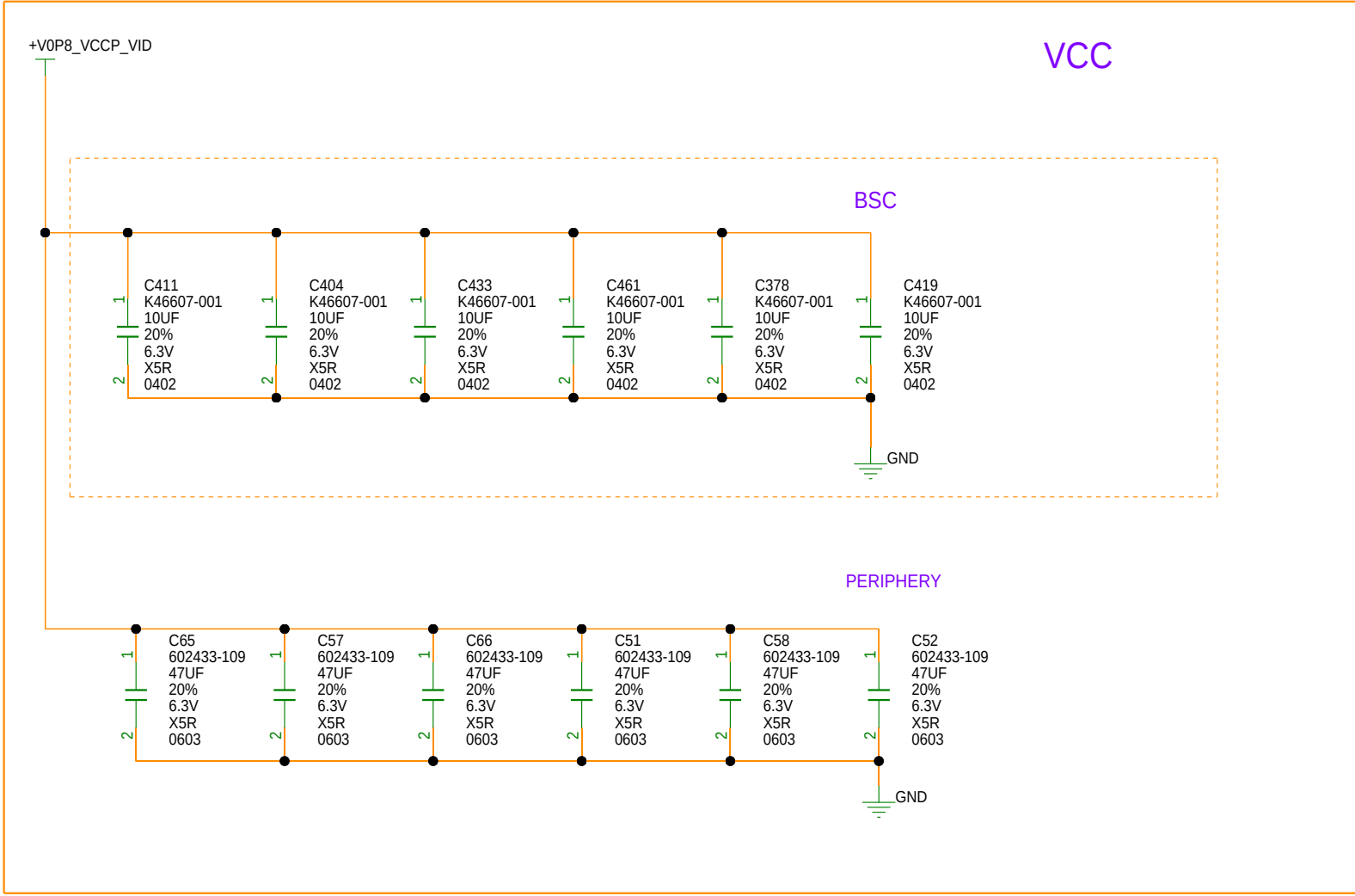
SCALE:

DO NOT SCALE DRAWING

SHEET

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Mon Jul 28 12:45:55 2025

AG5E - DECAP

DEPARTMENT



SIZE

CODE

DOCUMENT NUMBER

REV

C+

34649

150-0330723-B1

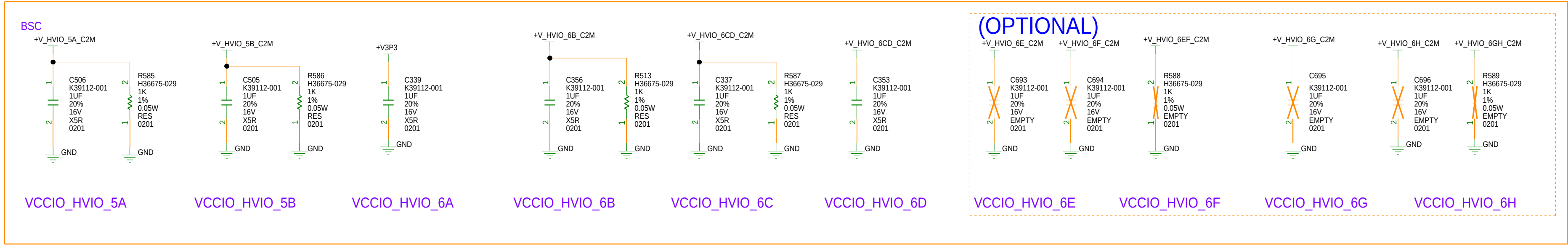
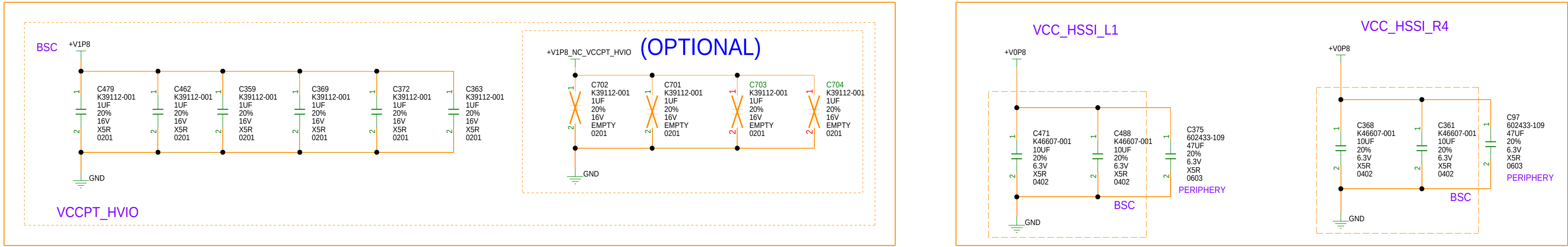
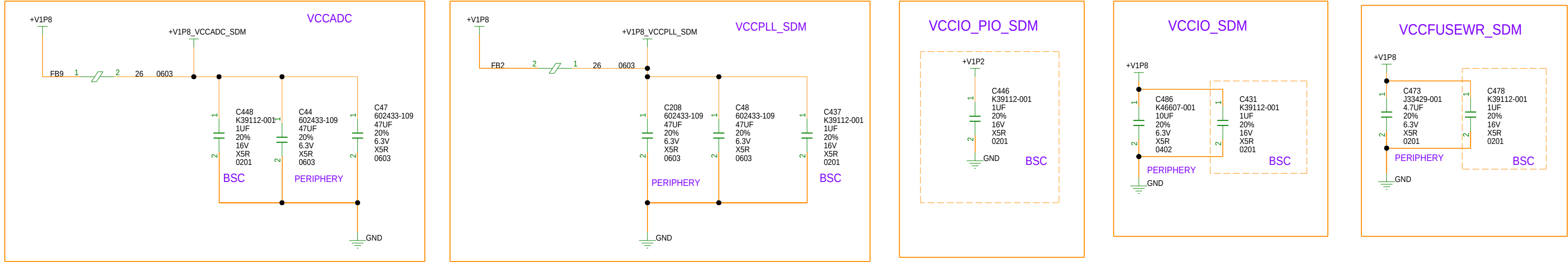
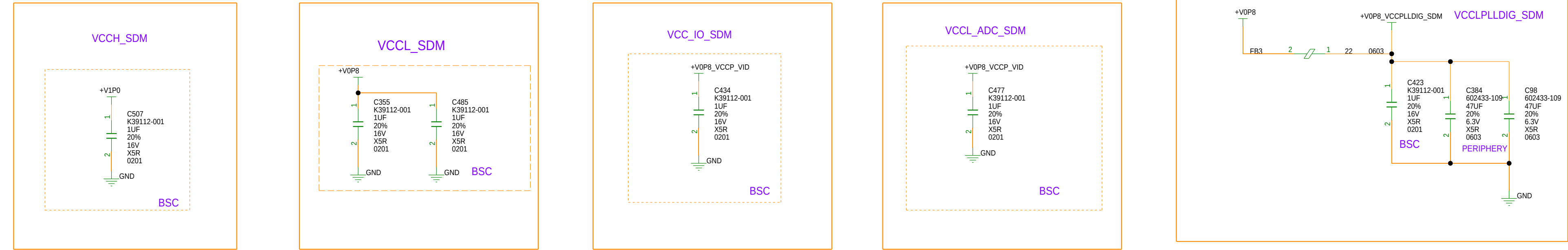
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SCALE:

DO NOT SCALE DRAWING

SHEET

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Mon Jul 28 12:45:57 2025

AG5E - DECAP

DEPARTMENT



SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

2.1

SCALE:

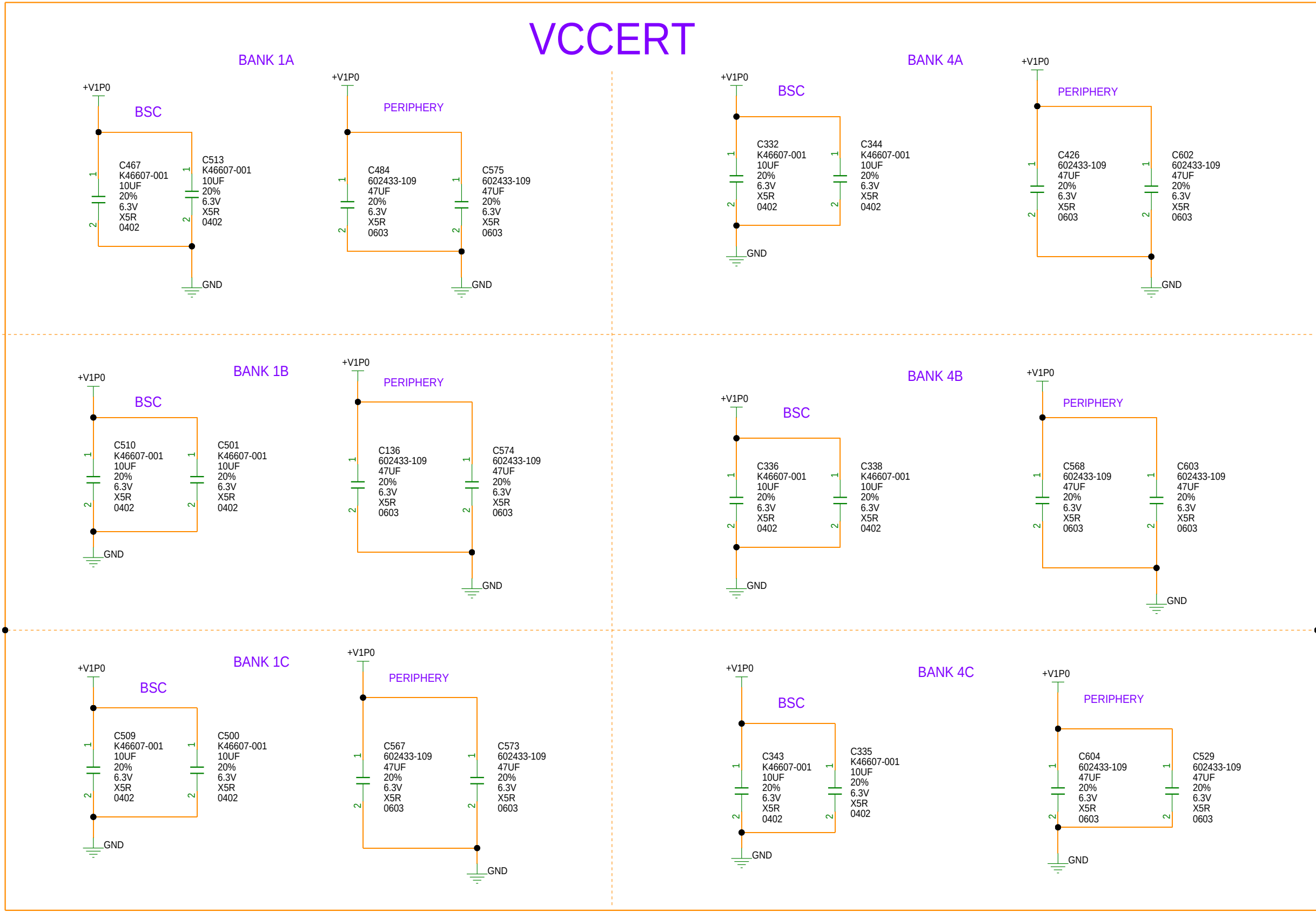
DO NOT SCALE DRAWING

SHEET

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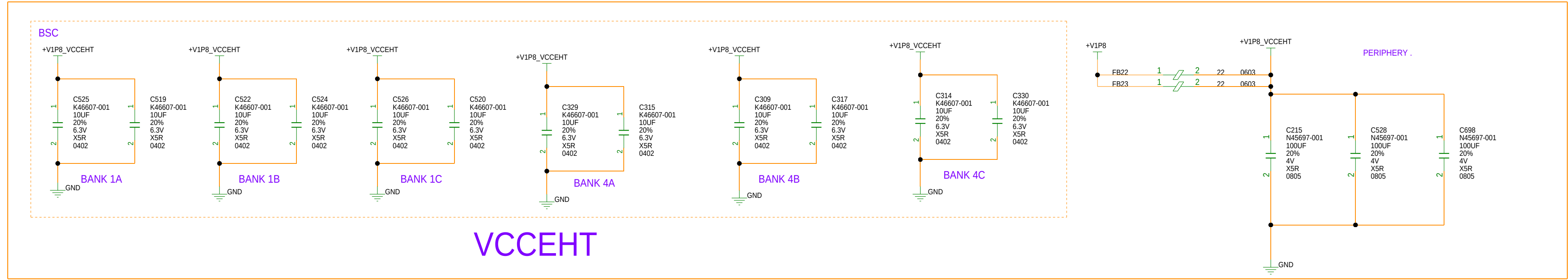
B

B



A

A



AG5E - DECAP

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330723-B1	2.1
		SCALE:	DO NOT SCALE DRAWING		SHEET 28 OF 62

CLAMSHELL TOPOLOGY

TOP SIDE

BL4 - DQ0

BL0 - DQ1

B

B

A

A

Address Mirroring

Memory Controller Pin	DRAM Pin (Non-Mirrored)	DRAM Pin (Mirrored)
A3	A3	A4
A4	A4	A3
A5	A5	A6
A6	A6	A5
A7	A7	A8
A8	A8	A7
A11	A11	A13
A13	A13	A11
BA0	BA0	BA1
BA1	BA1	BA0
BG0 ⁽¹⁾	BG0	BG1
BG1 ⁽¹⁾	BG1	BG0

⁽¹⁾ BG0 and BG1 can be mirrored only when pin BG1 is present on the memory device.

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DDR4 X40 BANK2B

DEPARTMENT

altera

SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

2.1

SCALE:

DO NOT SCALE DRAWING

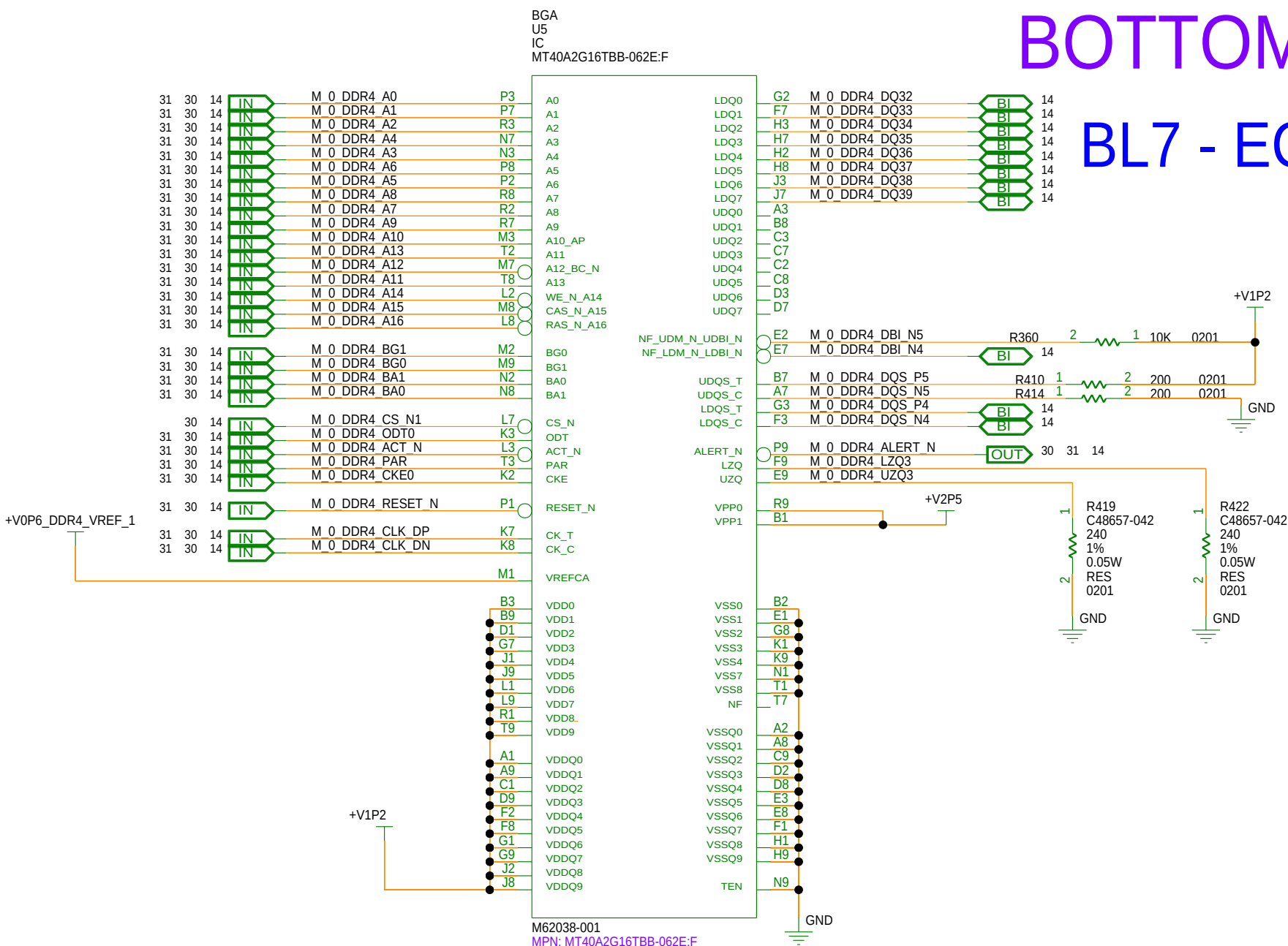
SHEET

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CLAMSHELL TOPOLOGY

BOTTOM SIDE

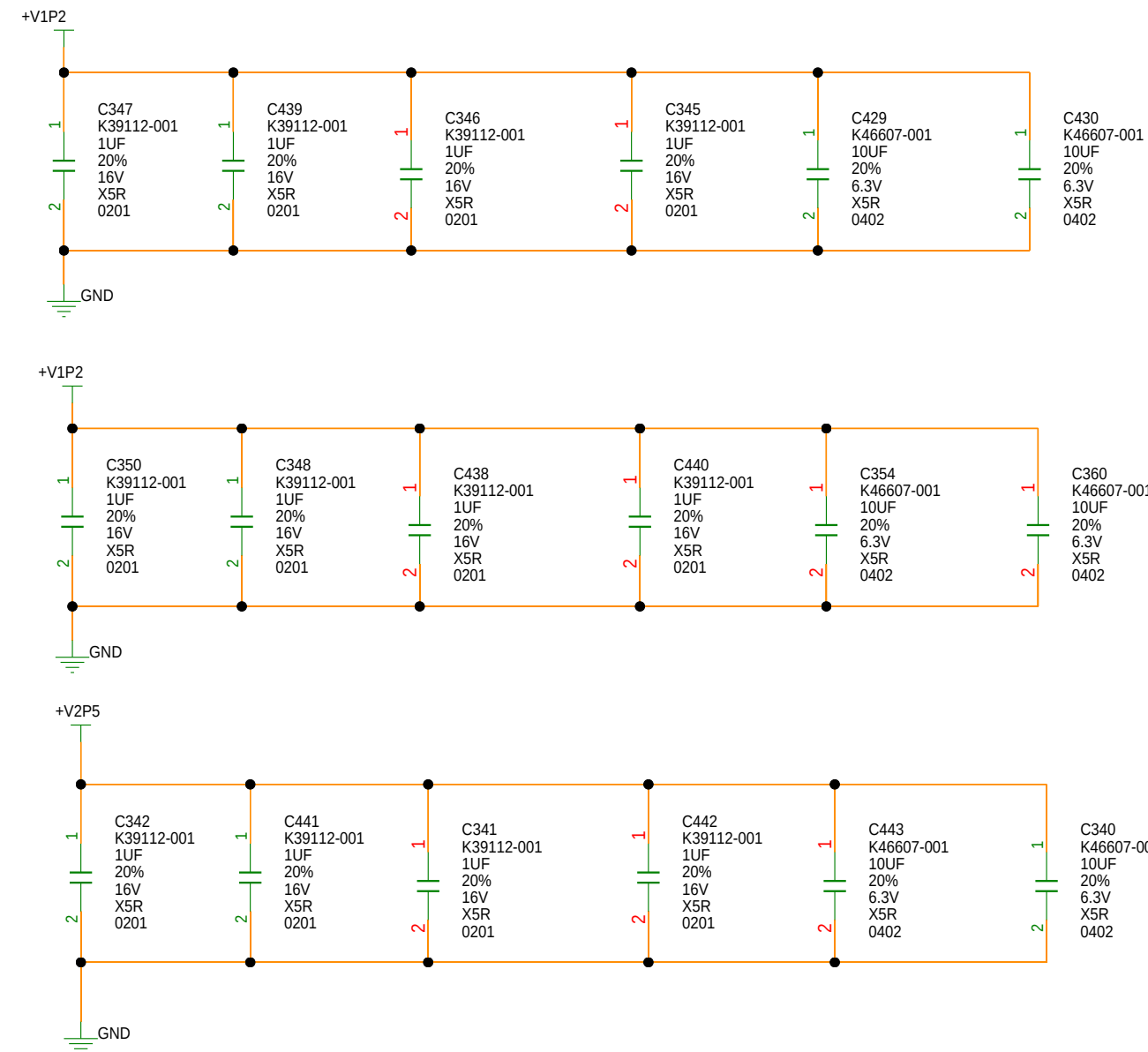
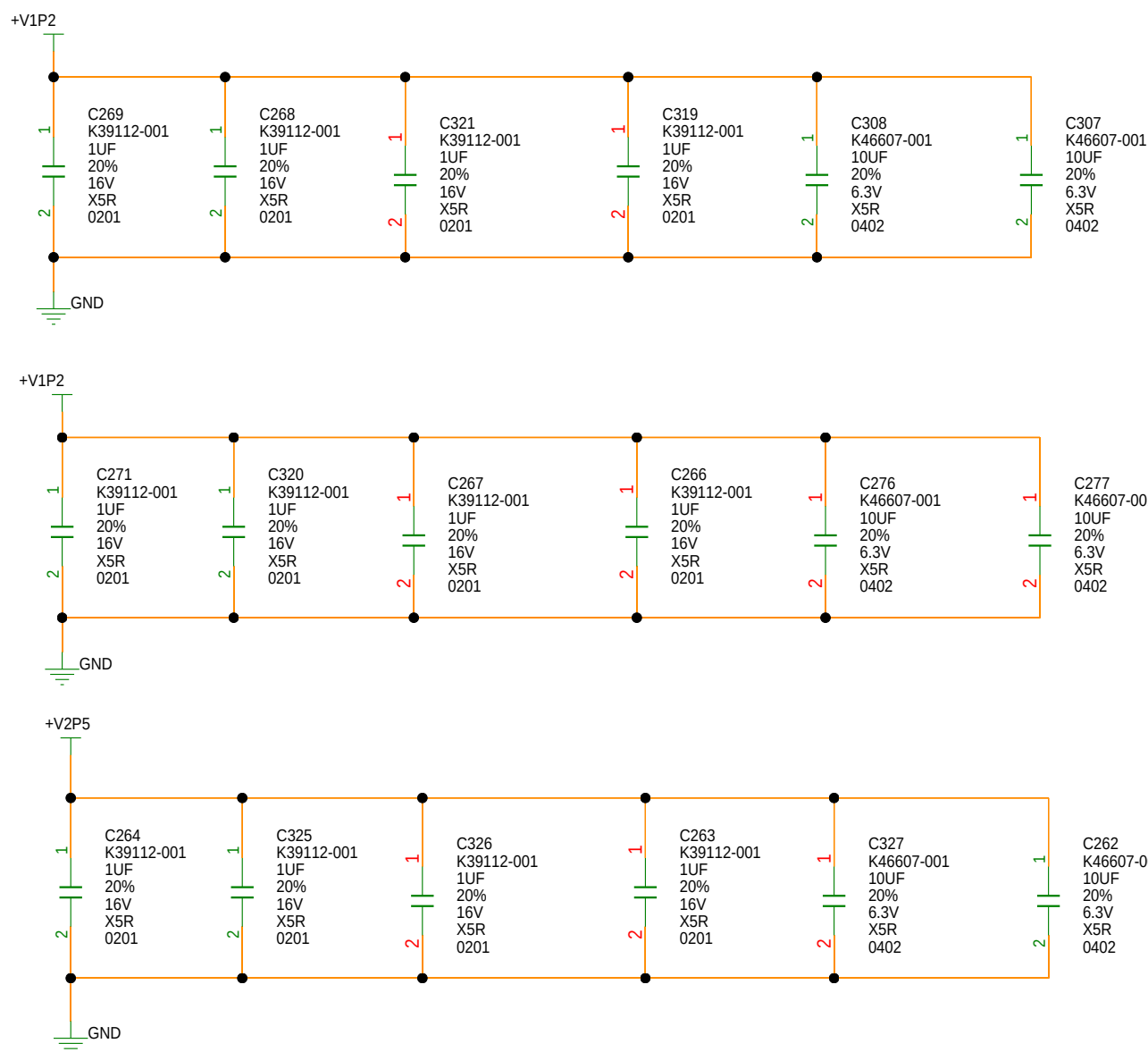
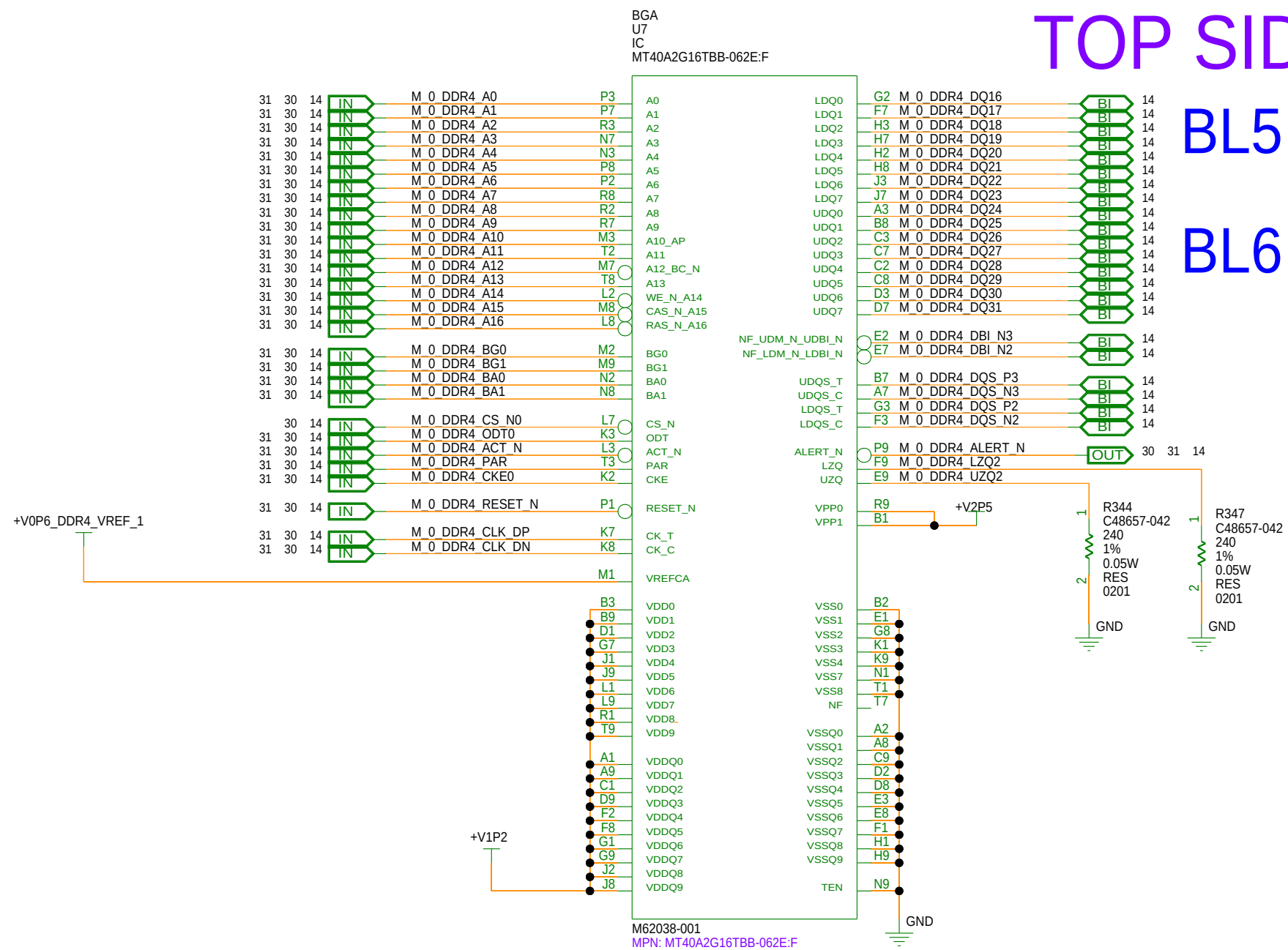
BL7 - ECC



TOP SIDE

BL5 - DQ2

BL6 - DQ3



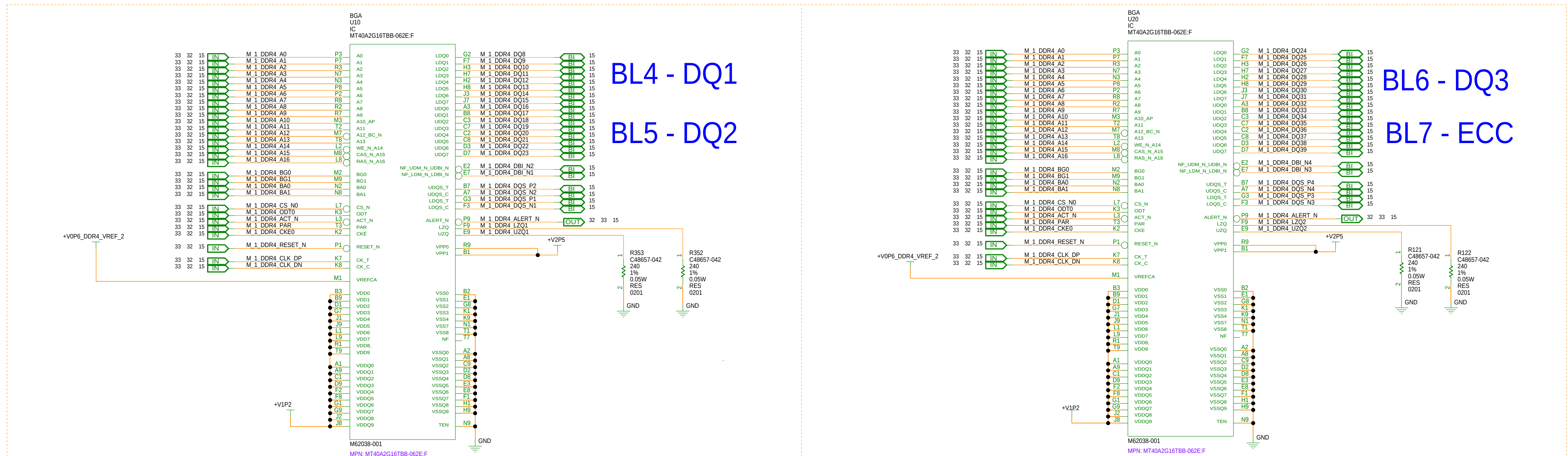
DDR4 X40 BANK2B

DEPARTMENT

altera

SIZE	CODE	DOCUMENT NUMBER	REV
C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING		SHEET 30 OF 62

B



B

A

A

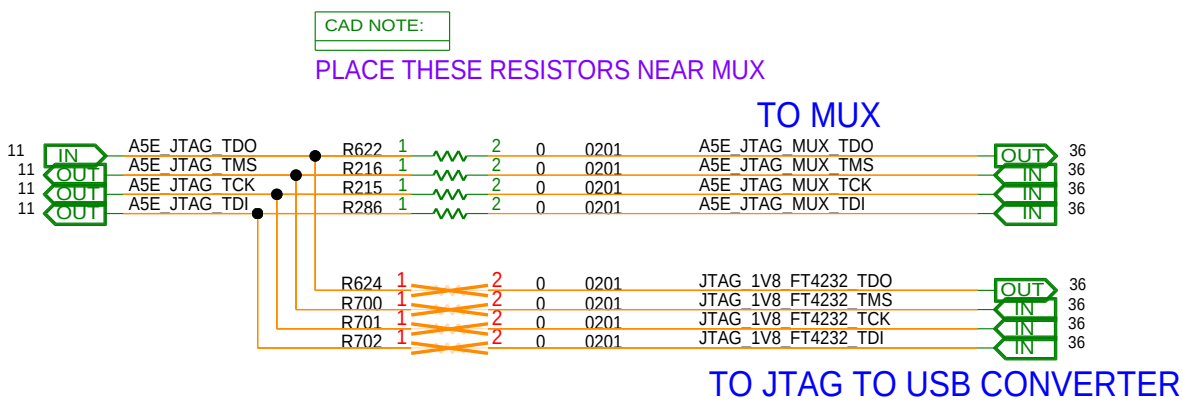
Mon Jul 28 12:46:03 2025

DDR4 X40 BANK3A

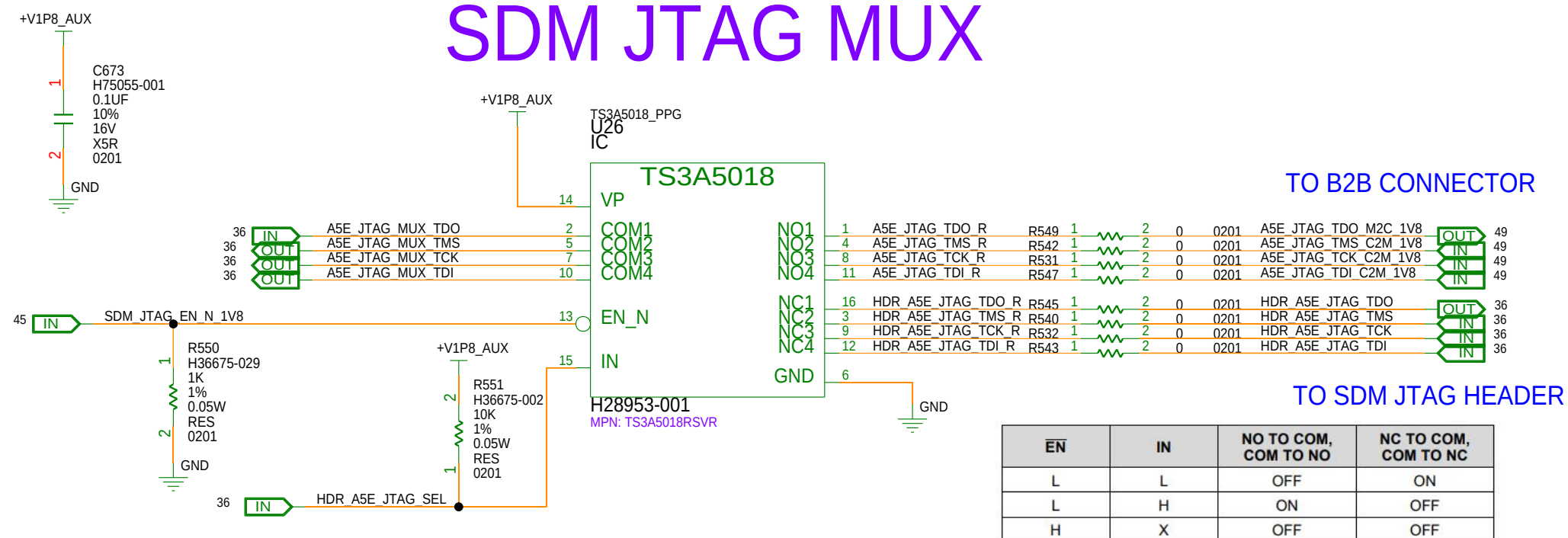
DEPARTMENT



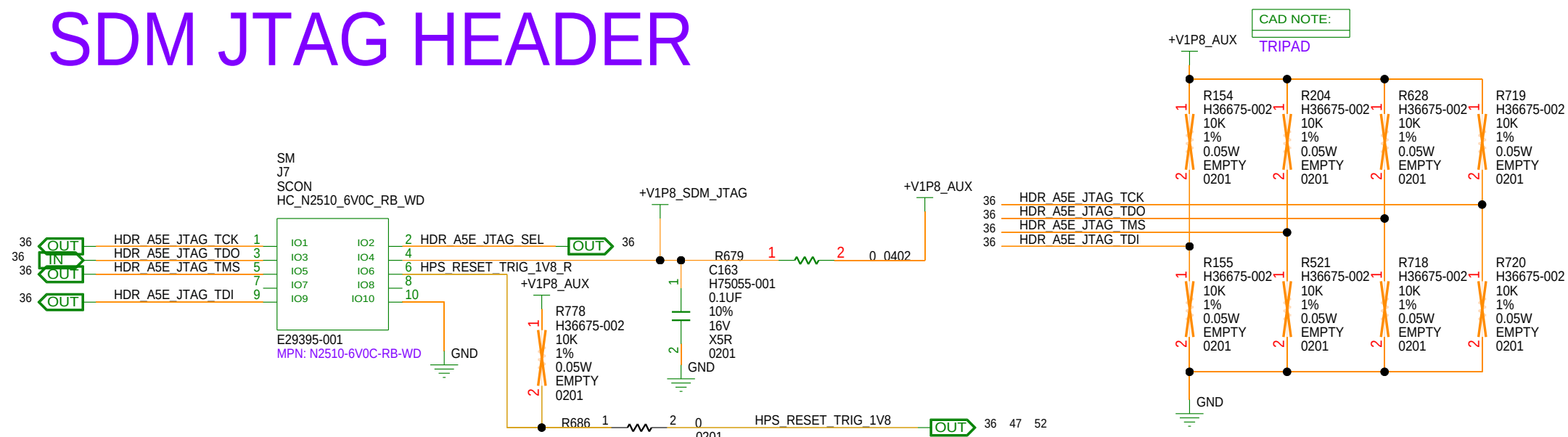
SIZE	CODE	DOCUMENT NUMBER	REV
C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING		SHEET 32 OF 62



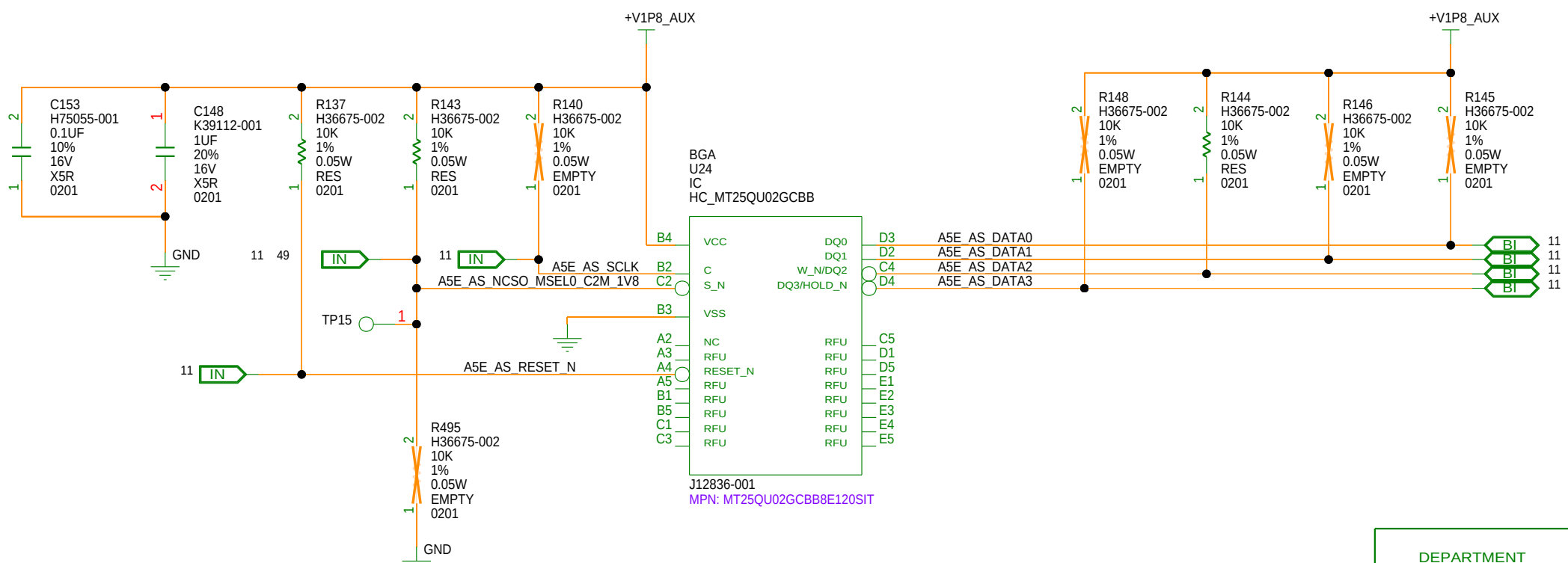
SDM JTAG MUX



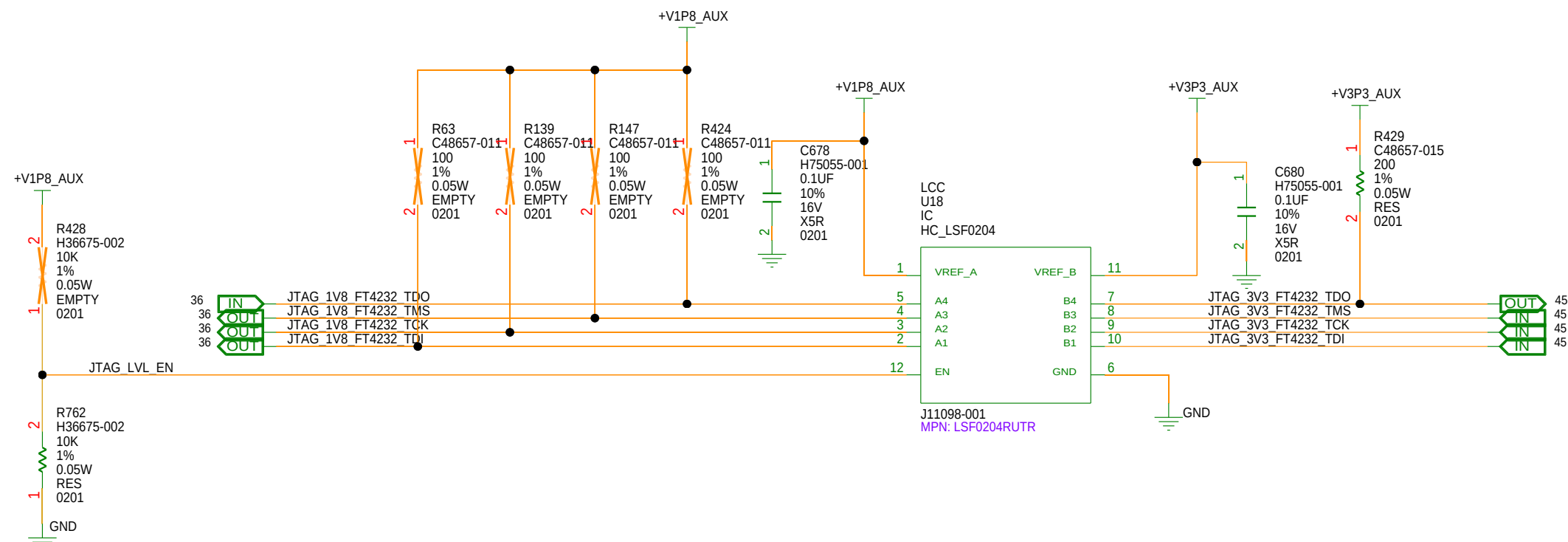
SDM JTAG HEADER



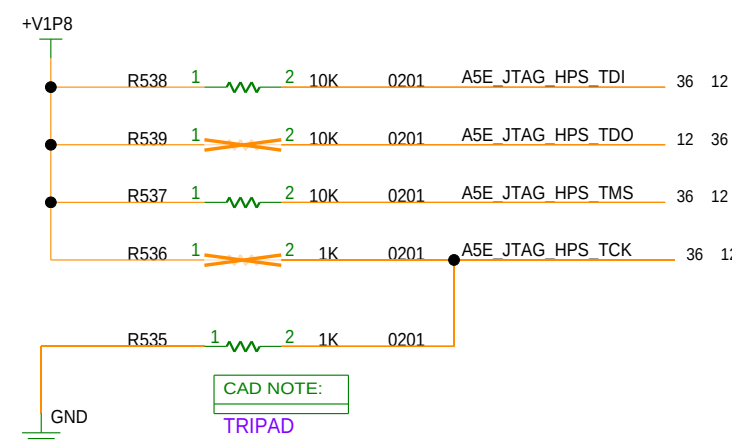
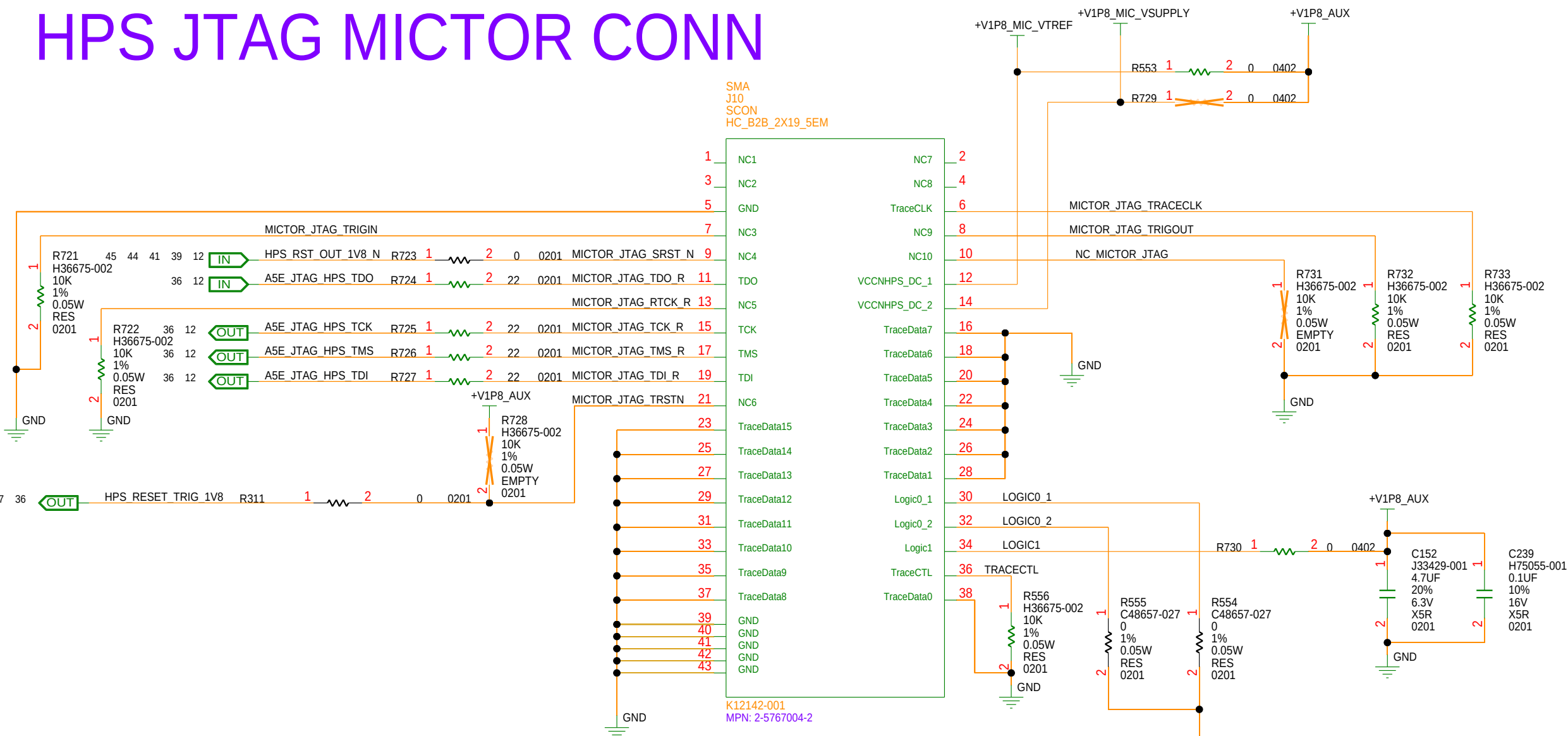
NOR FLASH



SDM JTAG LEVEL SHIFTER



HPS JTAG MICTOR CONN



AG5E - JTAG & NOR FLASH

DEPARTMENT

altera

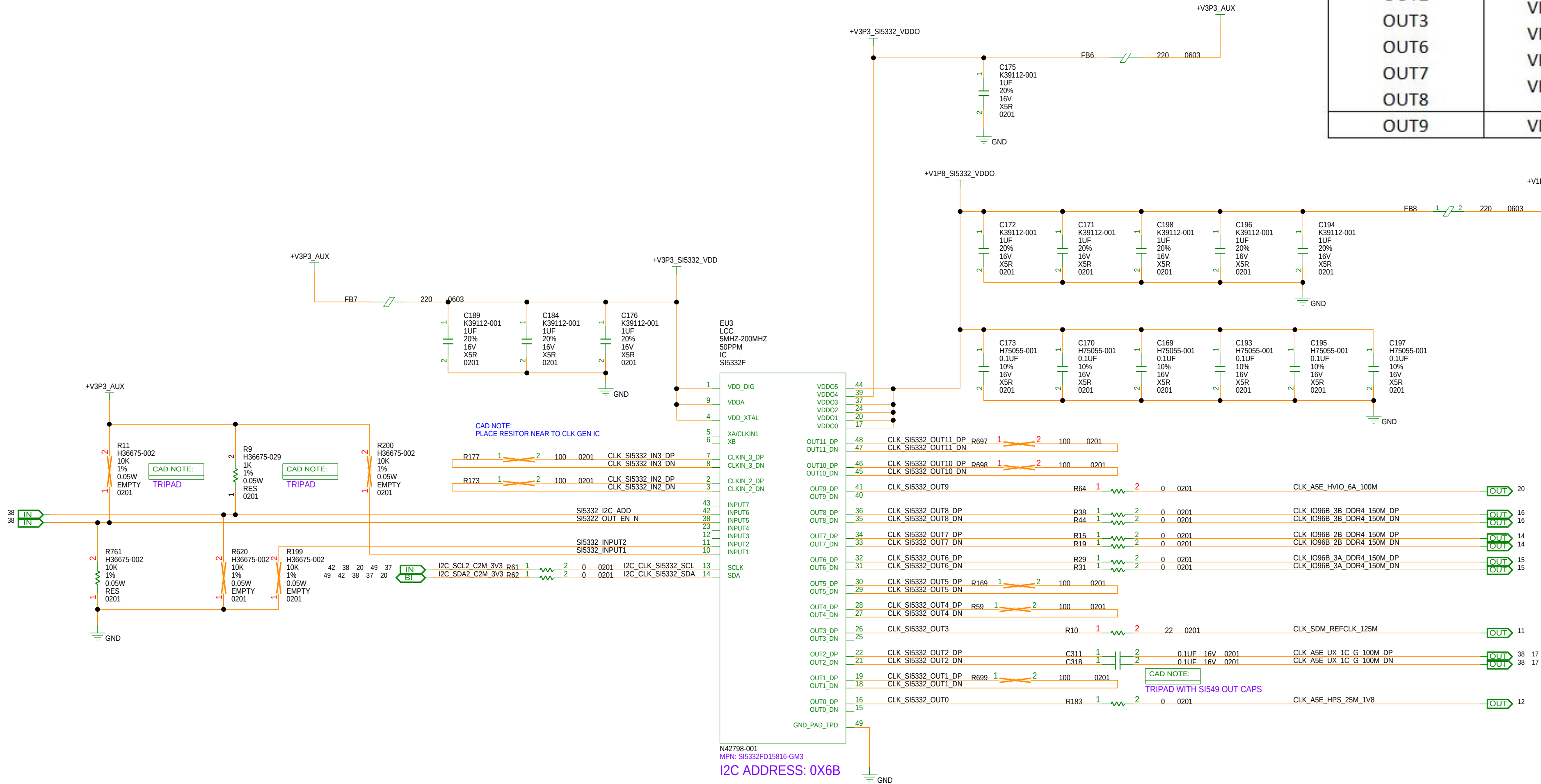
SIZE	CODE	DOCUMENT NUMBER	REV
C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING		SHEET 35 OF 62

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SI5332 Output Channel	SI5332 Output Supply Pin	Rail Name
OUT0	VDDO0	+V1P8_AUX
OUT2	VDDO1	
OUT3	VDDO2	
OUT6	VDDO3	
OUT7	VDDO5	
OUT8	VDDO5	+V3P3_AUX
OUT9	VDDO4	

B

B



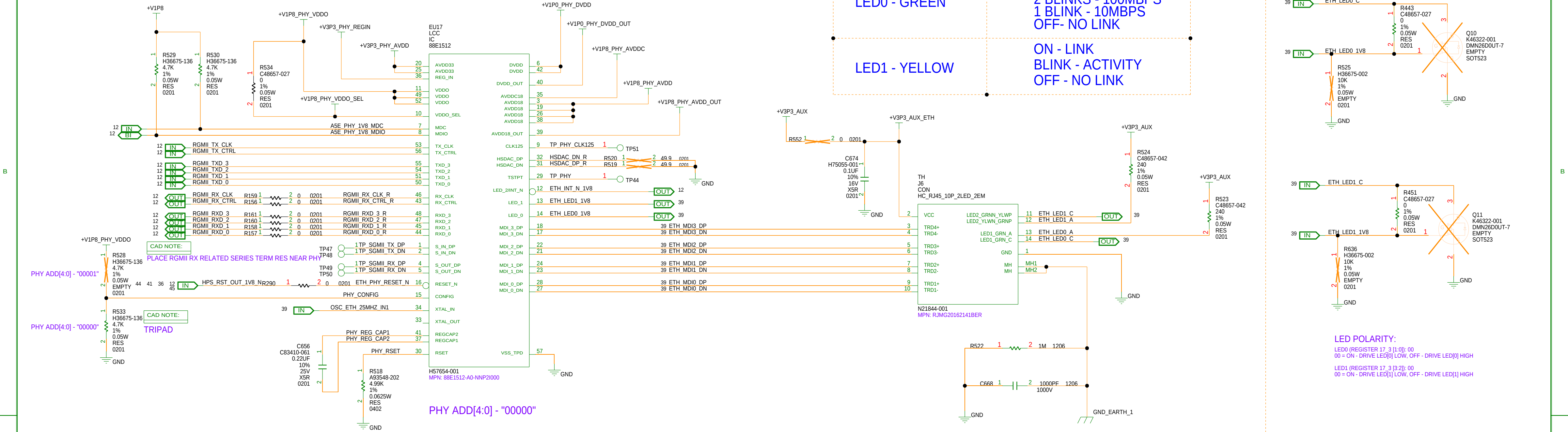
A

A

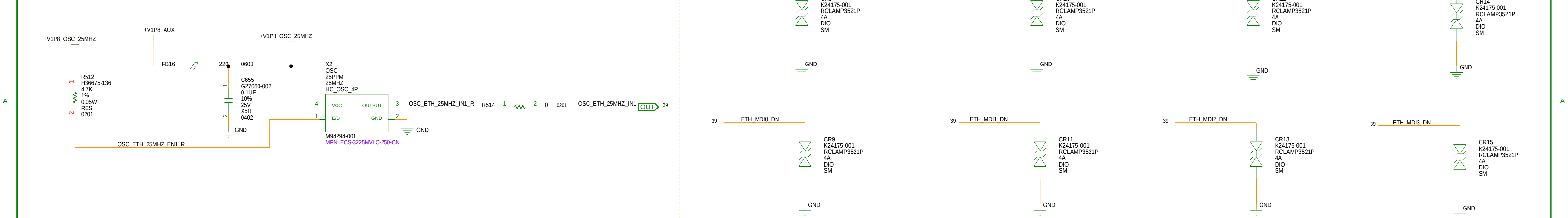
CLOCK_SI5332

Mon Jul 28 12:46:15 2025

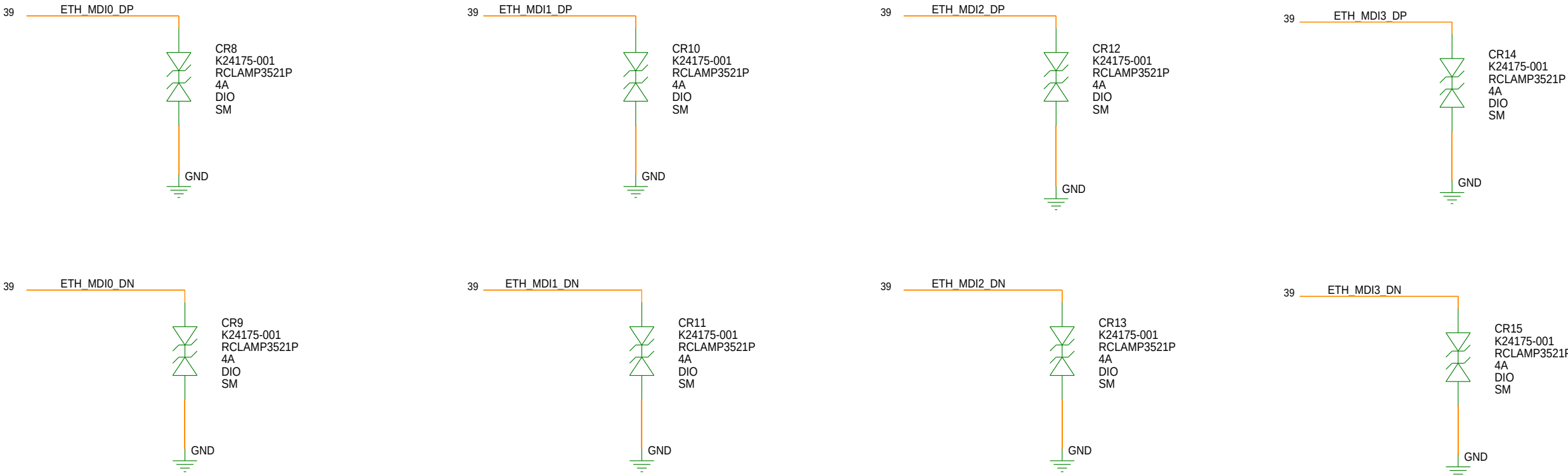
ETHERNET PHY



25MHZ CLOCK



ESD DIODE



Mon Jul 28 12:46:22 2025

ETHERNET

DEPARTMENT

altera

SIZE
C+

CODE
34649

DOCUMENT NUMBER

150-0330723-B1

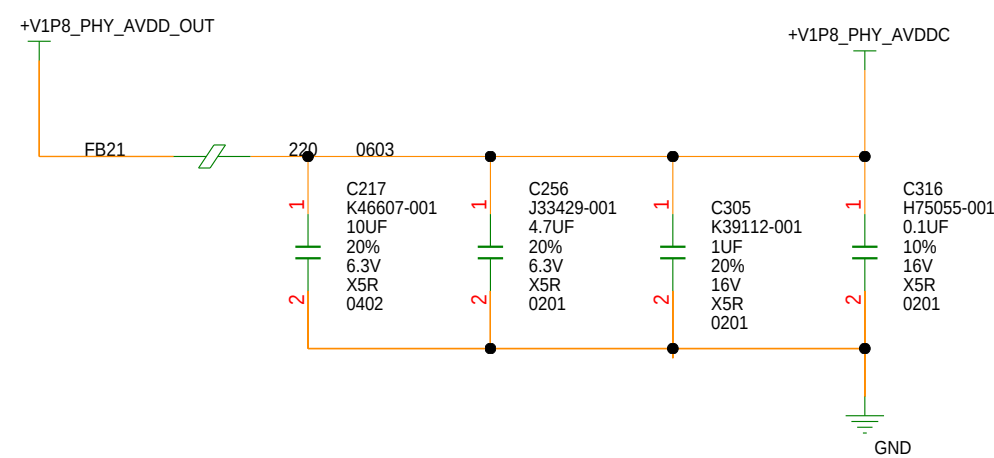
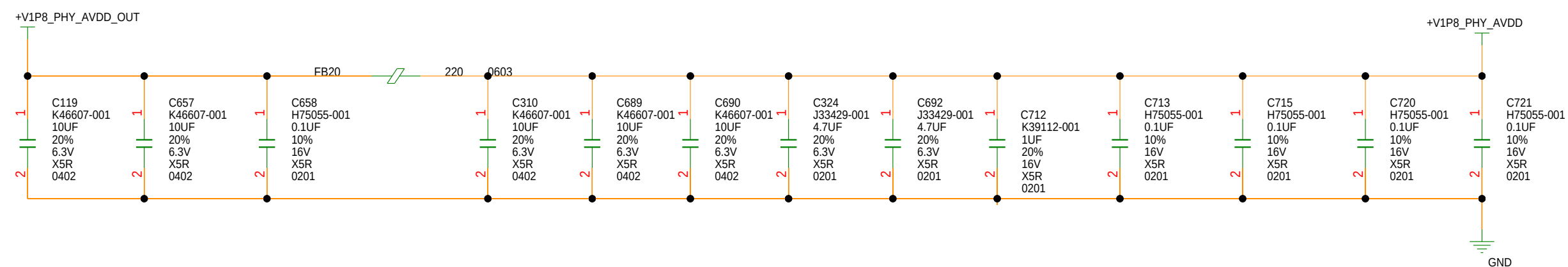
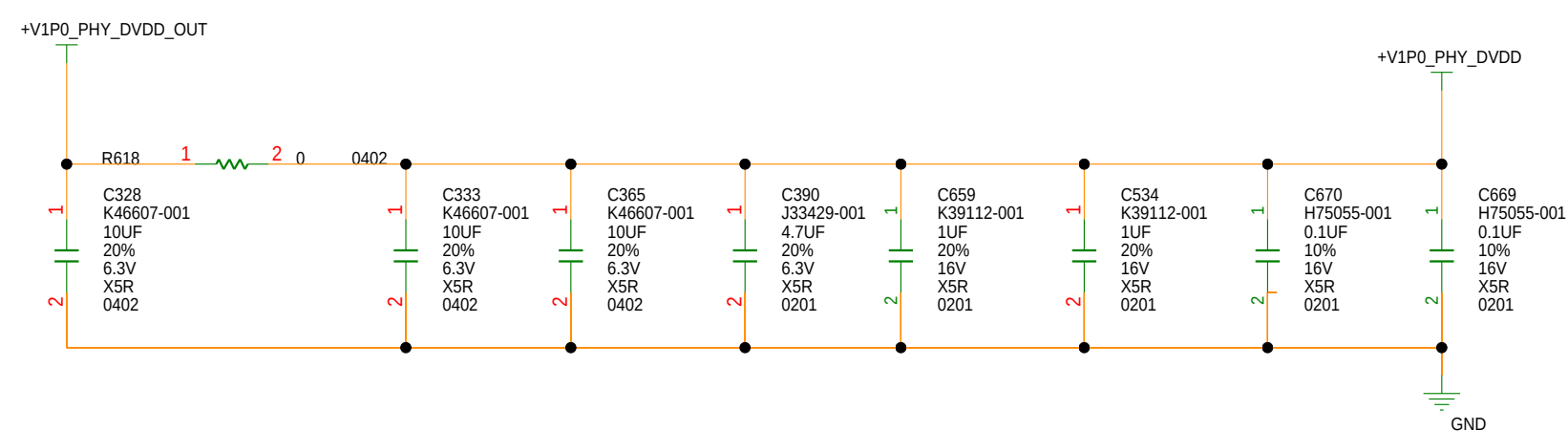
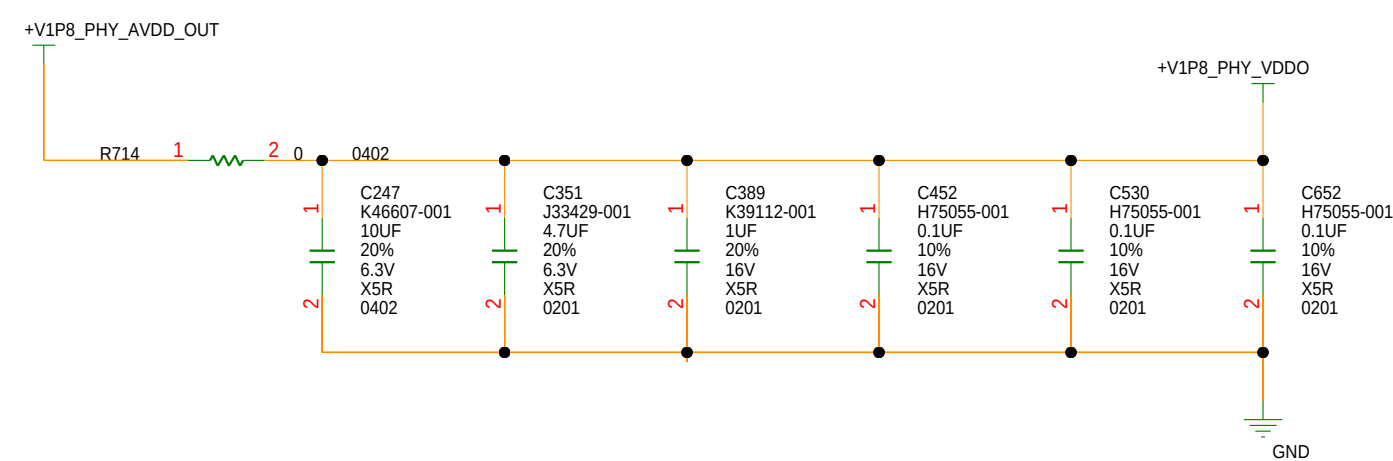
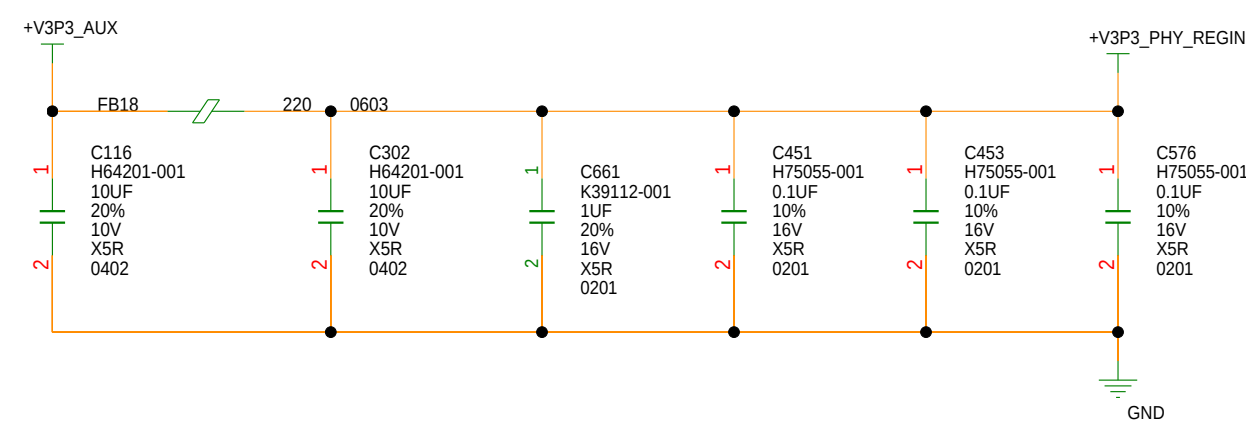
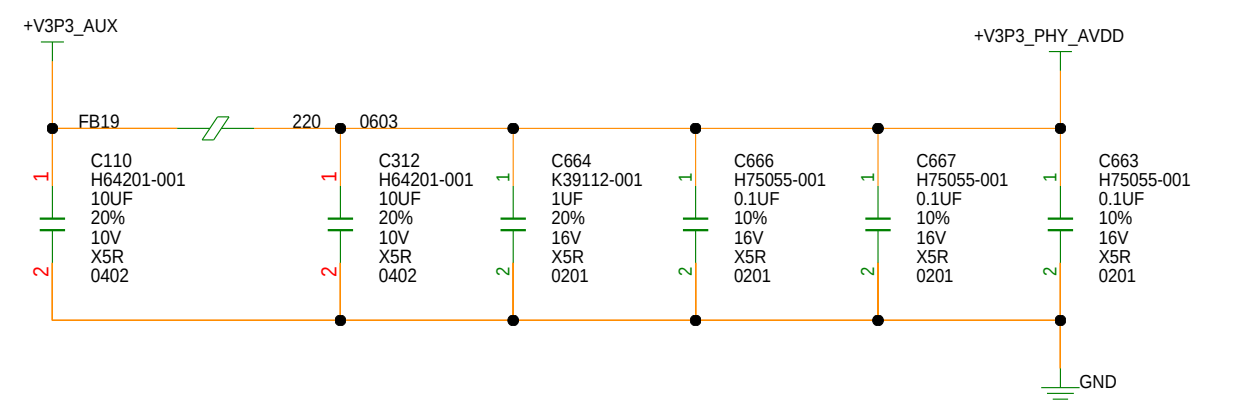
REV
2.1

SCALE:

DO NOT SCALE DRAWING

SHEET

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Mon Jul 28 12:46:23 2025

ETHERNET PHY DECAP

DEPARTMENT



SIZE

CODE

DOCUMENT NUMBER

150-0330723-B1

REV

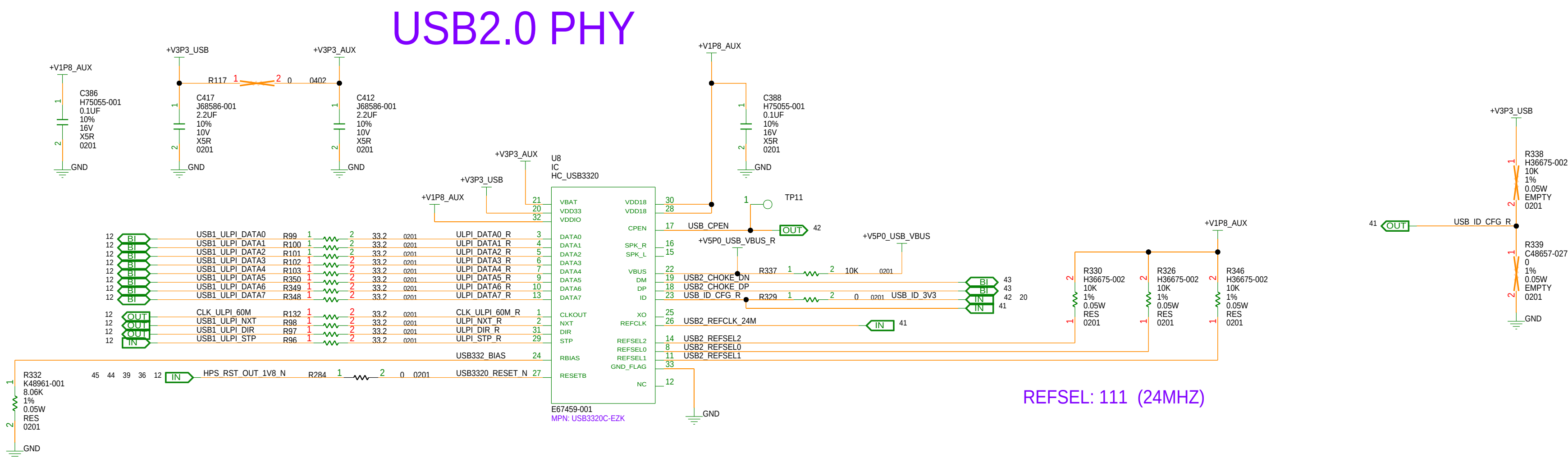
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SCALE:

DO NOT SCALE DRAWING

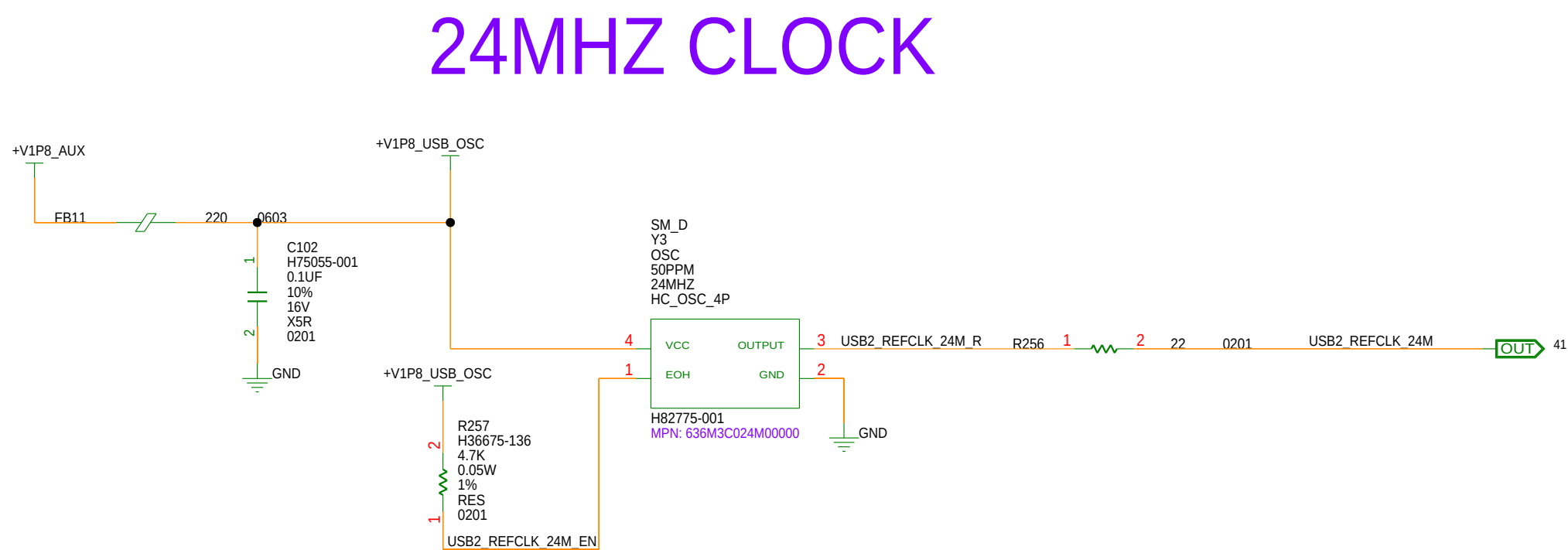
SHEET 39 OF 62

B



B

A



A

Mon Jul 28 12:46:25 2025

USB2.0PHY

DEPARTMENT



SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

2.1

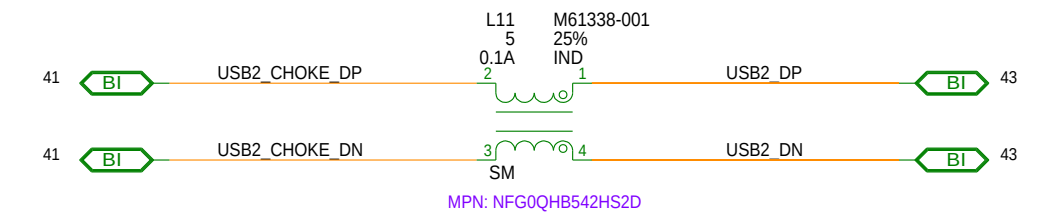
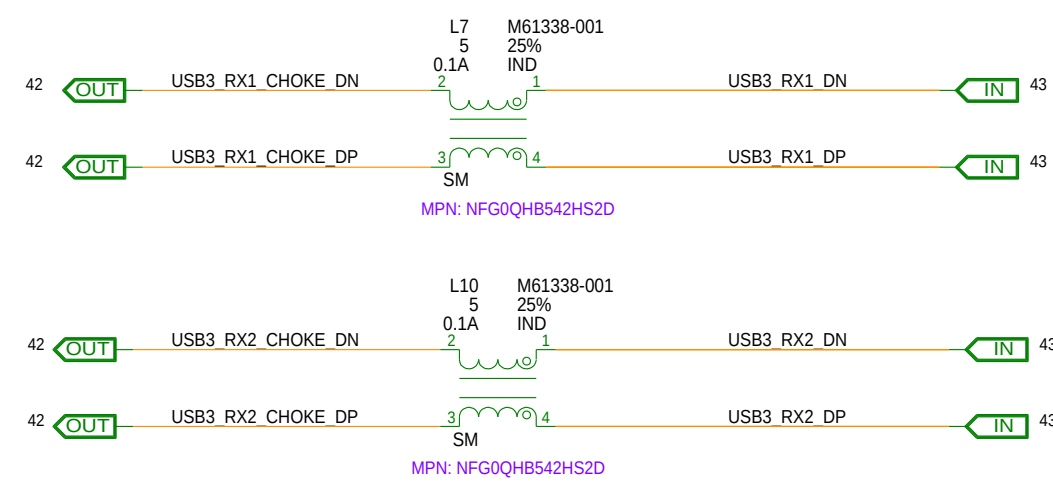
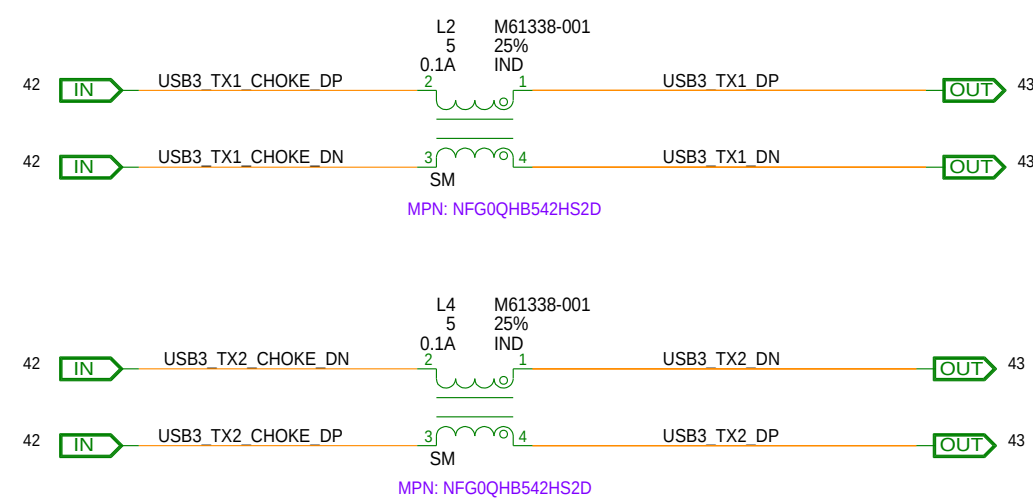
SCALE:

DO NOT SCALE DRAWING

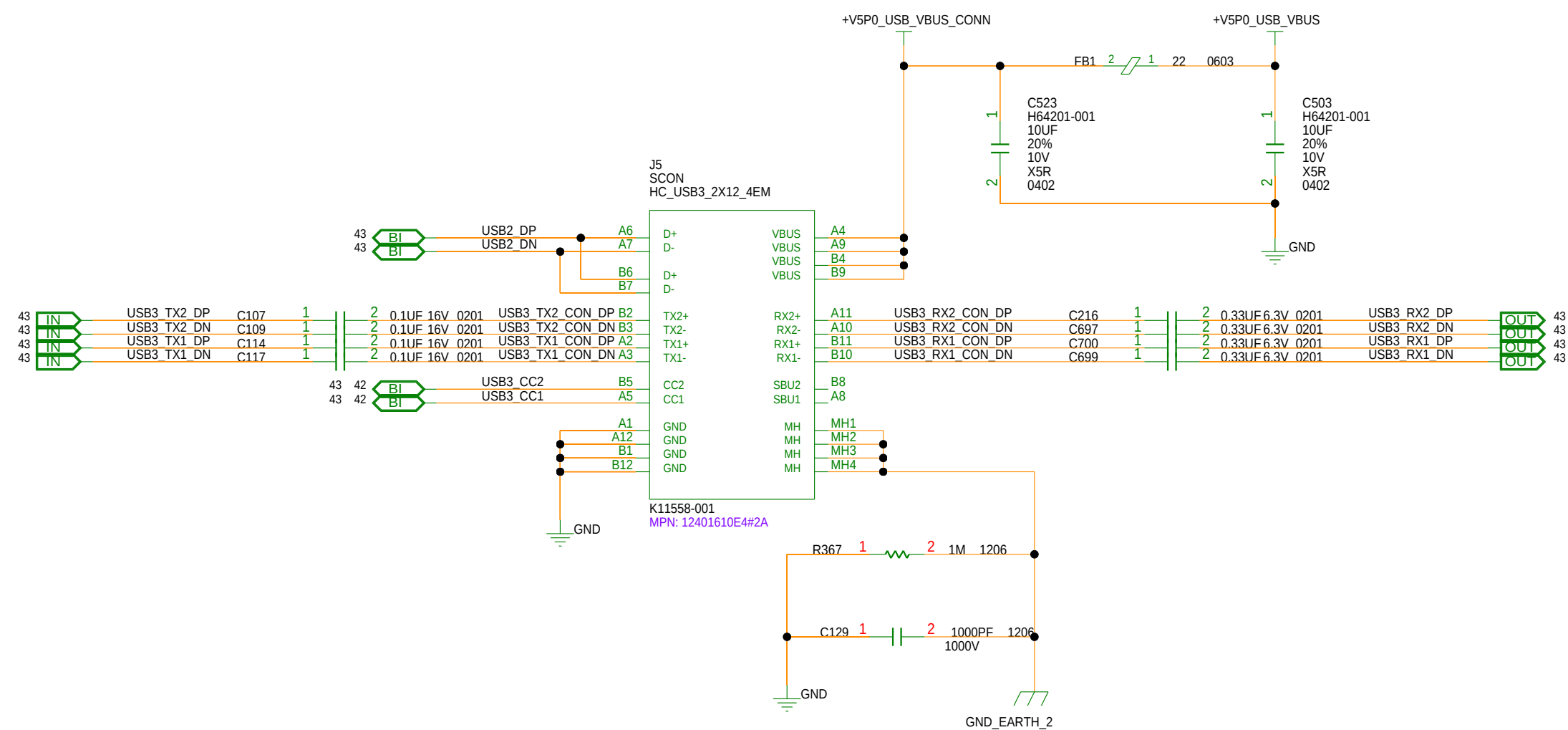
SHEET

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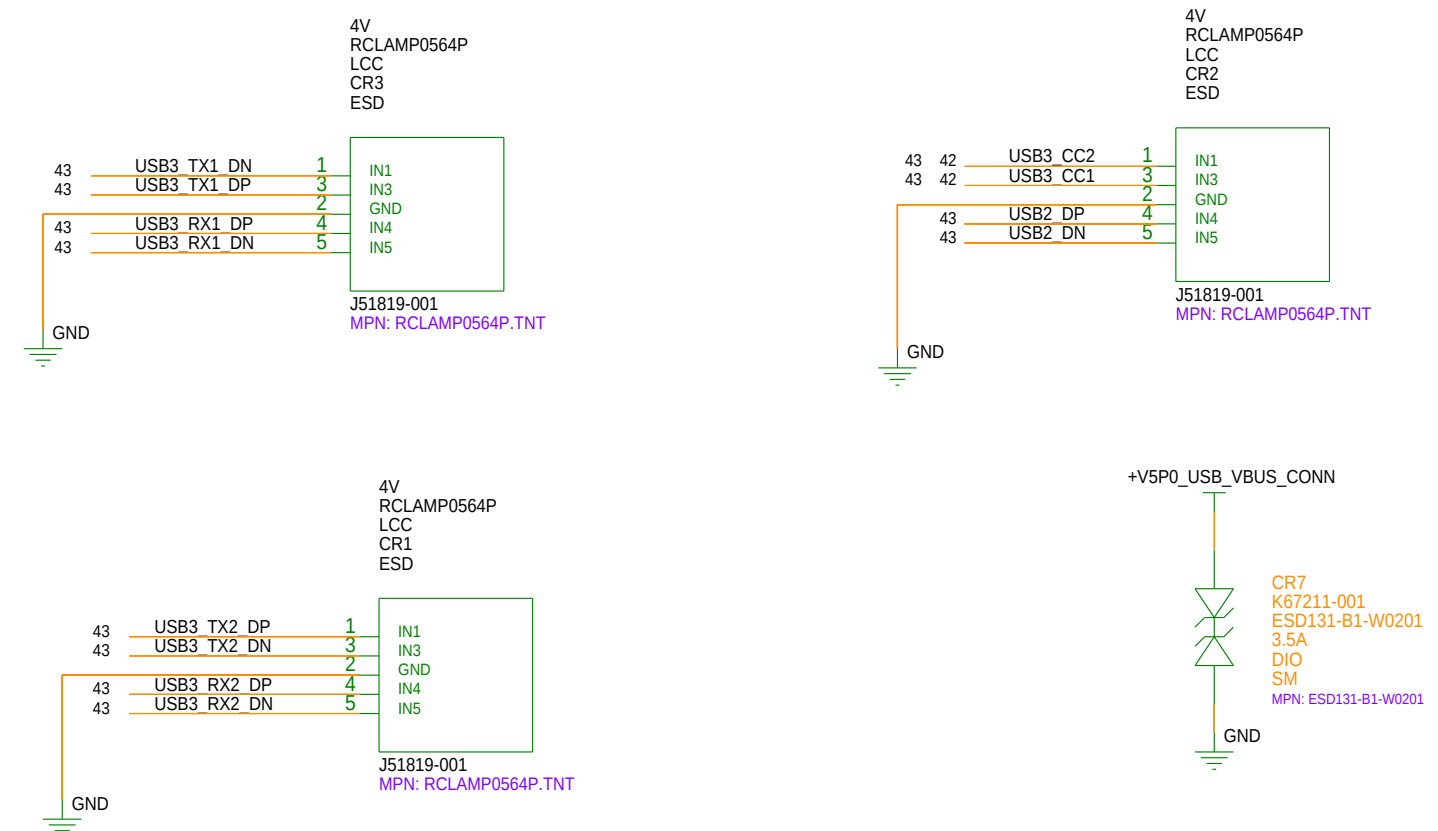
COMMON MODE CHOKE



TYPE C CONN



ESD DIODE



Mon Jul 28 12:46:33 2025

USB3.0 CONN

DEPARTMENT



SIZE

	COD
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34649

CODE	DOCUMENT NUMBER
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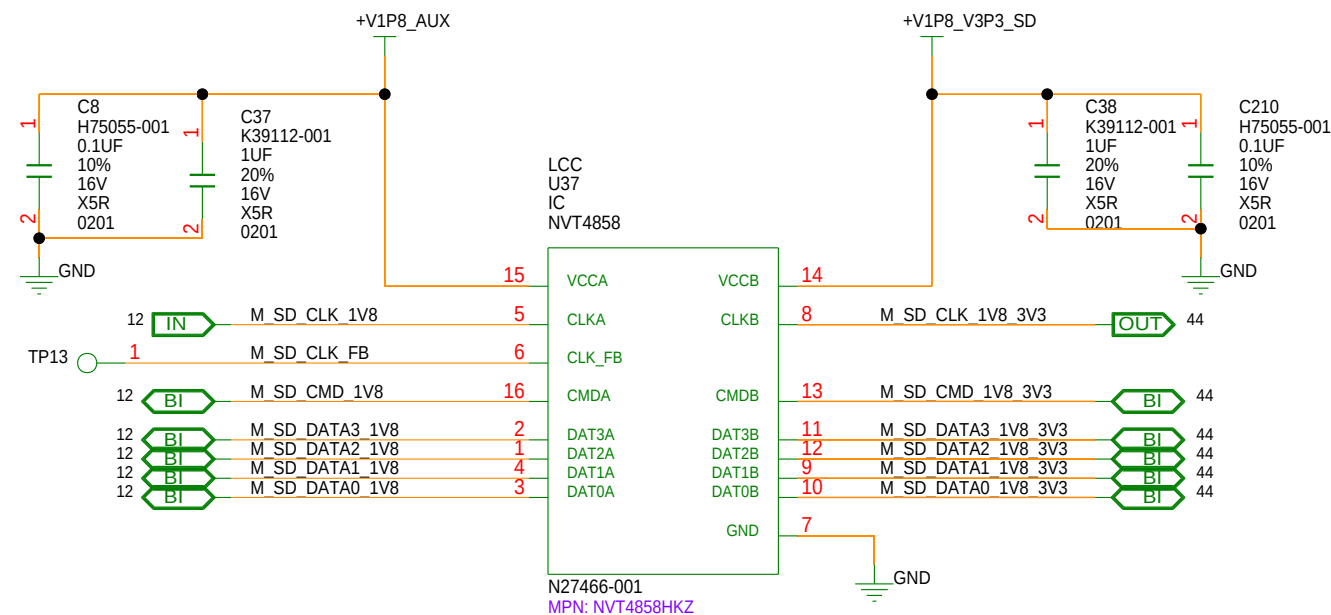
34649

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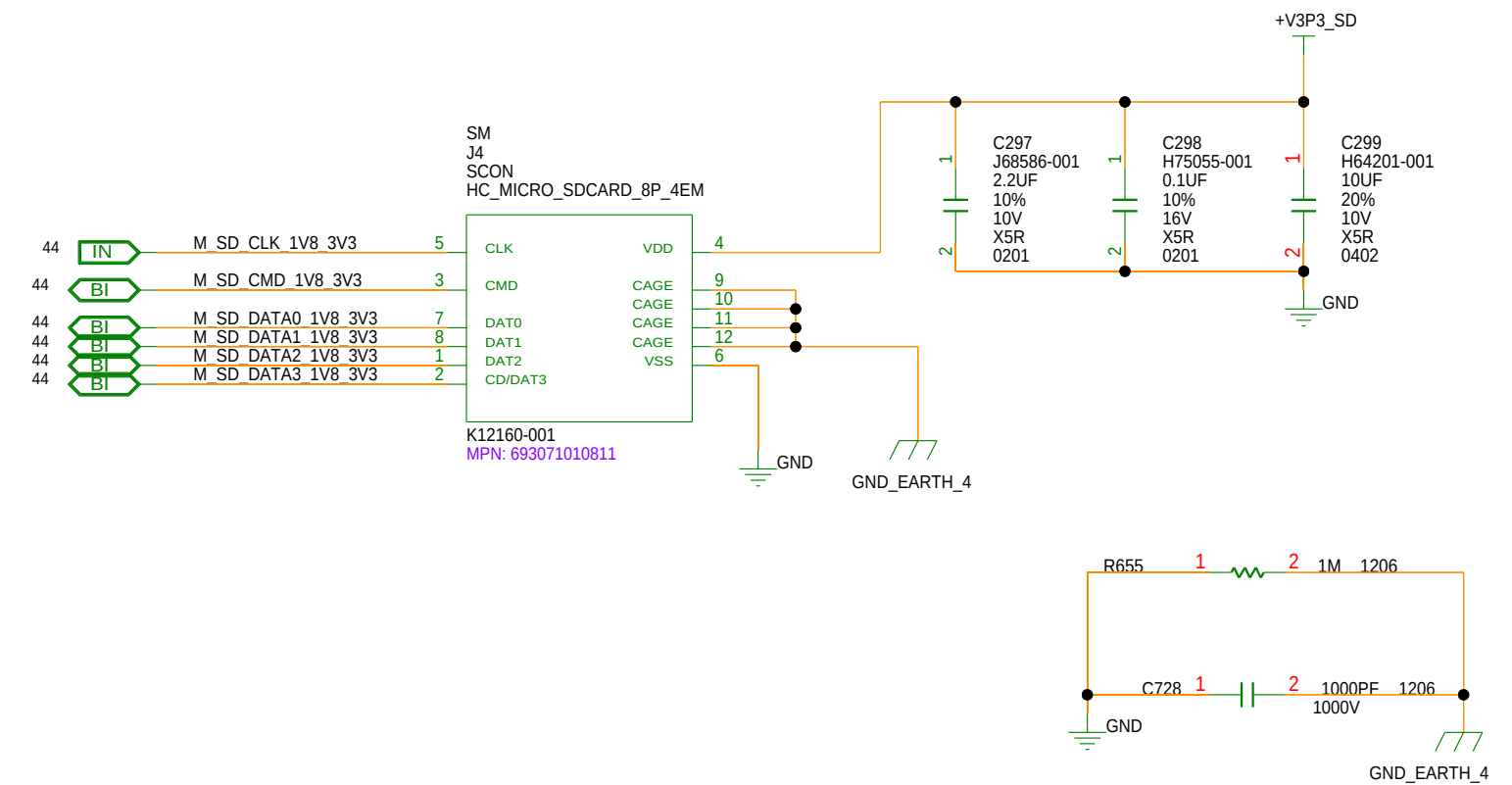
	REV
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42.05.62

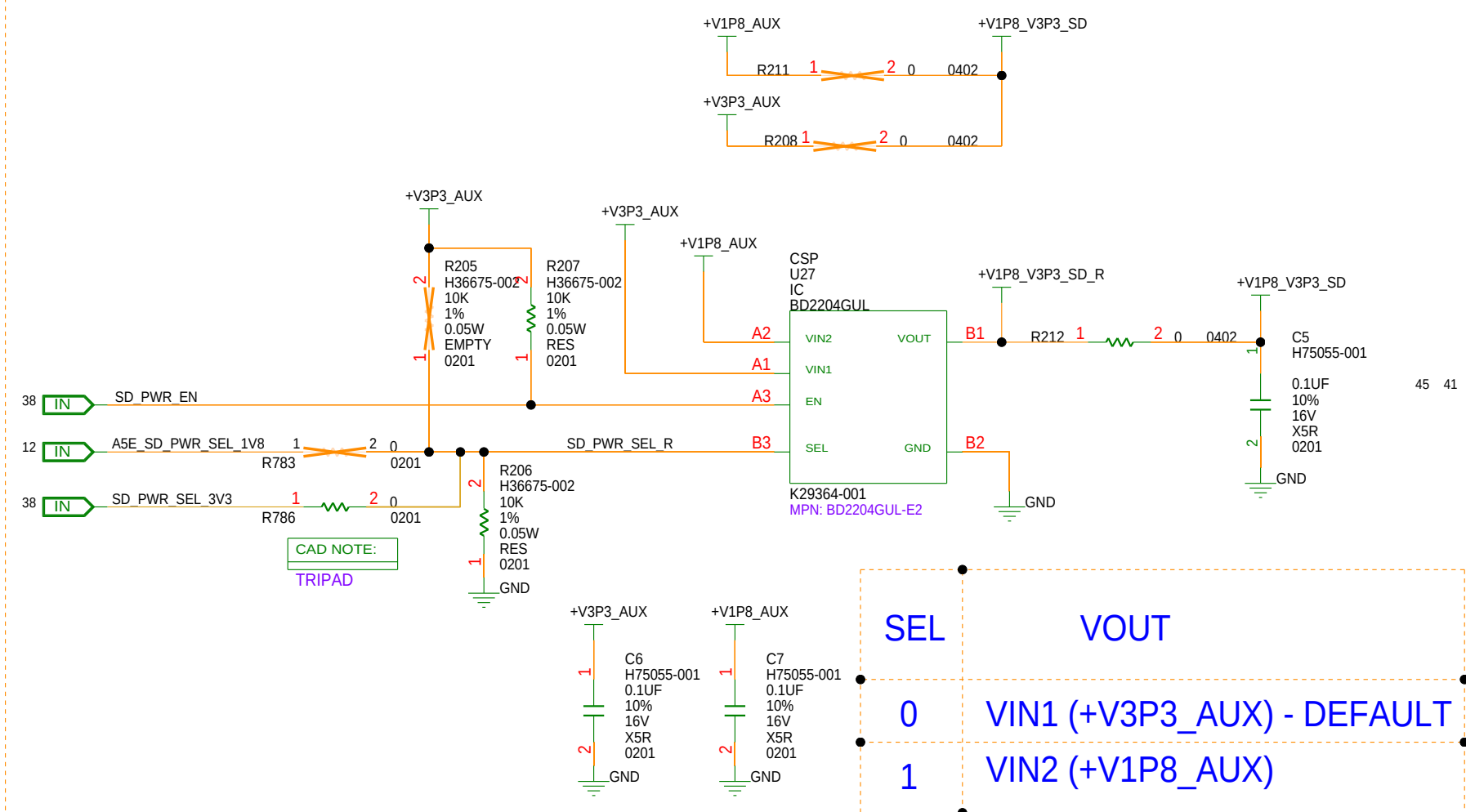
LEVEL SHIFTER



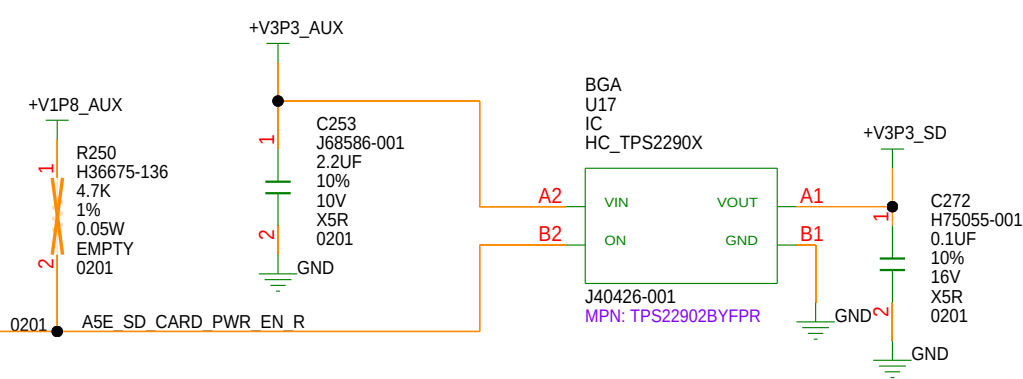
MICRO SD CARD CONNECTOR



POWER SELECTION



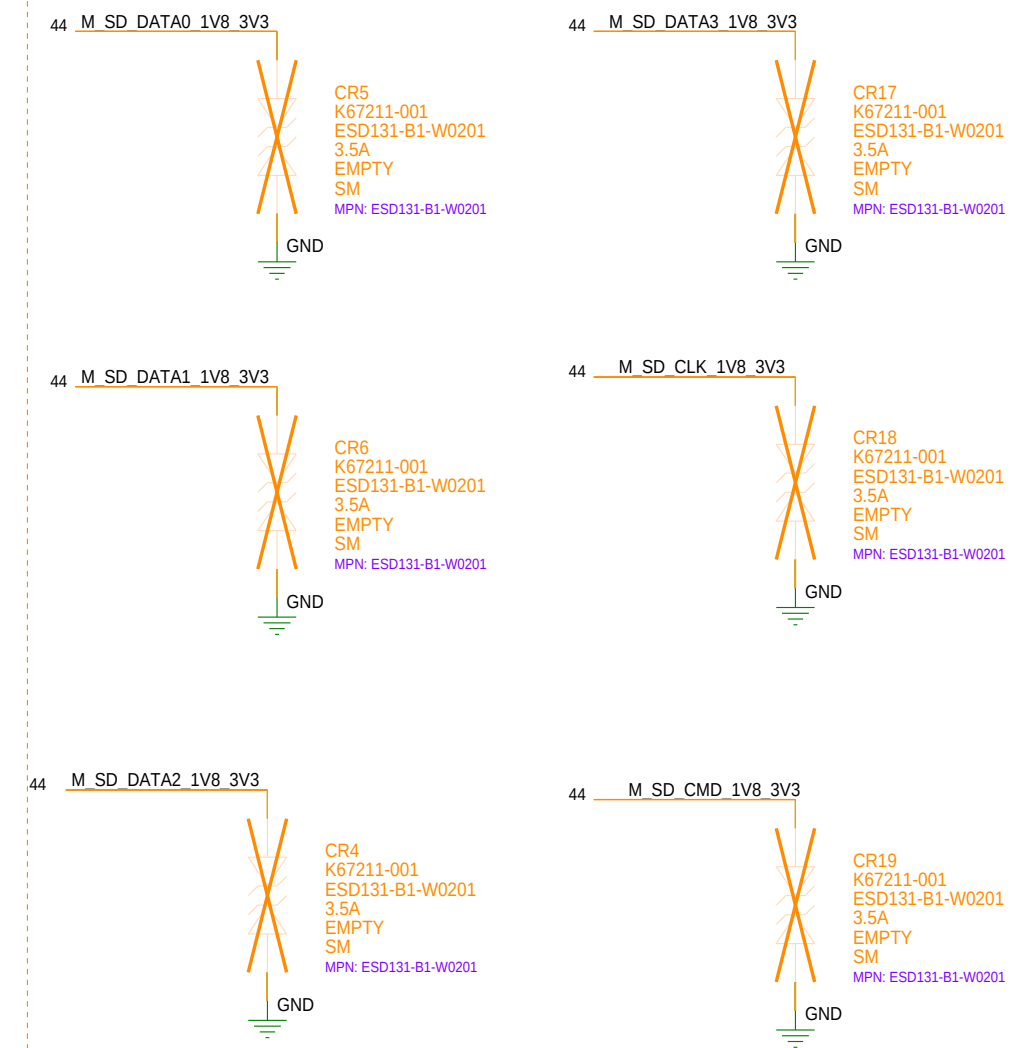
LOAD SWITCH



VOUT Function Table

ON (Control Input)	TPS22902/2B
L	GND
H	VIN

ESD DIODE



SD CARD

DEPARTMENT

altera

SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

2.1

SCALE:

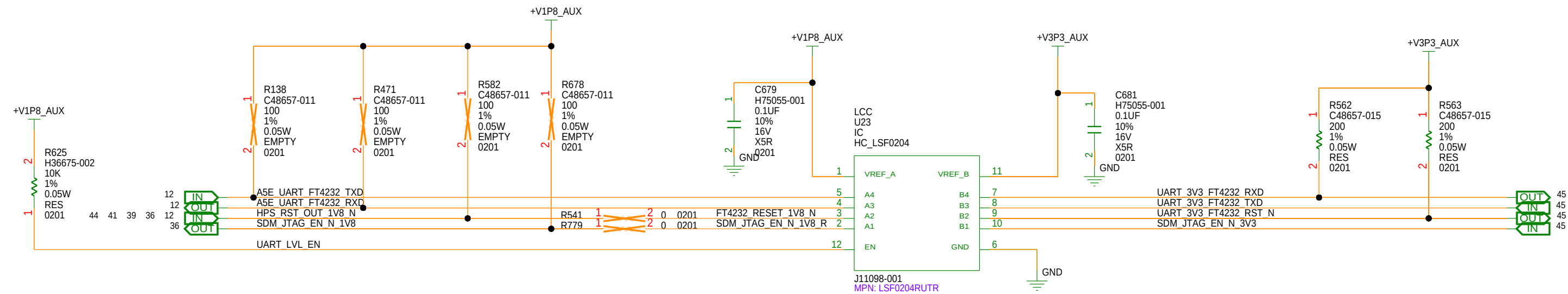
DO NOT SCALE DRAWING

SHEET

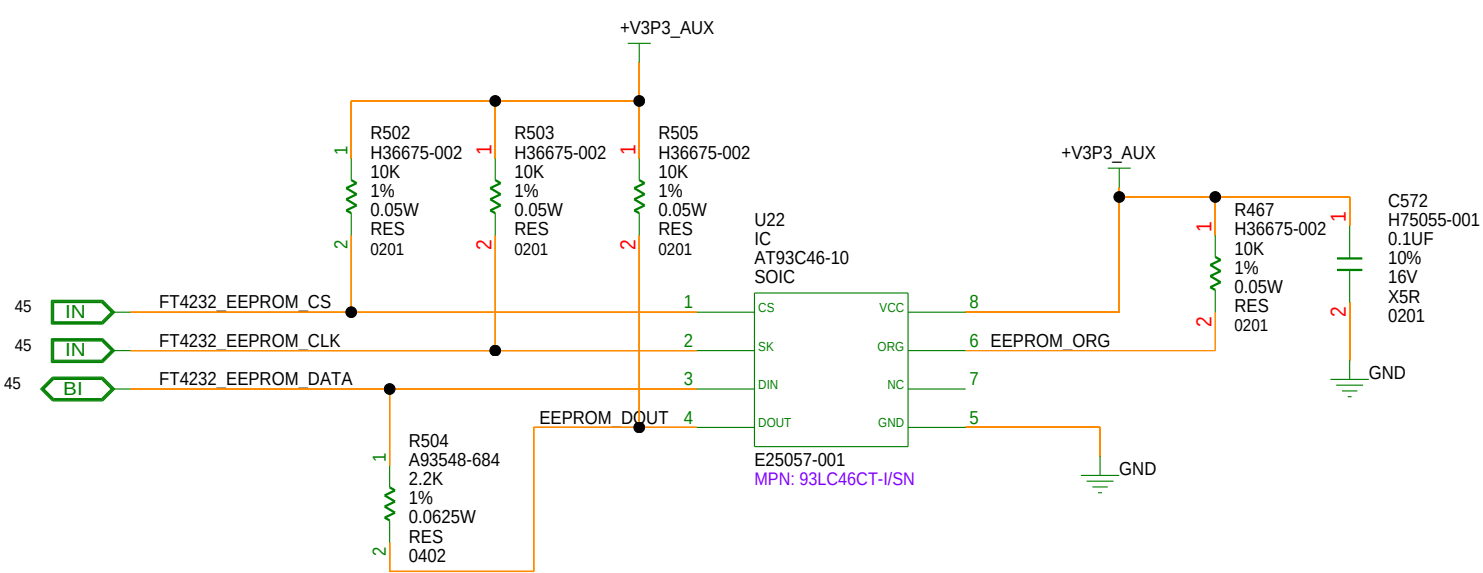
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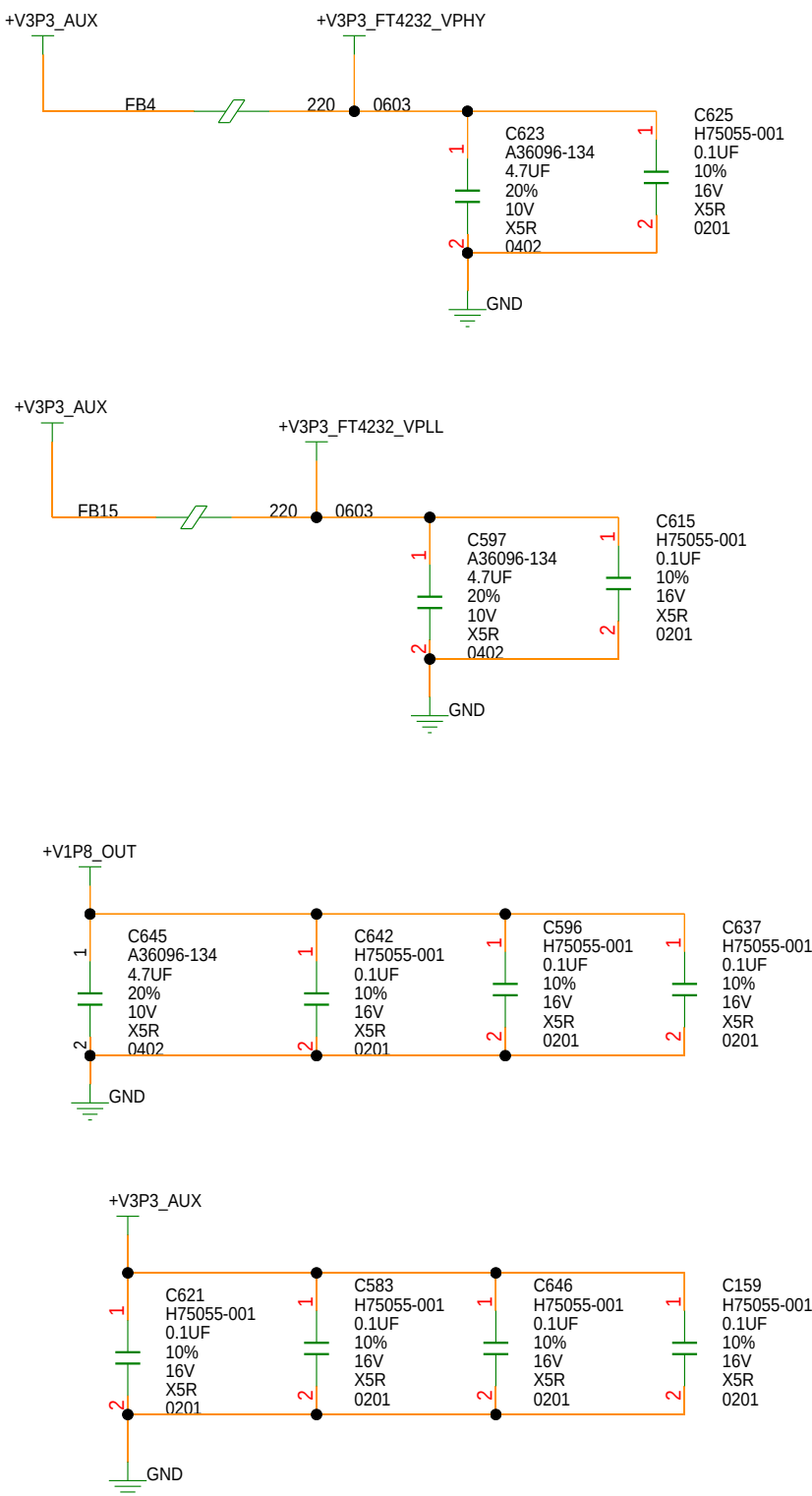
LEVEL SHIFTER



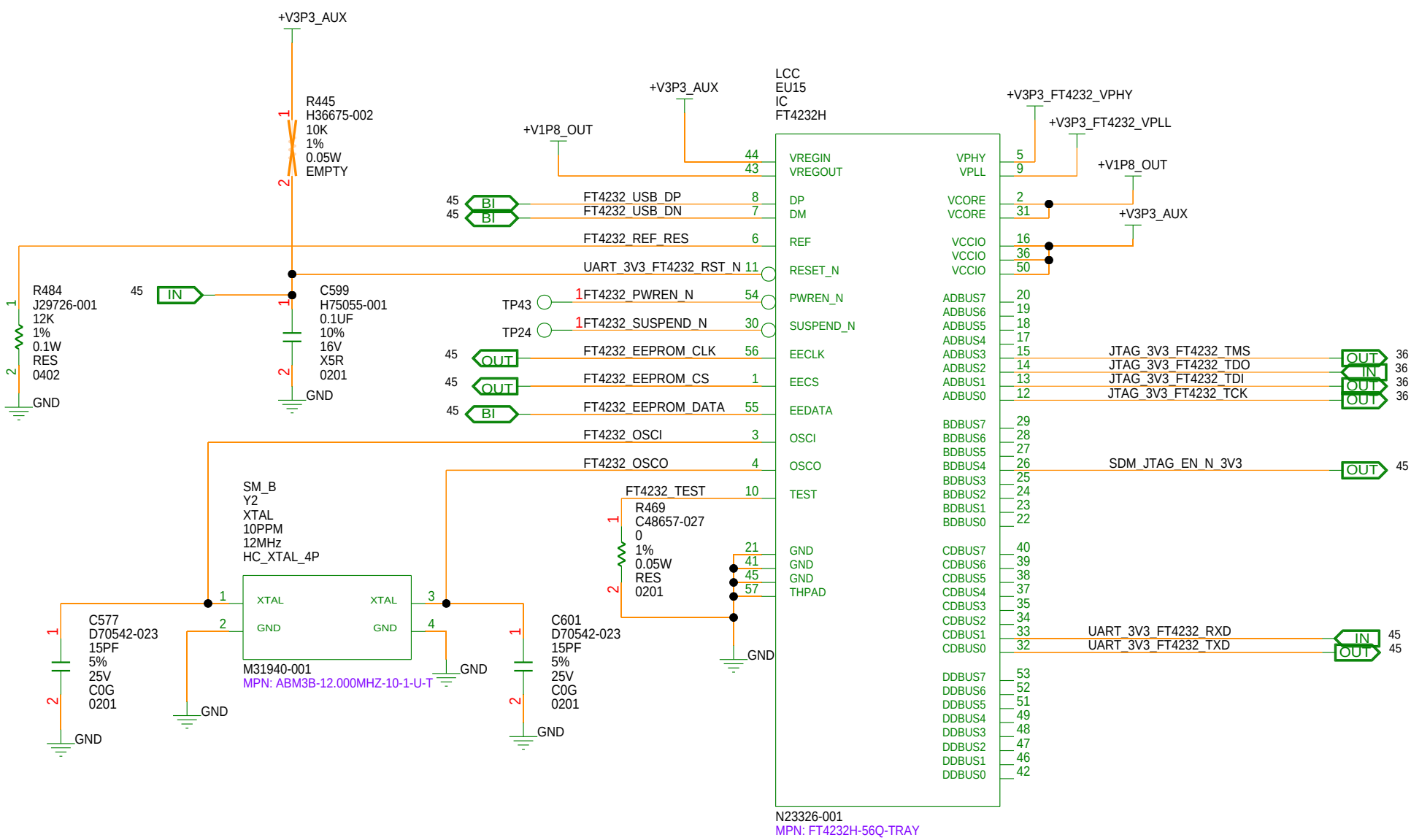
1Kbit EEPROM



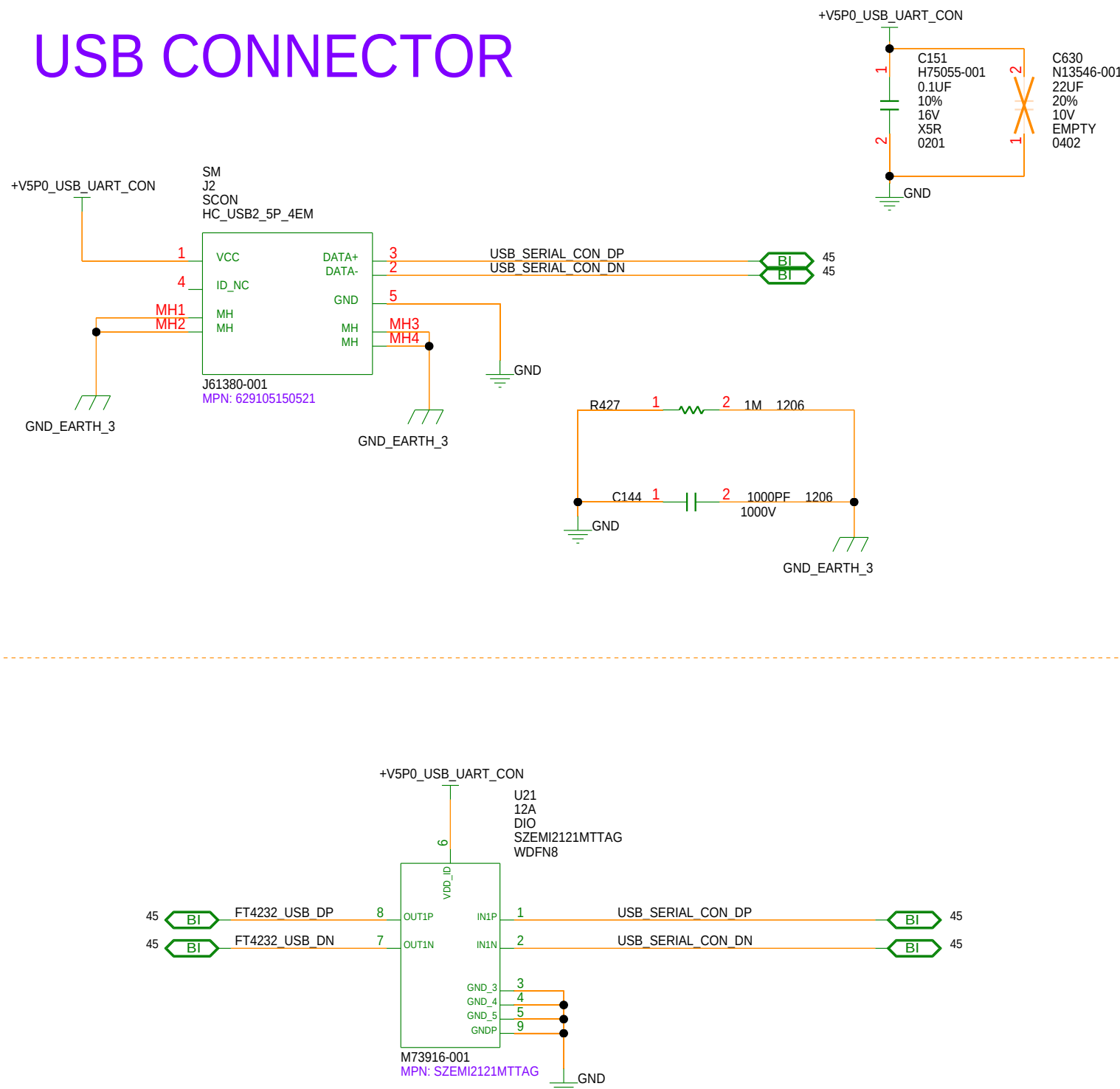
DECAPS



FT4232H



USB CONNECTOR



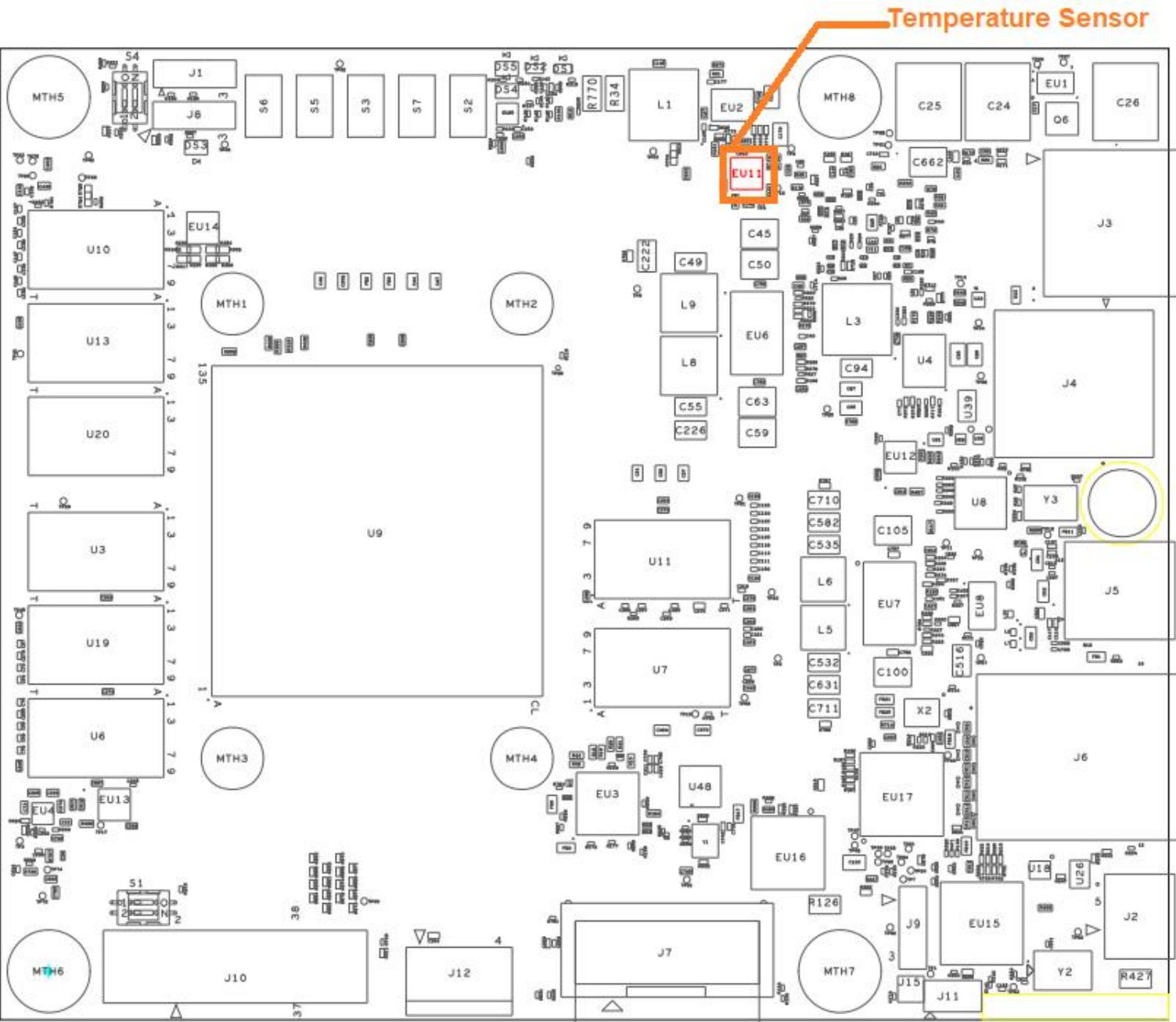
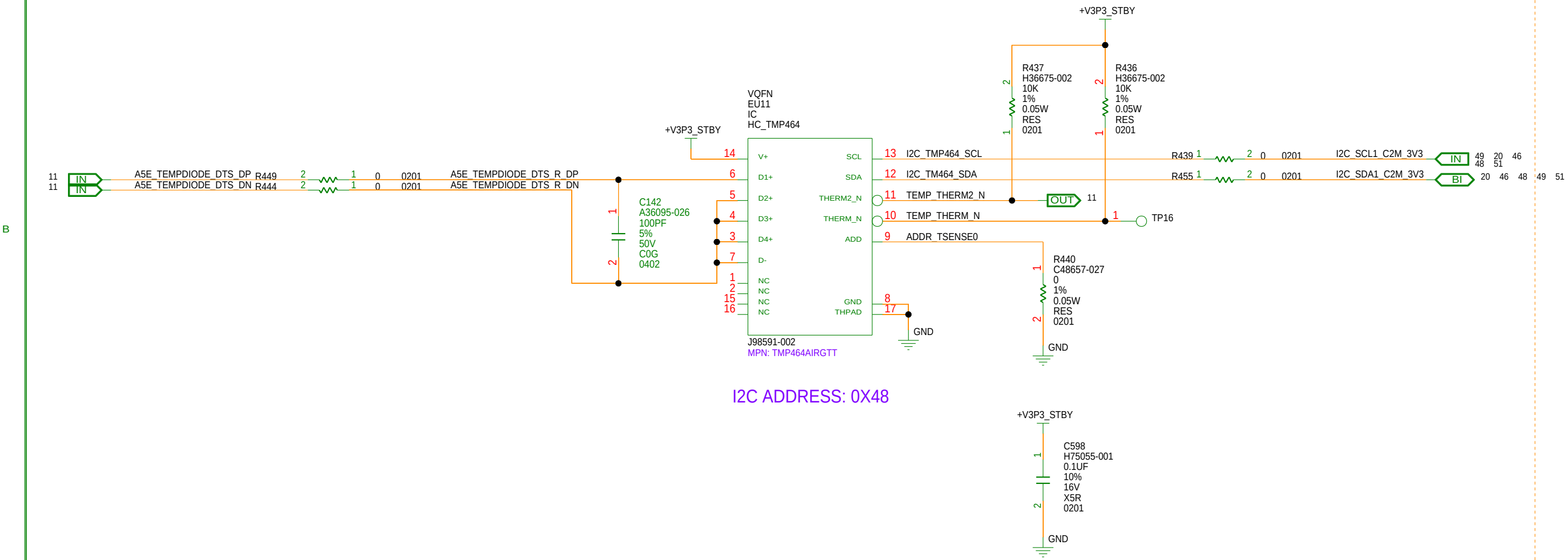
USB - JTAG/UART CONVERTER



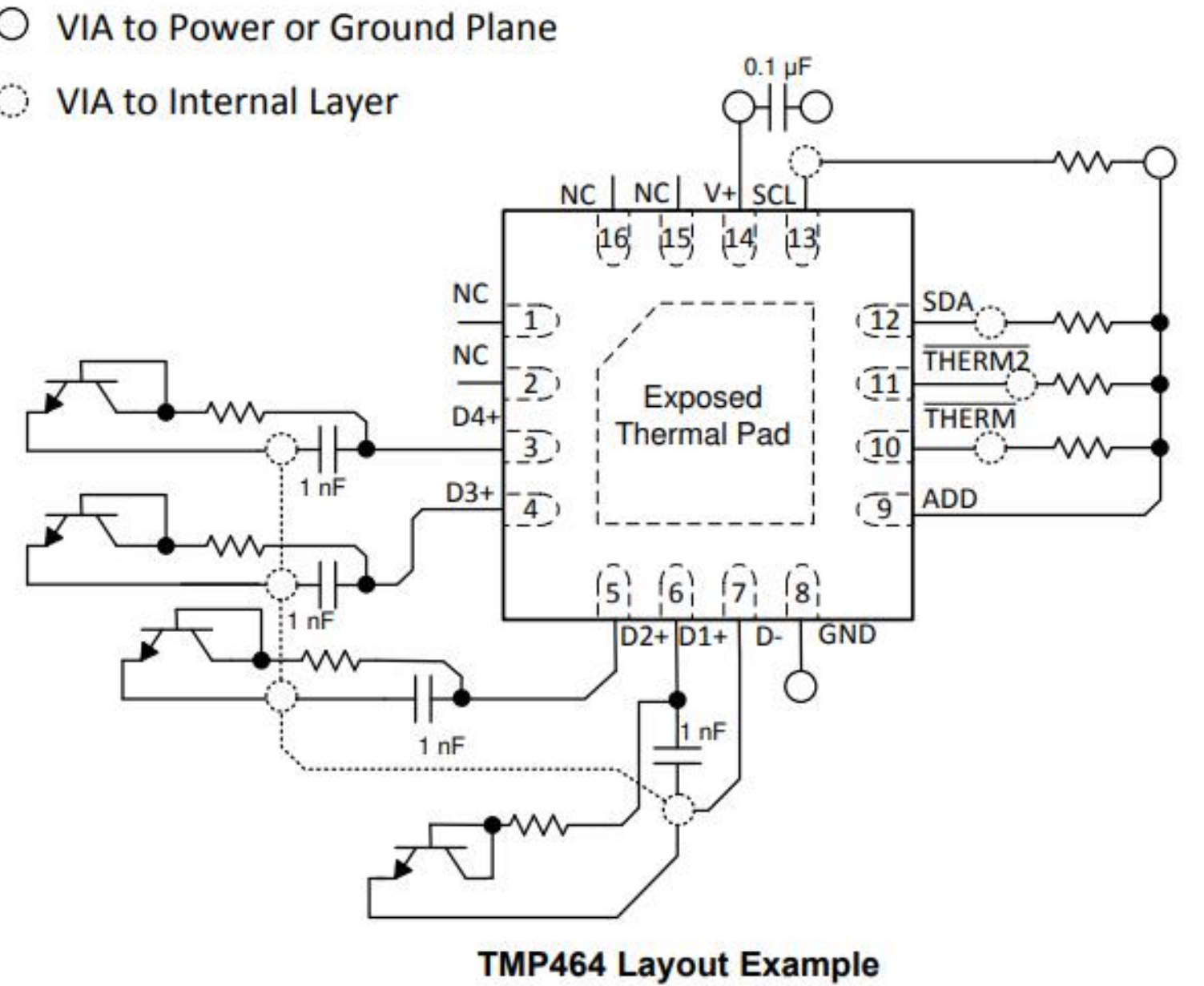
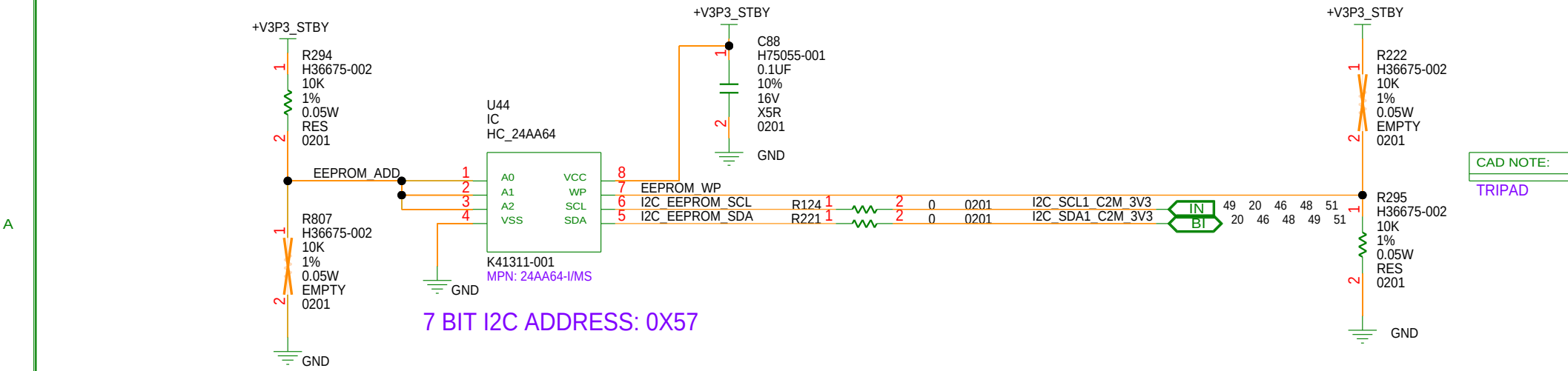
Mon Jul 28 12:46:41 2025

DEPARTMENT	SIZE	CODE	DOCUMENT NUMBER	REV
	C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING			SHEET 44 OF 62

TEMPERATURE SENSOR



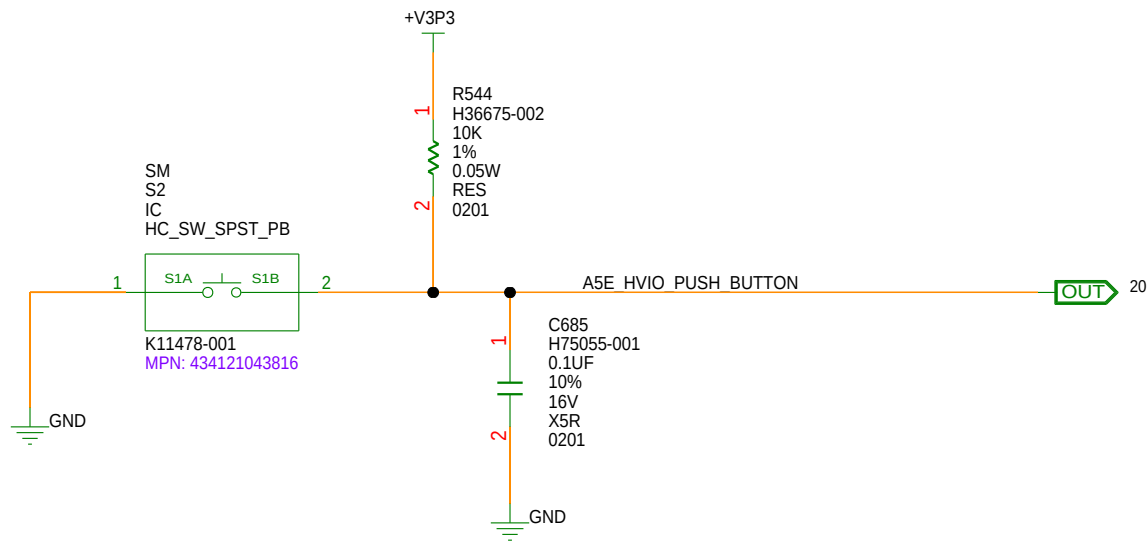
EEPROM



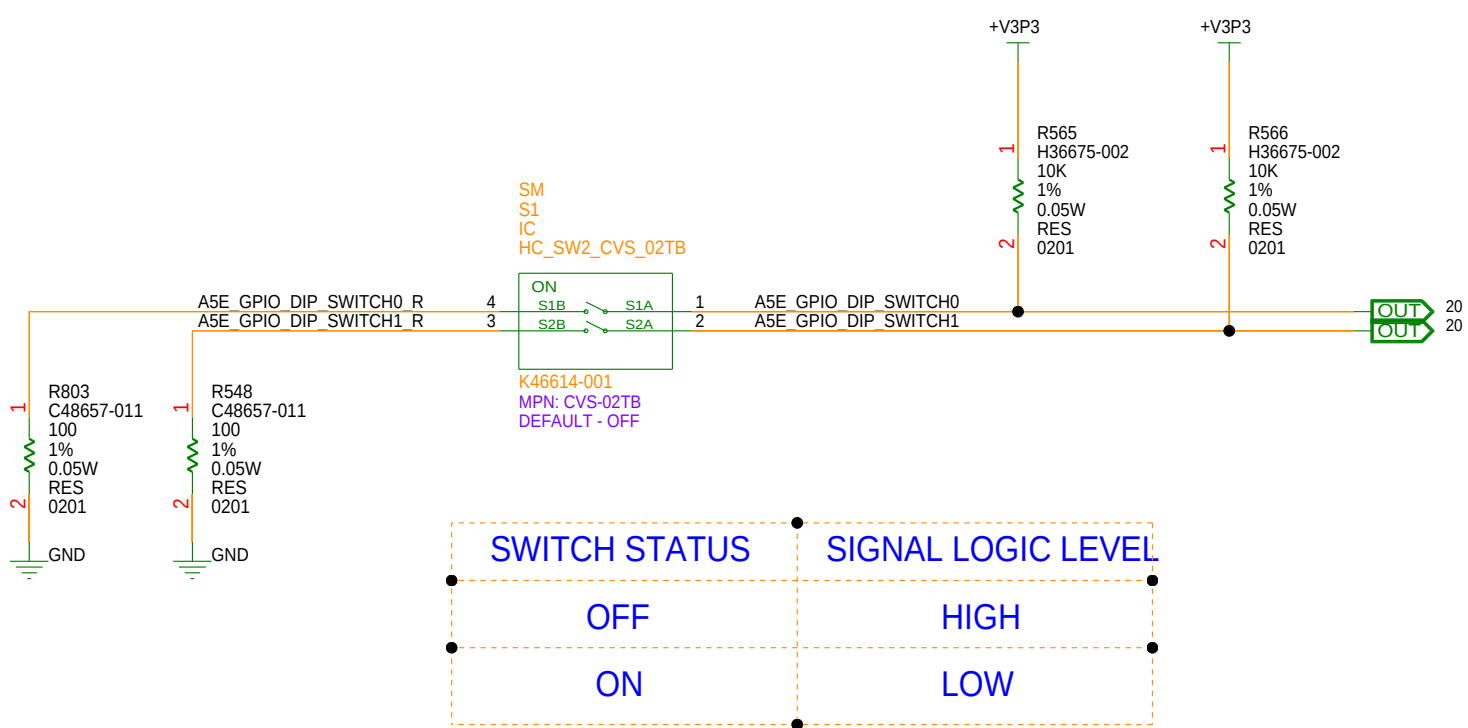
TEMP_SENSOR & EEPROM

DEPARTMENT	altera	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING	SHEET	45 OF 62		

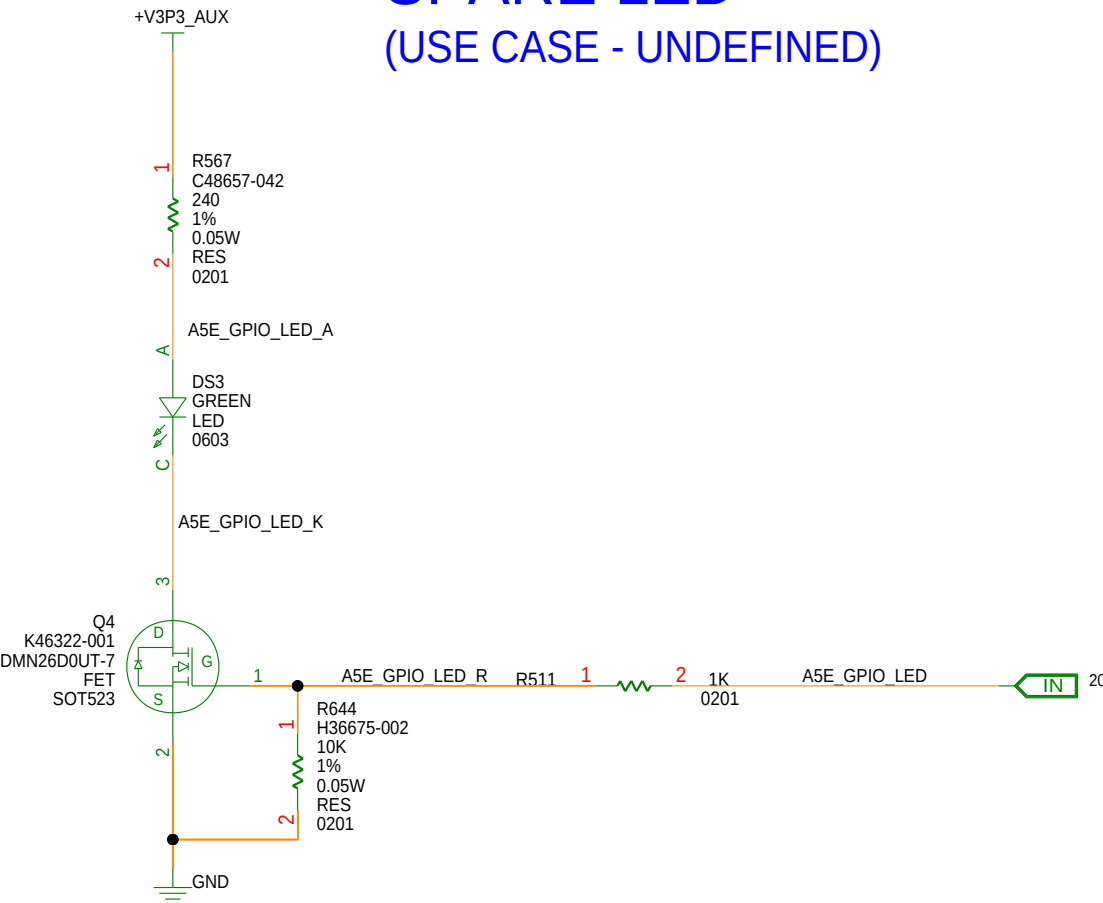
HVIO PUSH BUTTON SWITCH



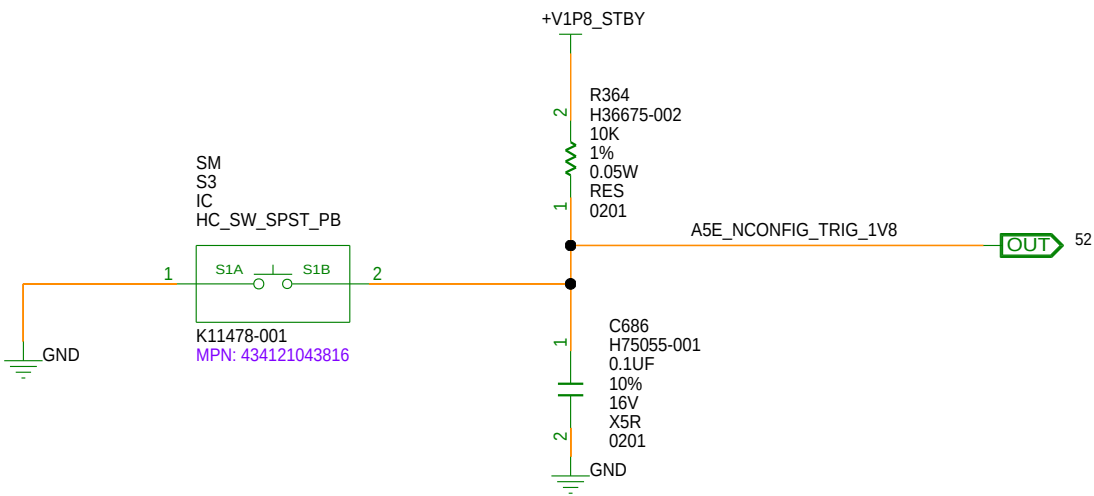
SPARE DIP SWITCH



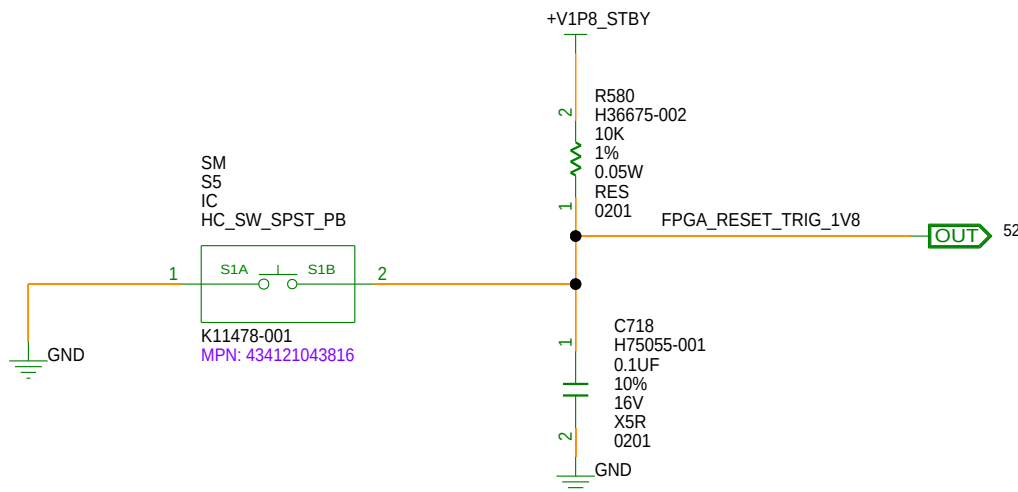
SPARE LED (USE CASE - UNDEFINED)



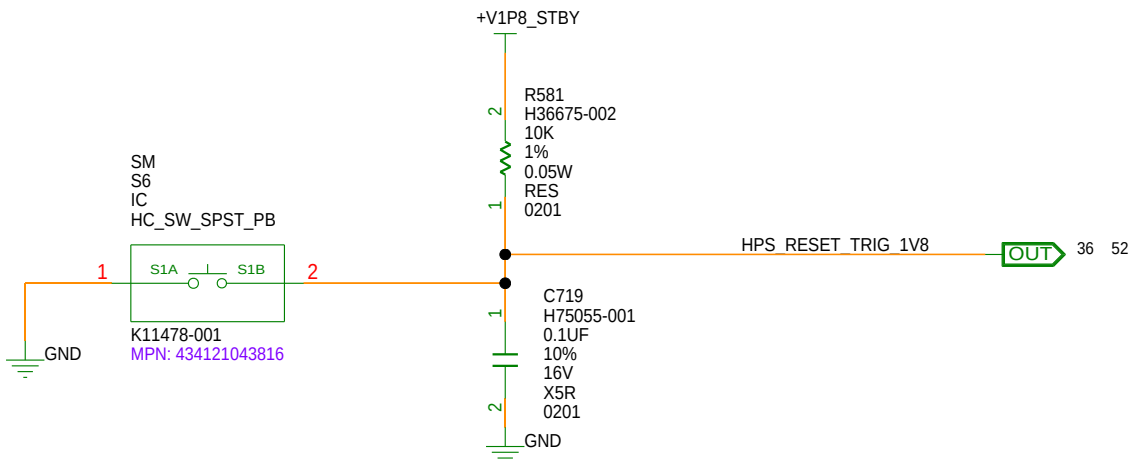
FPGA RESET SWITCH



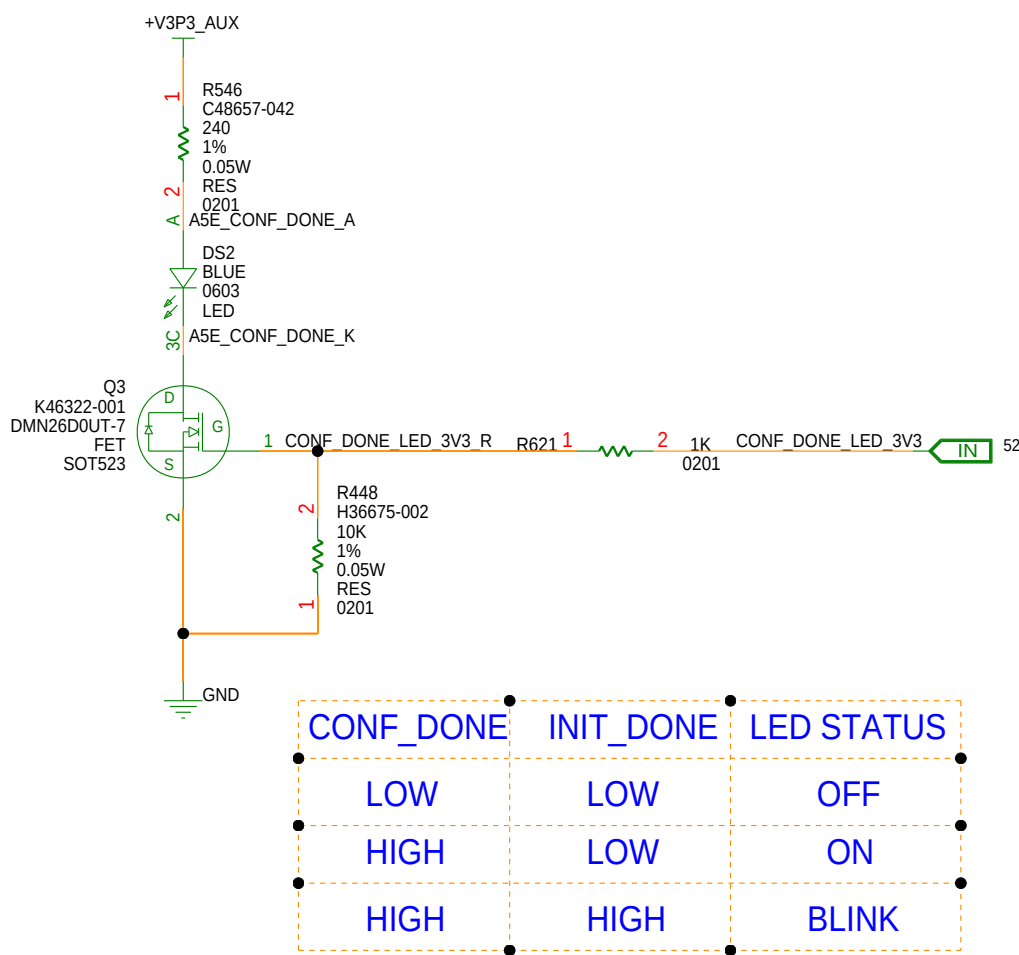
WARM RESET SWITCH



COLD RESET SWITCH



CONFIG DONE LED



ROW A

ROW B

ROW C

ROW D

B2B PRIMARY CONNN



Mon Jul 28 12:46:49 2025

DEPARTMENT

SIZE
C+

CODE
34649

DOCUMENT NUMBER
150-0330723-B1

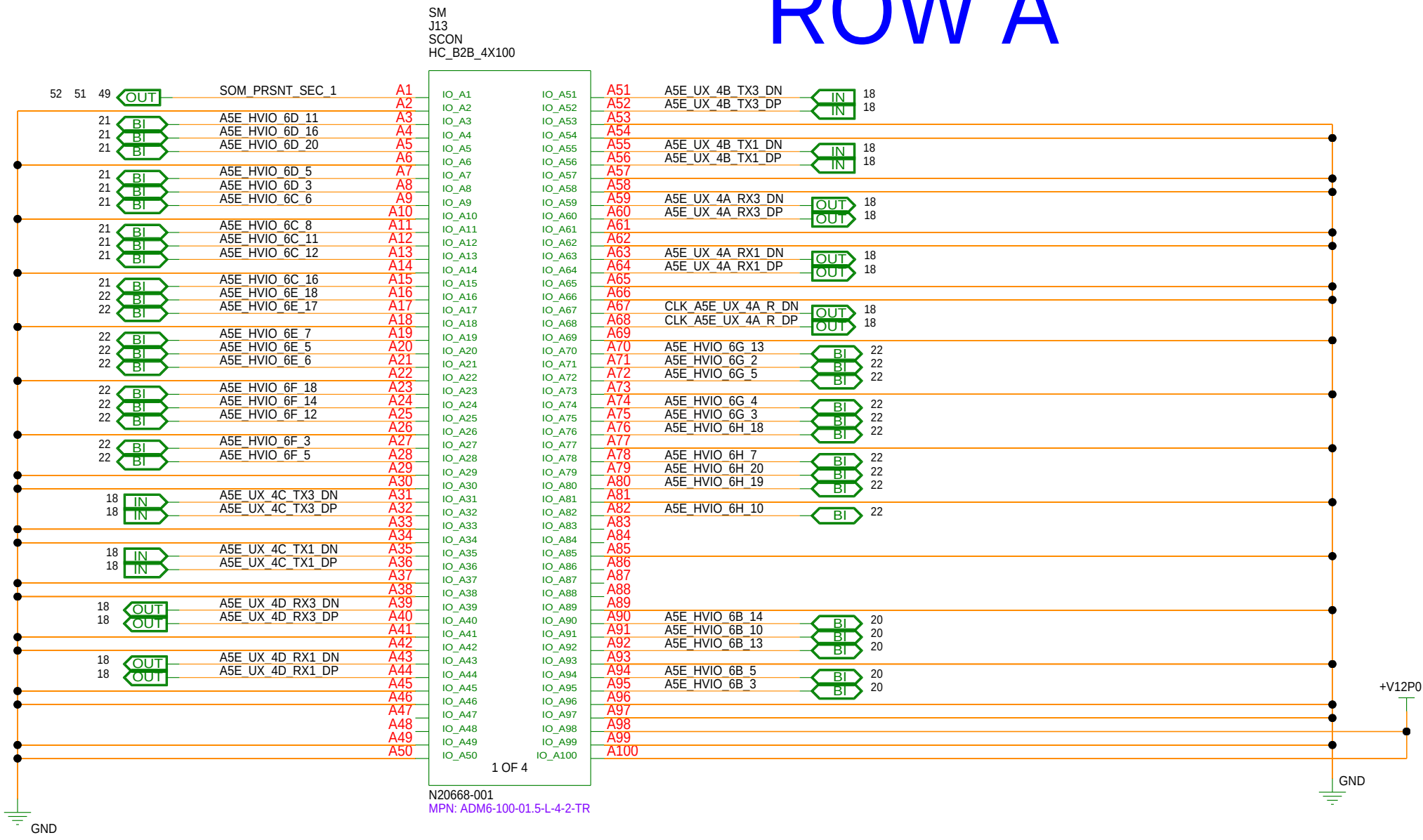
REV
2.1

SCALE:

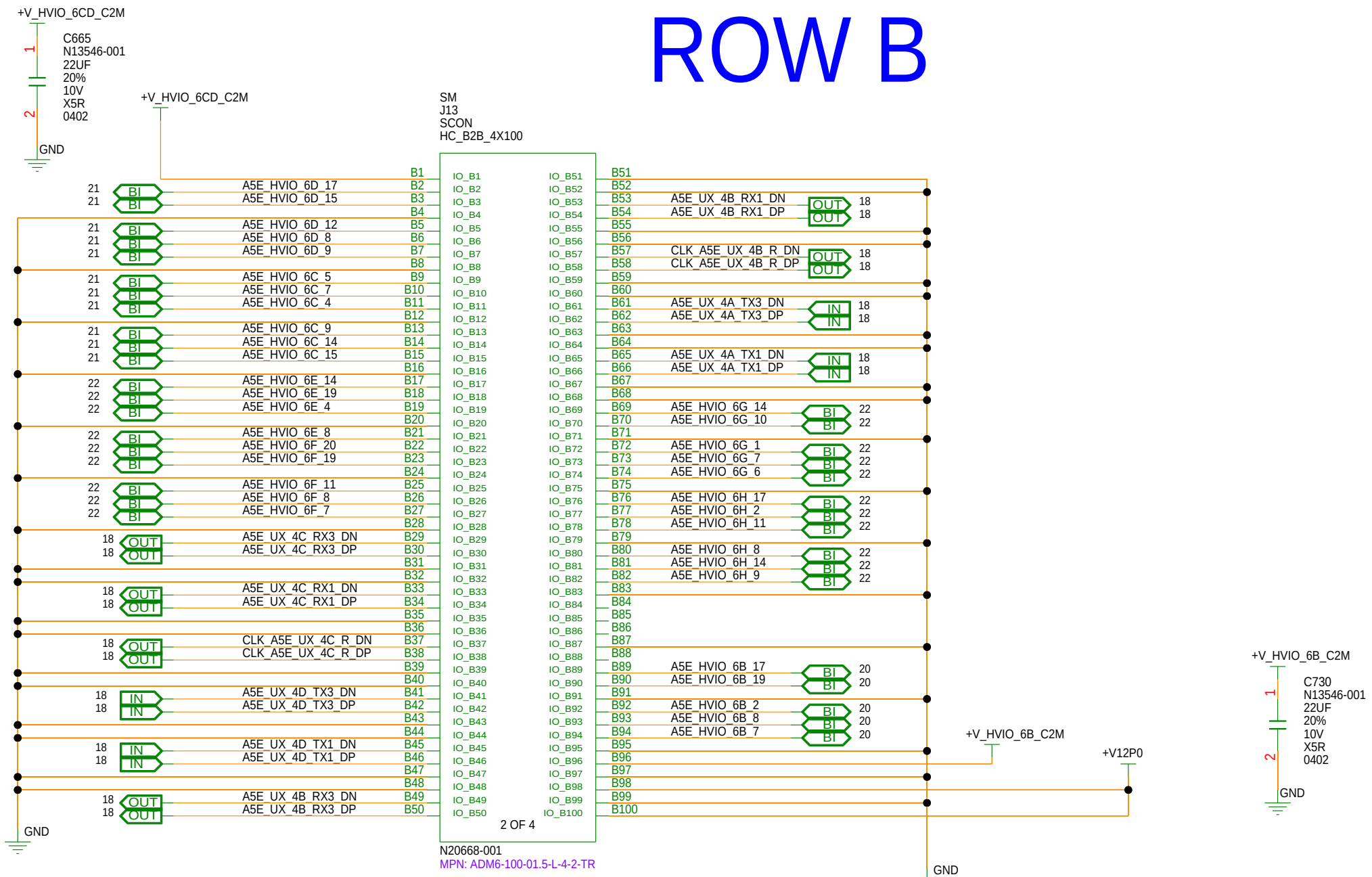
DO NOT SCALE DRAWING

SHEET
48 OF 62

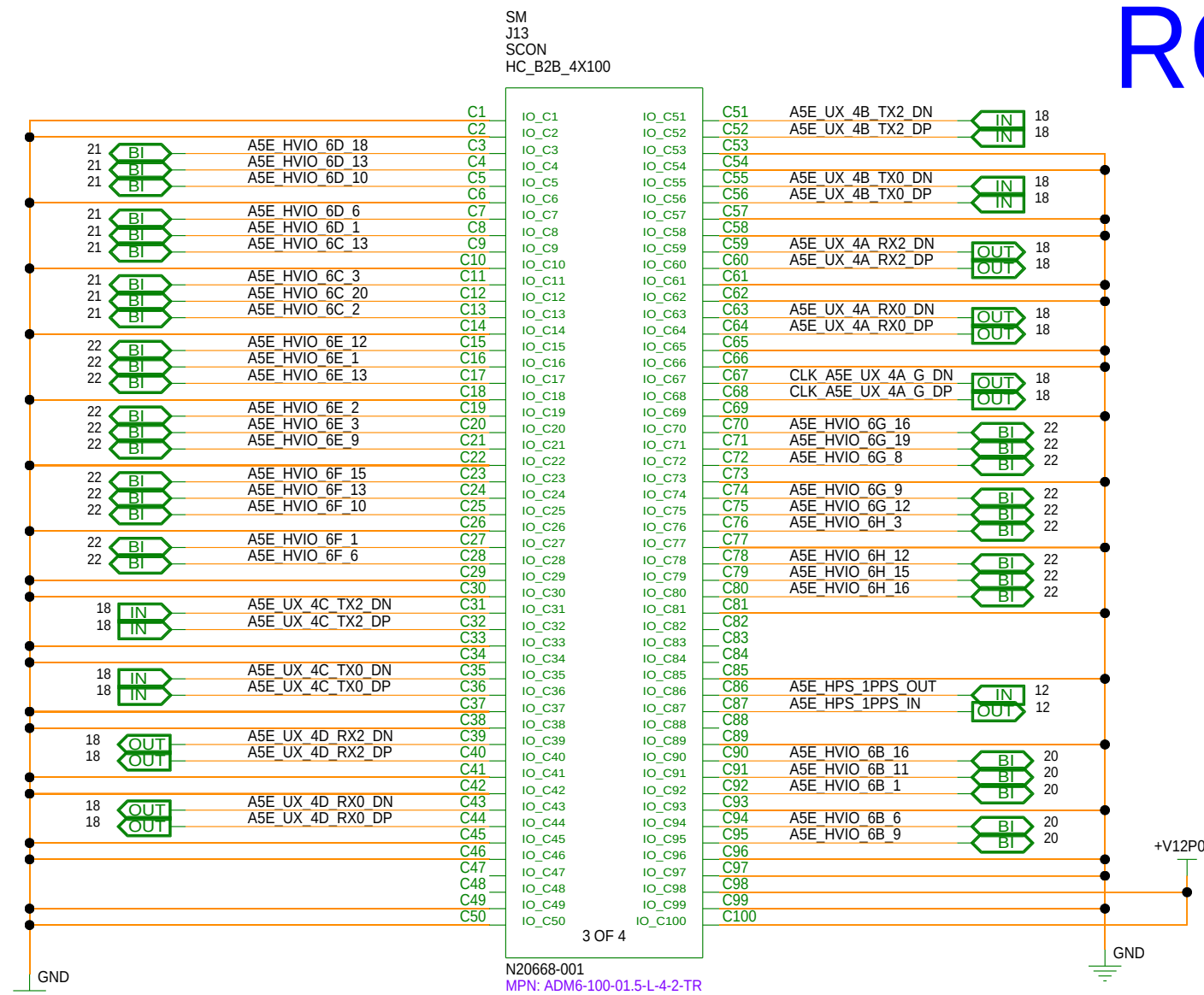
ROW A



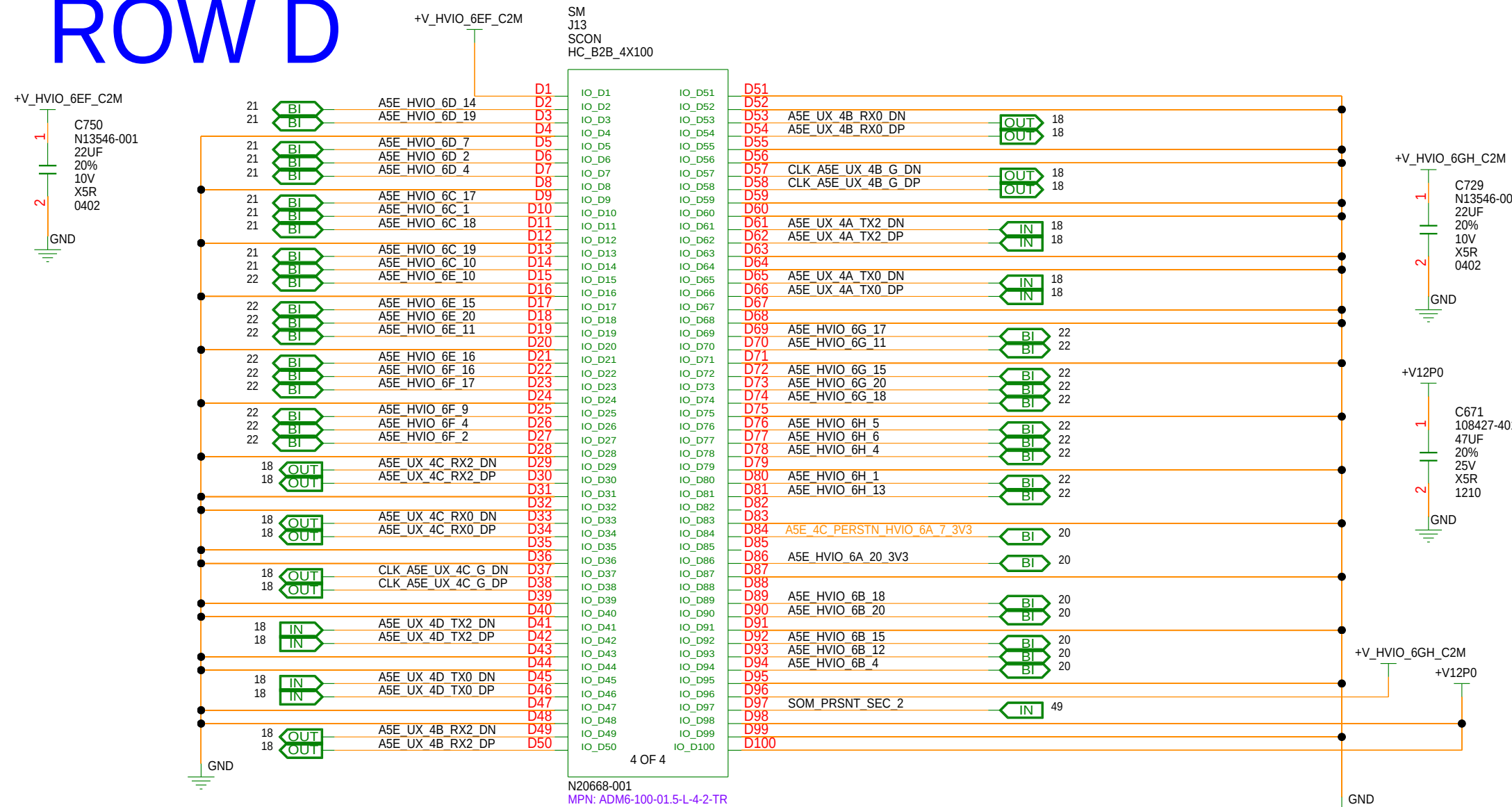
ROW B



ROW C



ROW D



B2B_SECONDARY_CONN

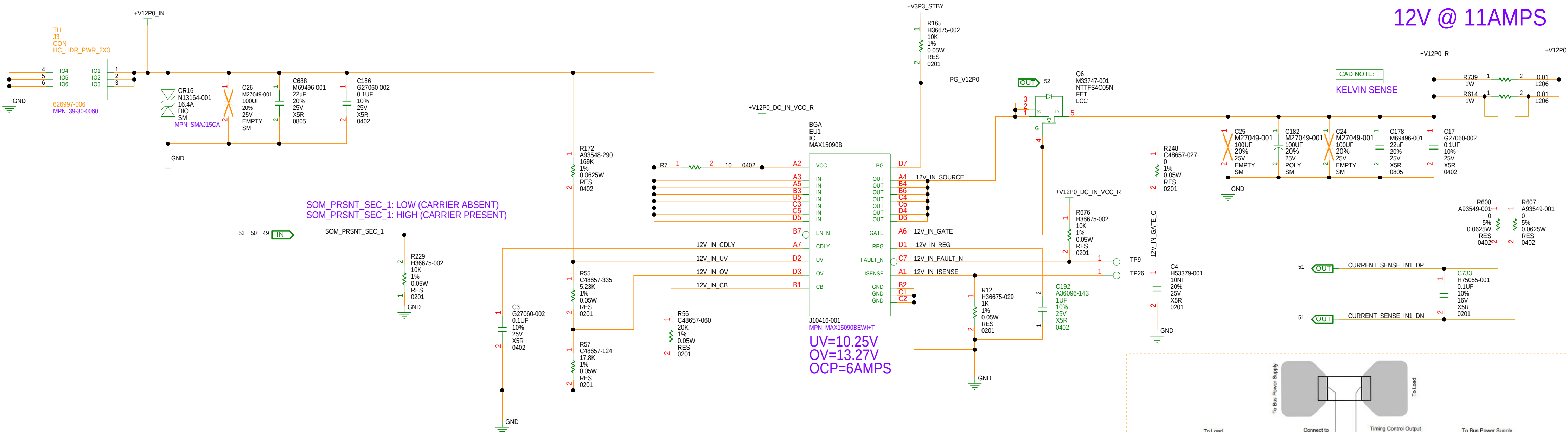
DEPARTMENT



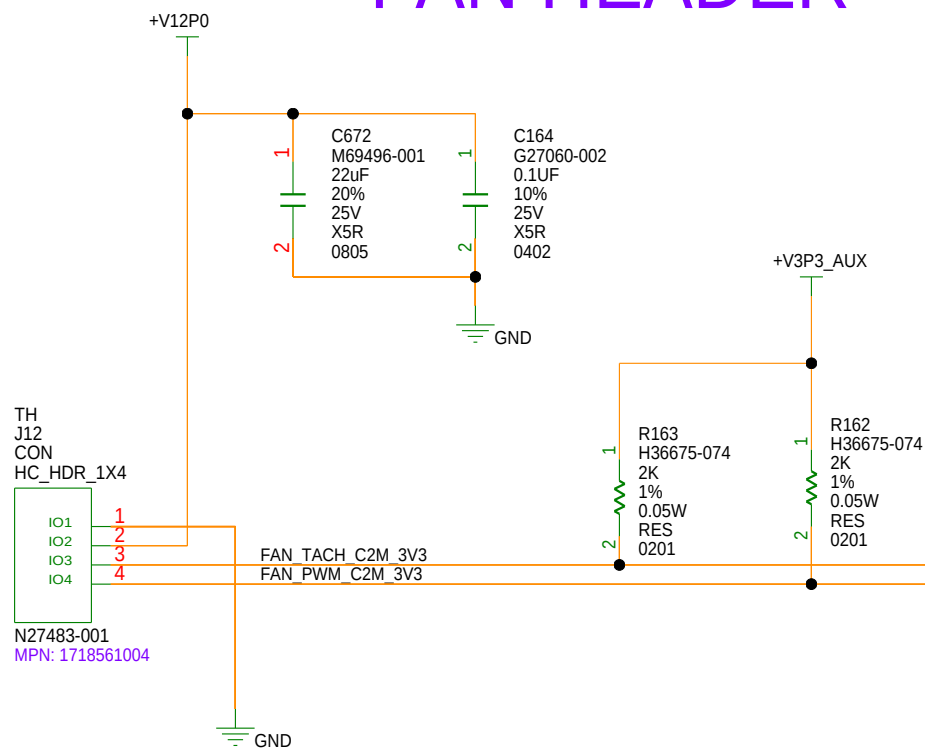
SIZE	CODE	DOCUMENT NUMBER	REV
C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING		SHEET 49 OF 62

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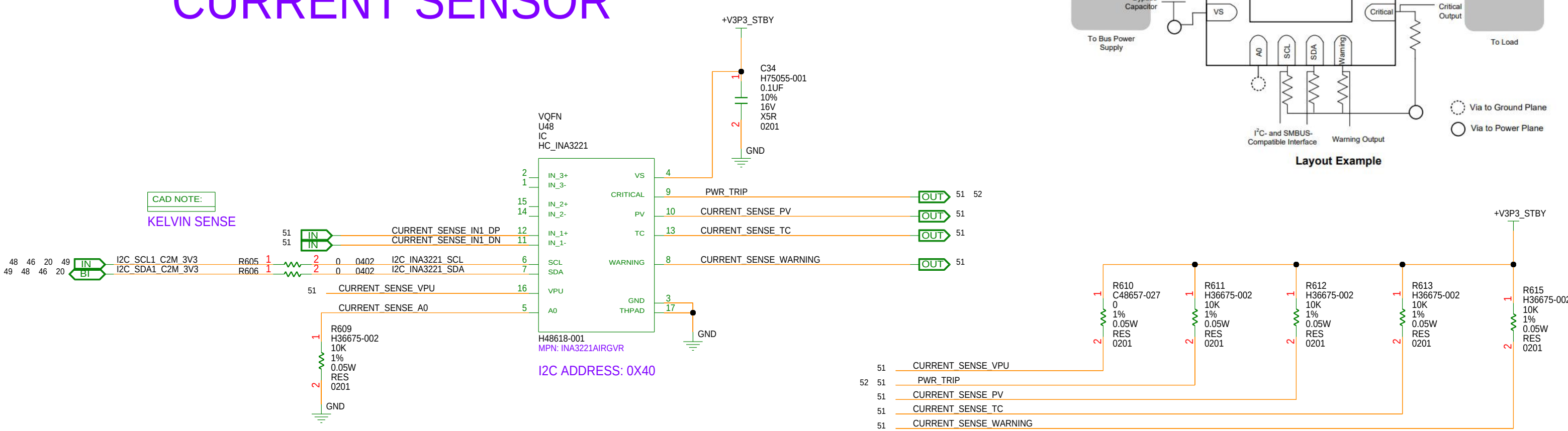
12V ATX CONN



FAN HEADER



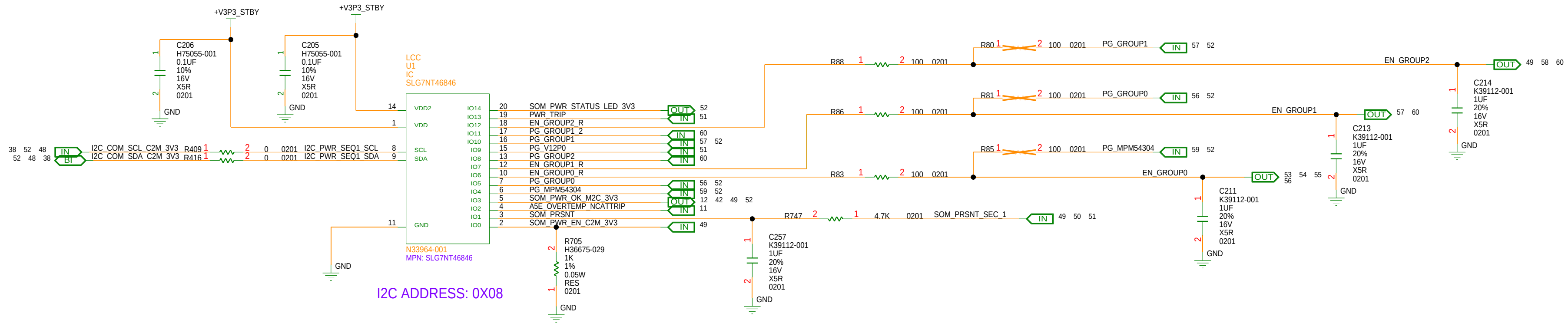
CURRENT SENSOR



12V_POWER INPUT

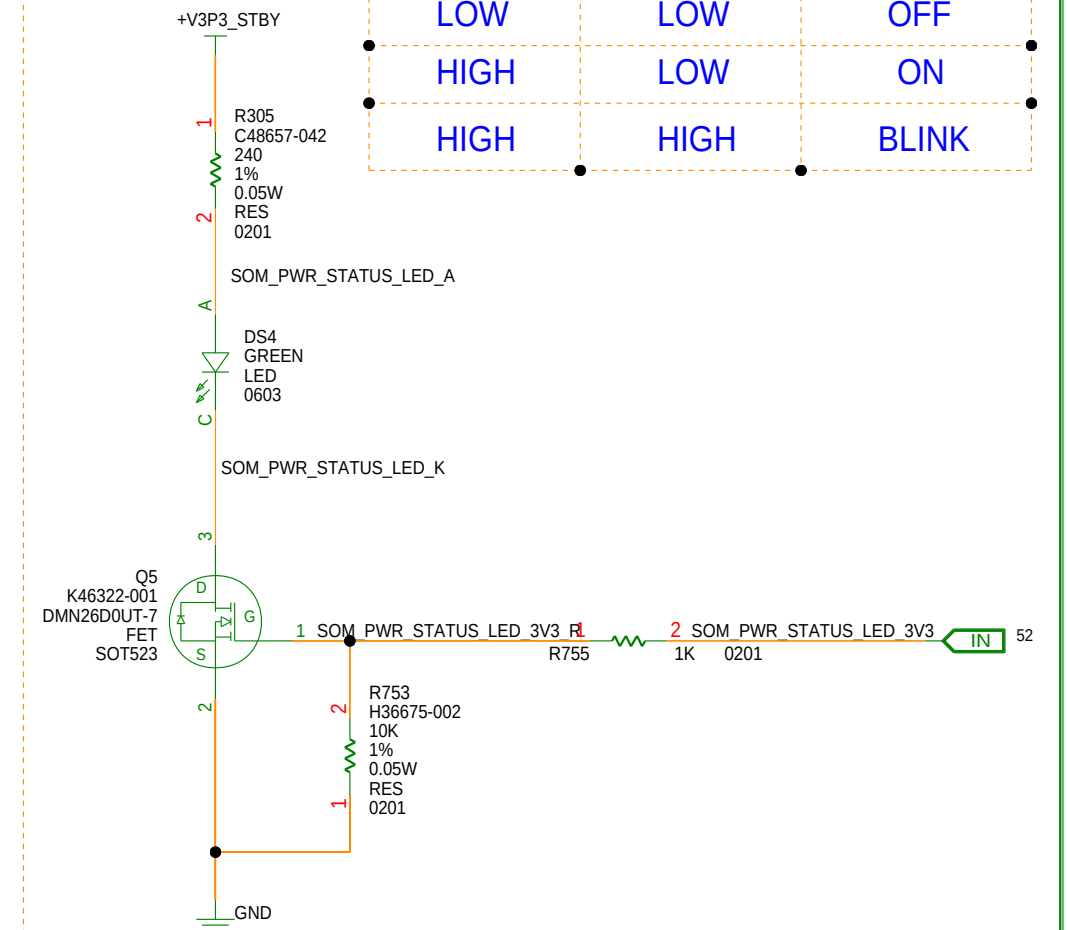
DEPARTMENT	altera	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330723-B1	2.1
SCALE:	DO NOT SCALE DRAWING			SHEET	50 OF 62

POWER SEQUENCER

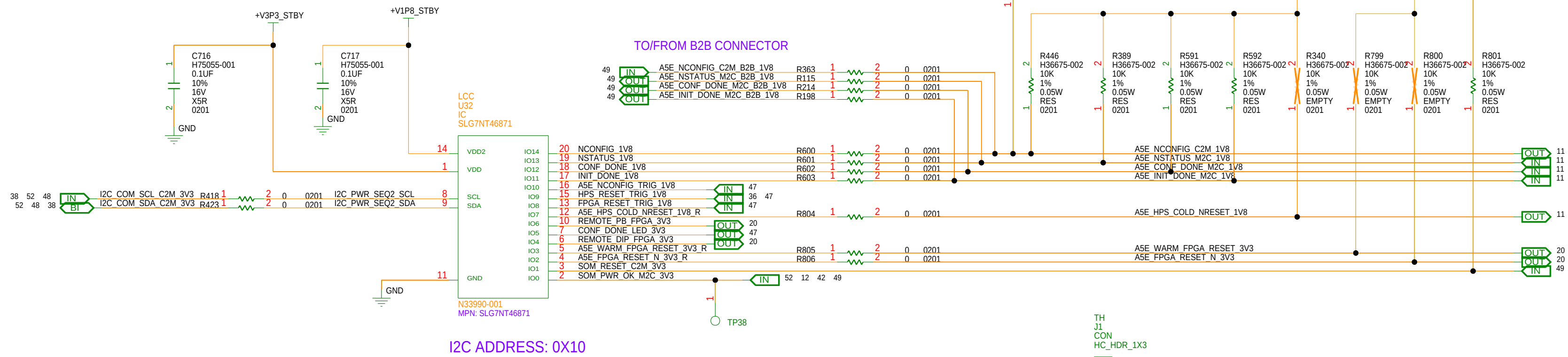


POWER STATUS LED

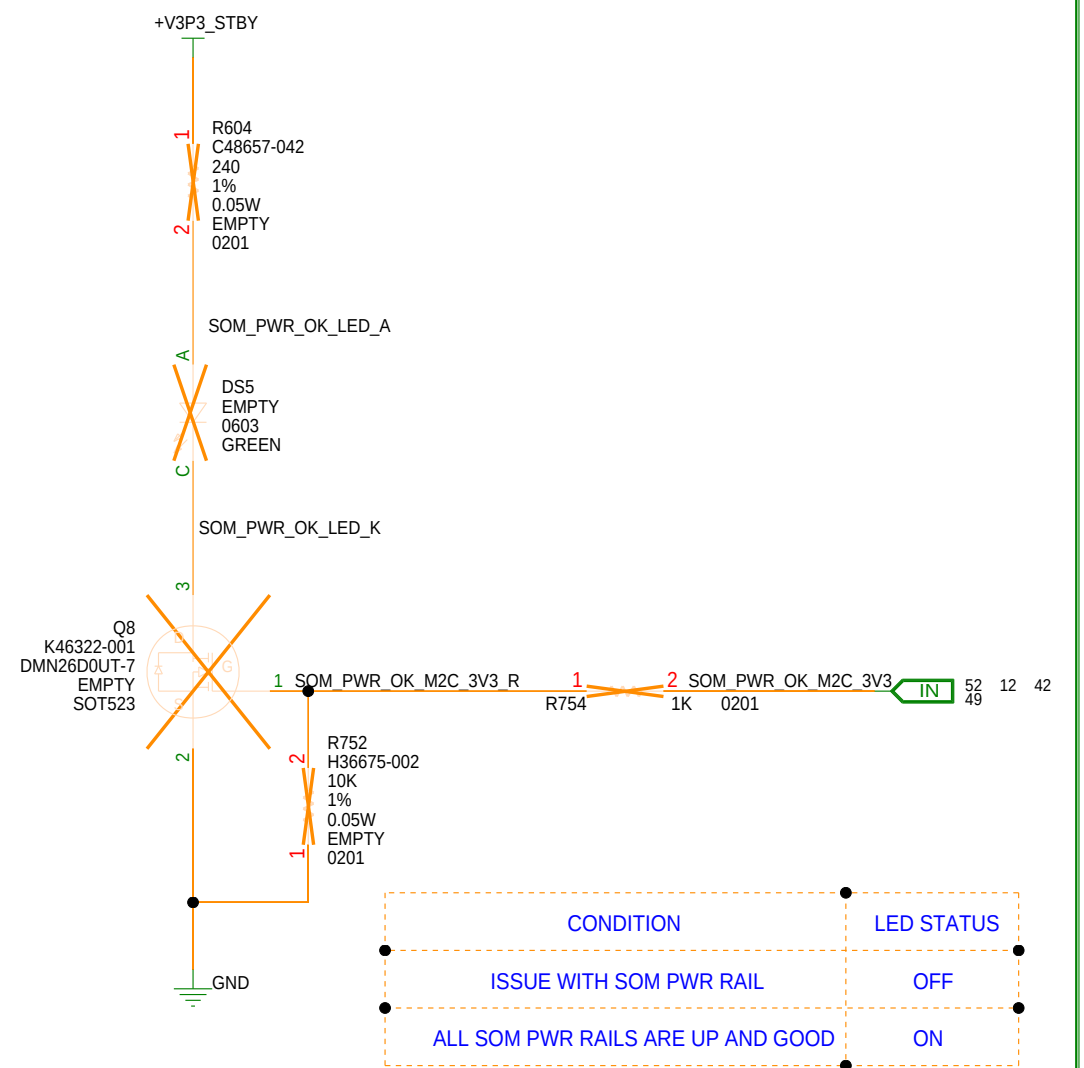
PG_V12P0	PG_GROUP2	LED STATUS
LOW	LOW	OFF
HIGH	LOW	ON
HIGH	HIGH	BLINK



RESET AND CONFIG SEQUENCER



RESET SIGNAL	RESET TRIGGER CONDITION
FPGA FABRIC	WHEN PUSH BUTTON SWITCH "S5" IS PRESSED
HPS WARM	WHEN PUSH BUTTON SWITCH "S6" IS PRESSED FOR <3S
HPS COLD	WHEN PUSH BUTTON SWITCH "S6" IS PRESSED FOR >3S OR/AND WHEN "RST_RST_C2M" SIGNAL GOES LOW



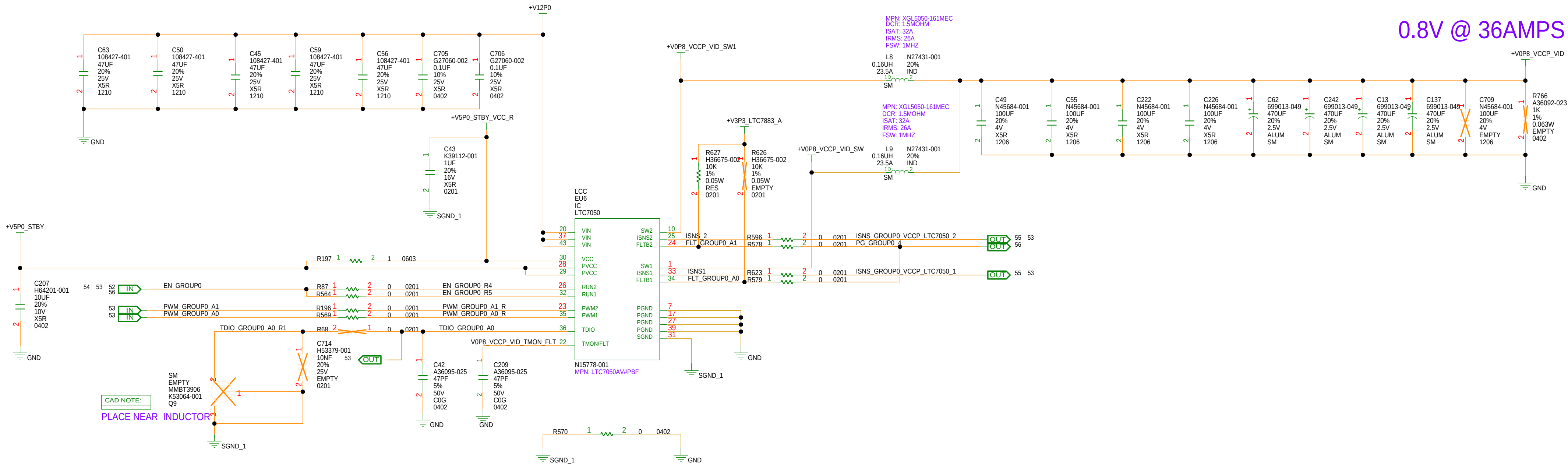
Mon Jul 28 12:46:59 2025

POWER SEQUENCER



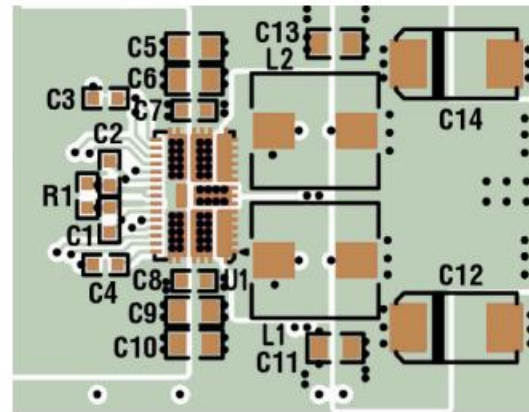
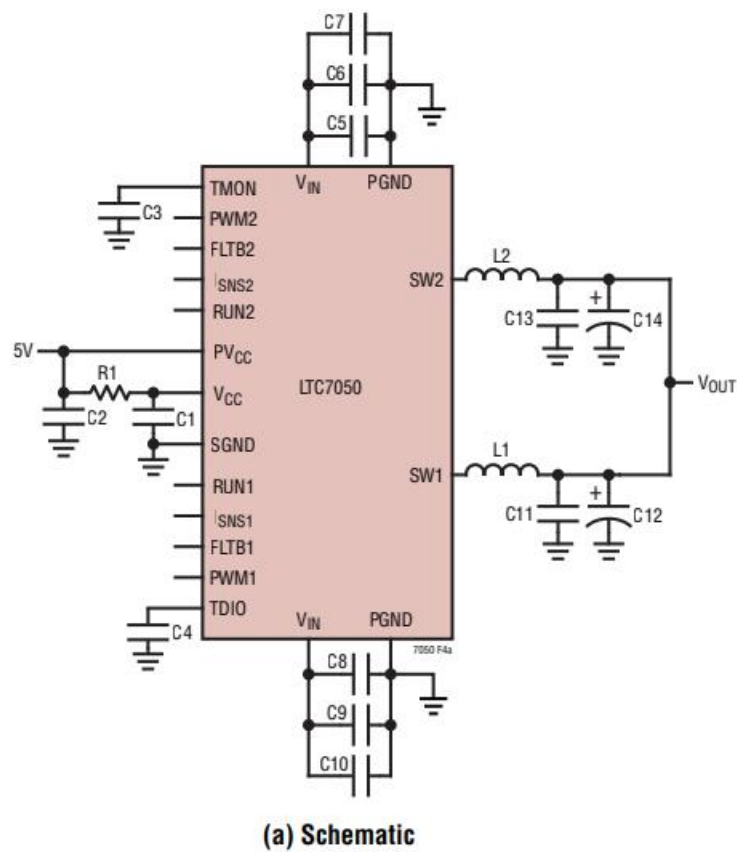
SIZE C+	CODE 34649	DOCUMENT NUMBER 150-0330723-B1	REV 2.1
SCALE:	DO NOT SCALE DRAWING		SHEET 51 OF 62

B

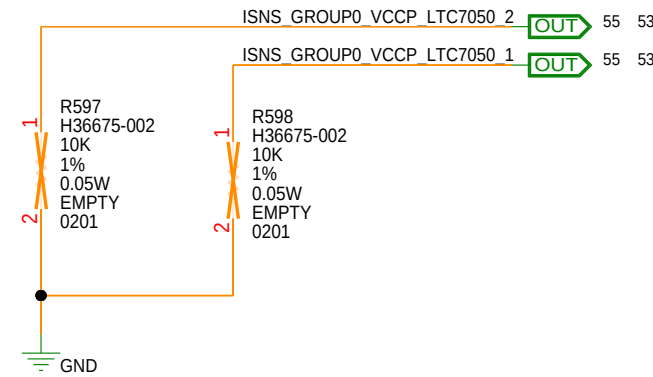


B

A



PROVISION TO REDUCE AN OFFSET ON THE CURRENT SENSE AT MID TO HEAVY LOADS DUE TO SWITCHING SPIKES



A

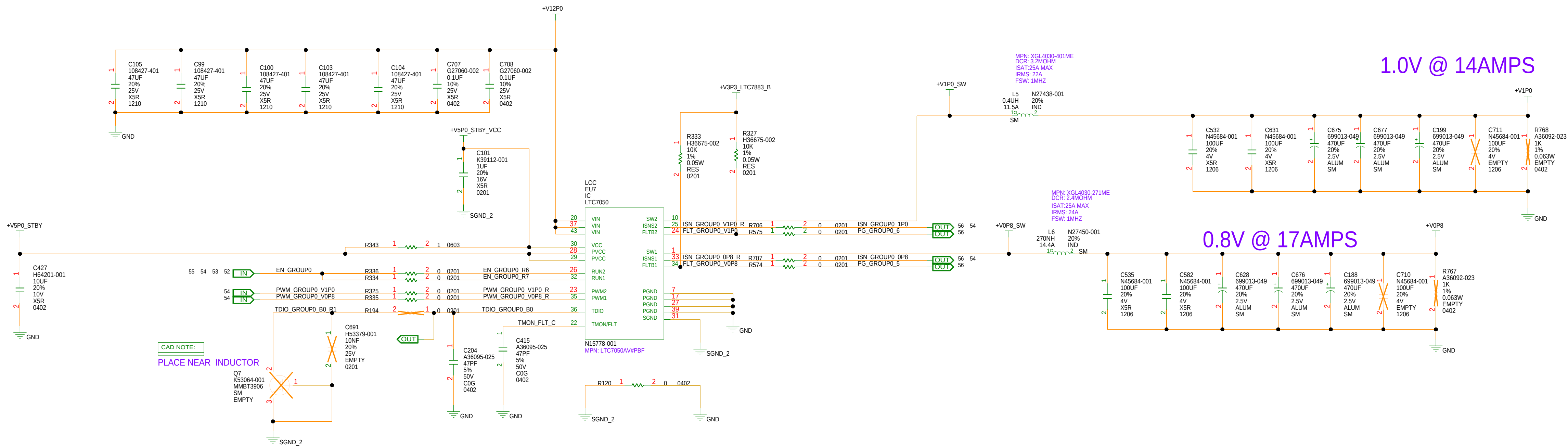
POWER-VCCP_VID

Mon Jul 28 12:47:05 2025

DEPARTMENT		SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330723-B1	2.1
		SCALE:		DO NOT SCALE DRAWING	SHEET 54 OF 62

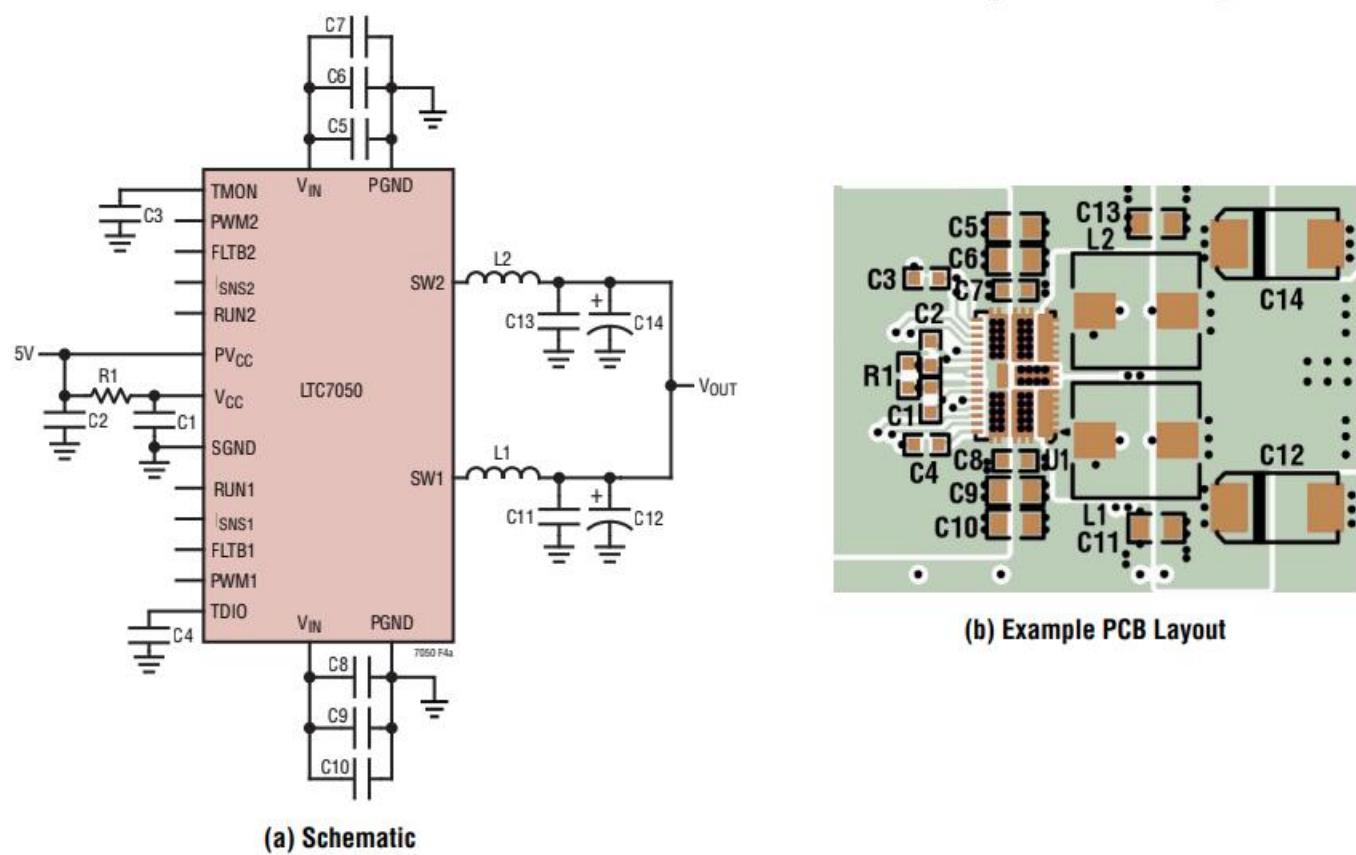
B

B

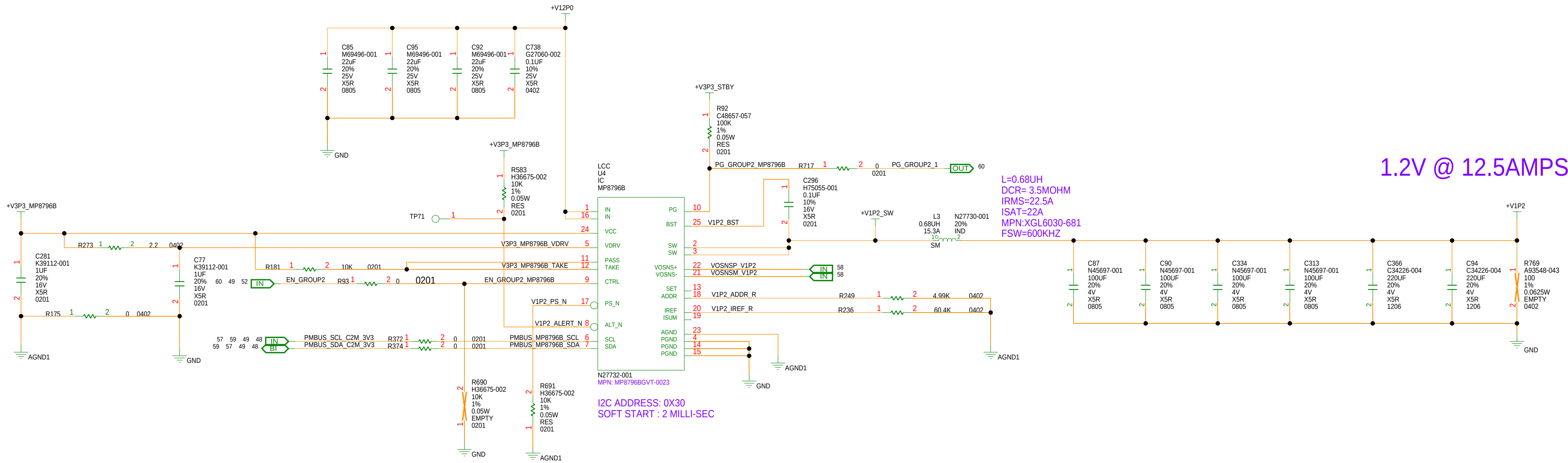


A

A

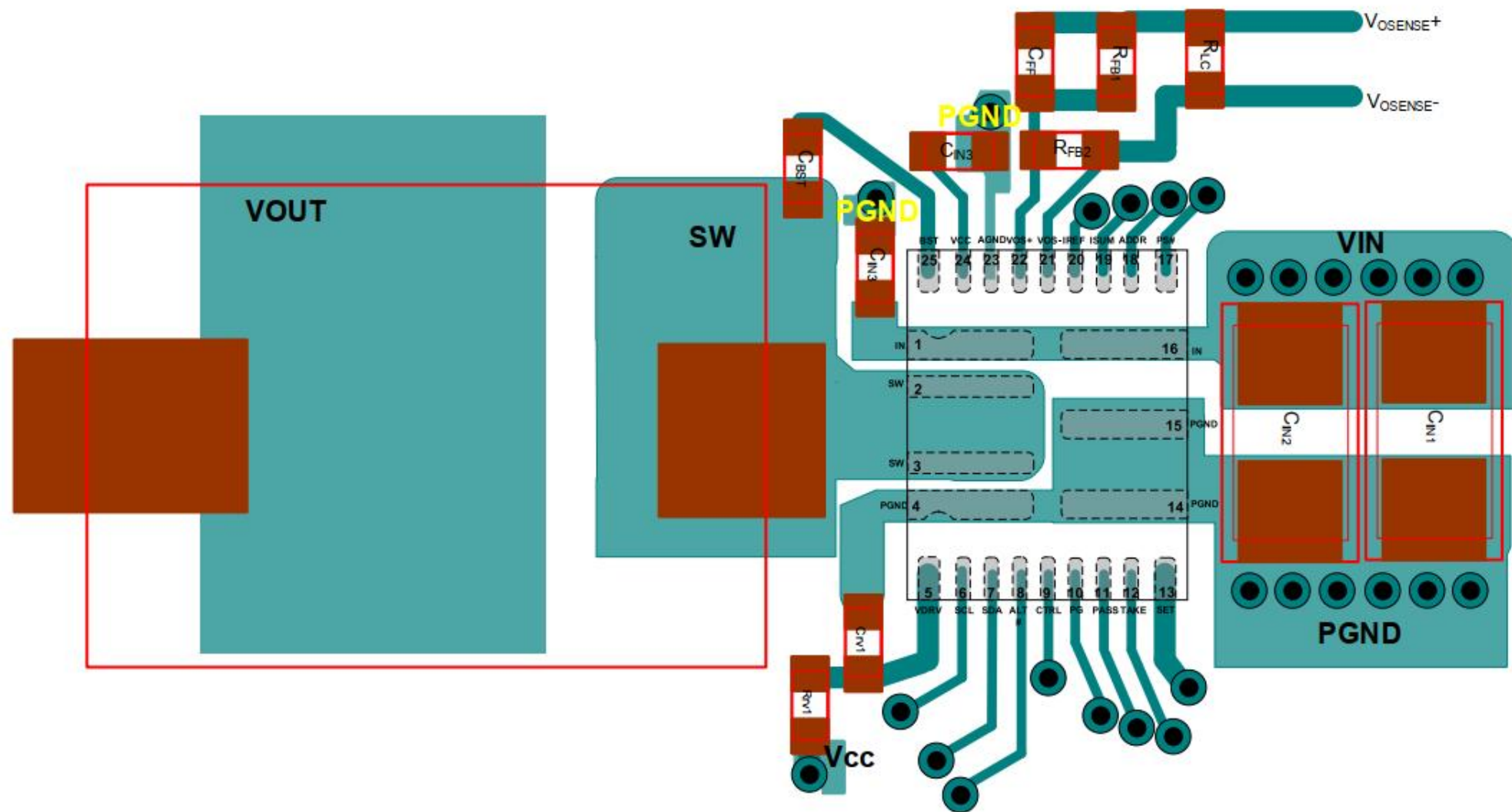


B



B

A

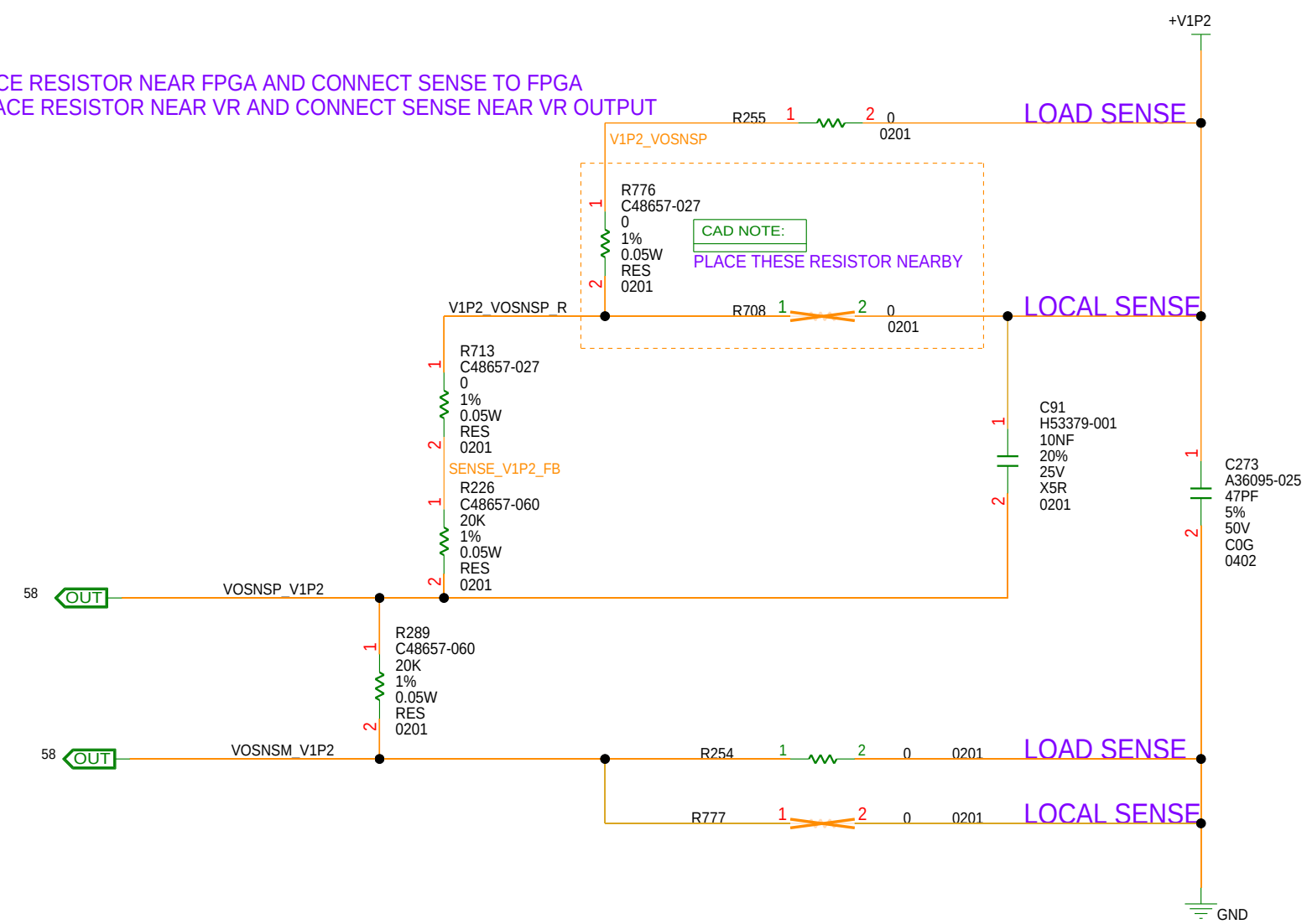


CAD NOTE:

ROUTE THEM AS DIFFERENTIAL

CAD NOTE:

LOAD SENSE - PLACE RESISTOR NEAR FPGA AND CONNECT SENSE TO FPGA
LOCAL SENSE - PLACE RESISTOR NEAR VR AND CONNECT SENSE NEAR VR OUTPUT



Mon Jul 28 12:47:08 2025

POWER - +V1P2

DEPARTMENT

altera

SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

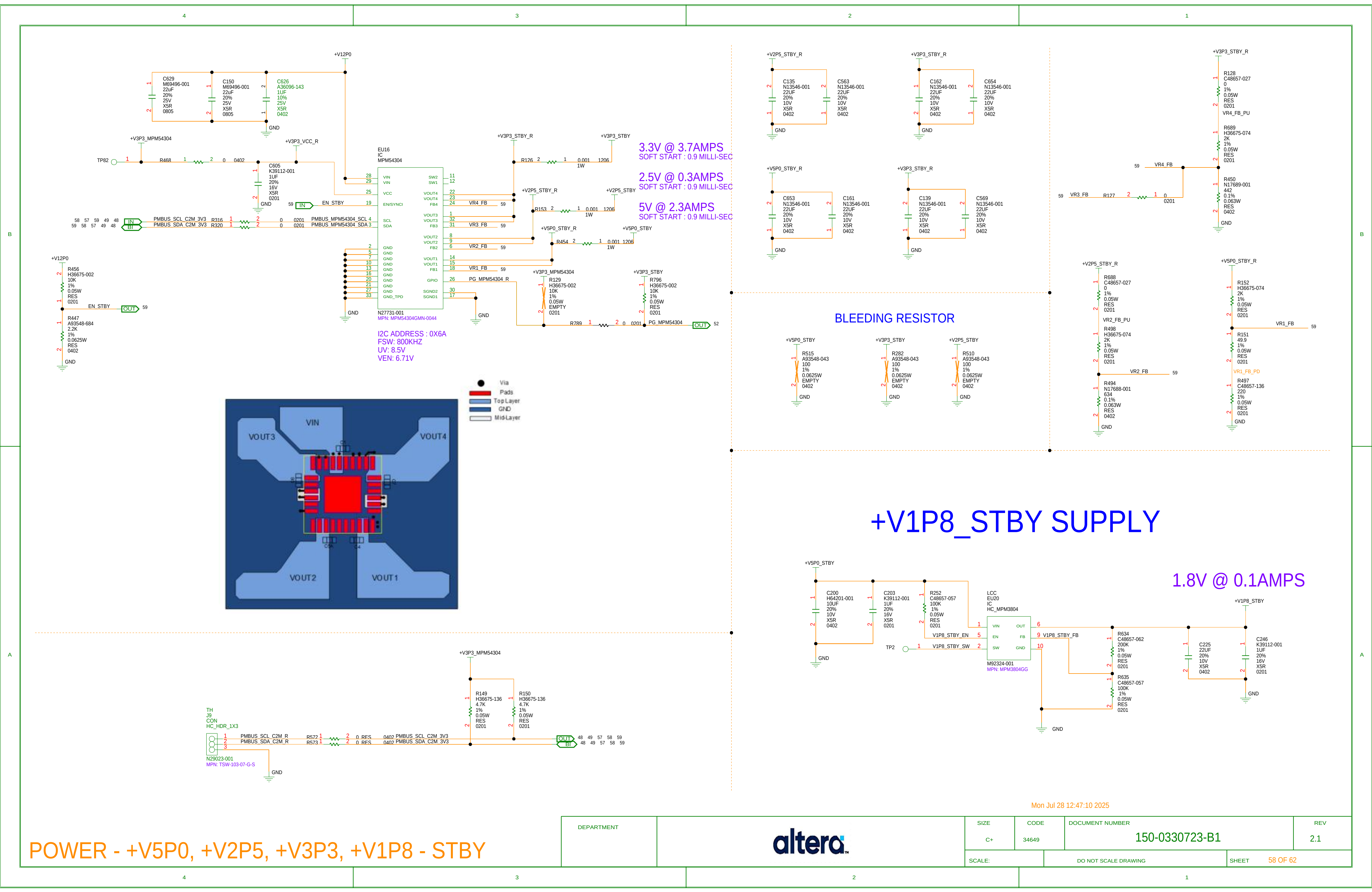
2.1

SCALE:

DO NOT SCALE DRAWING

SHEET

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POWER - +V5P0, +V2P5, +V3P3, +V1P8 - STBY

DEPARTMENT

altera

SIZE

C+

CODE

34649

DOCUMENT NUMBER

150-0330723-B1

REV

2.1

SCALE:

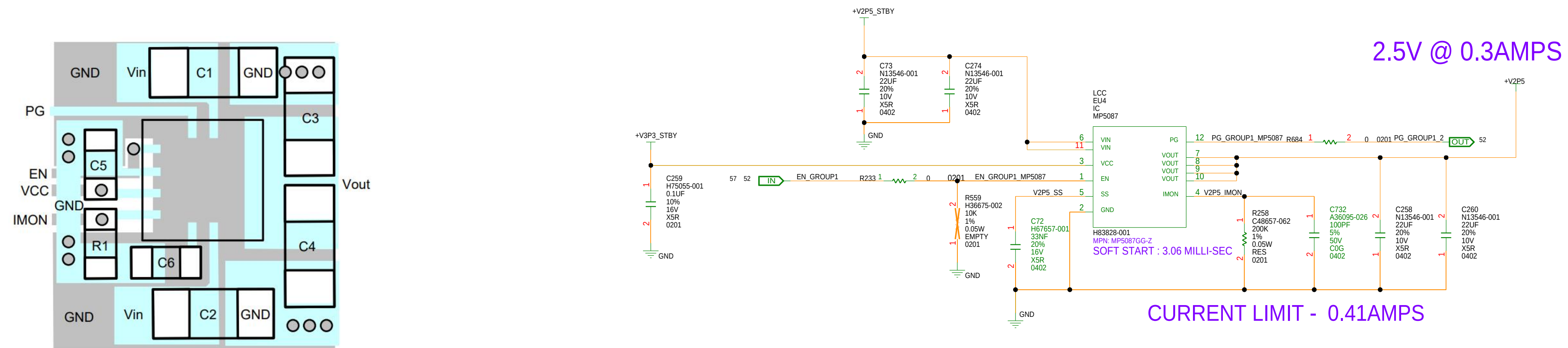
DO NOT SCALE DRAWING

SHEET

58 OF 62

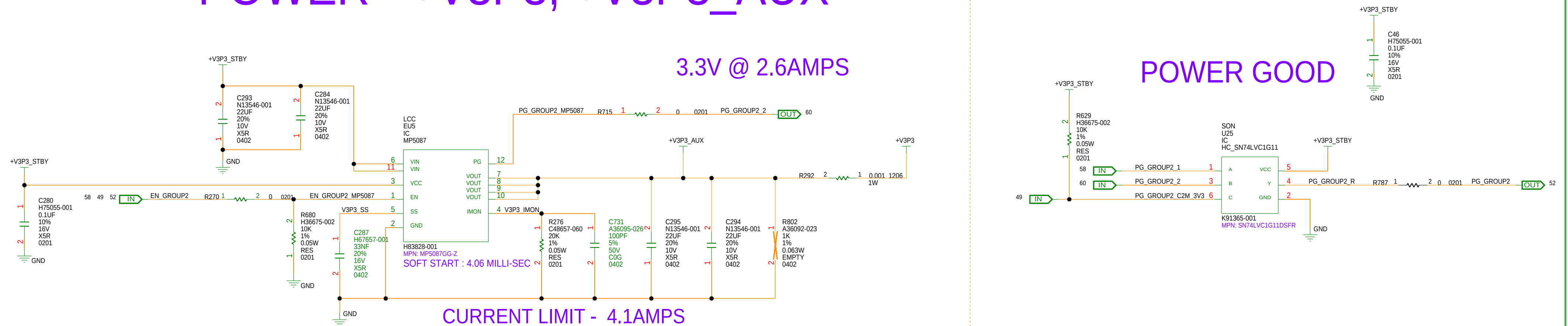
Mon Jul 28 12:47:10 2025

POWER - +V2P5



Recommended Layout

POWER - +V3P3, +V3P3_AUX



POWER GOOD

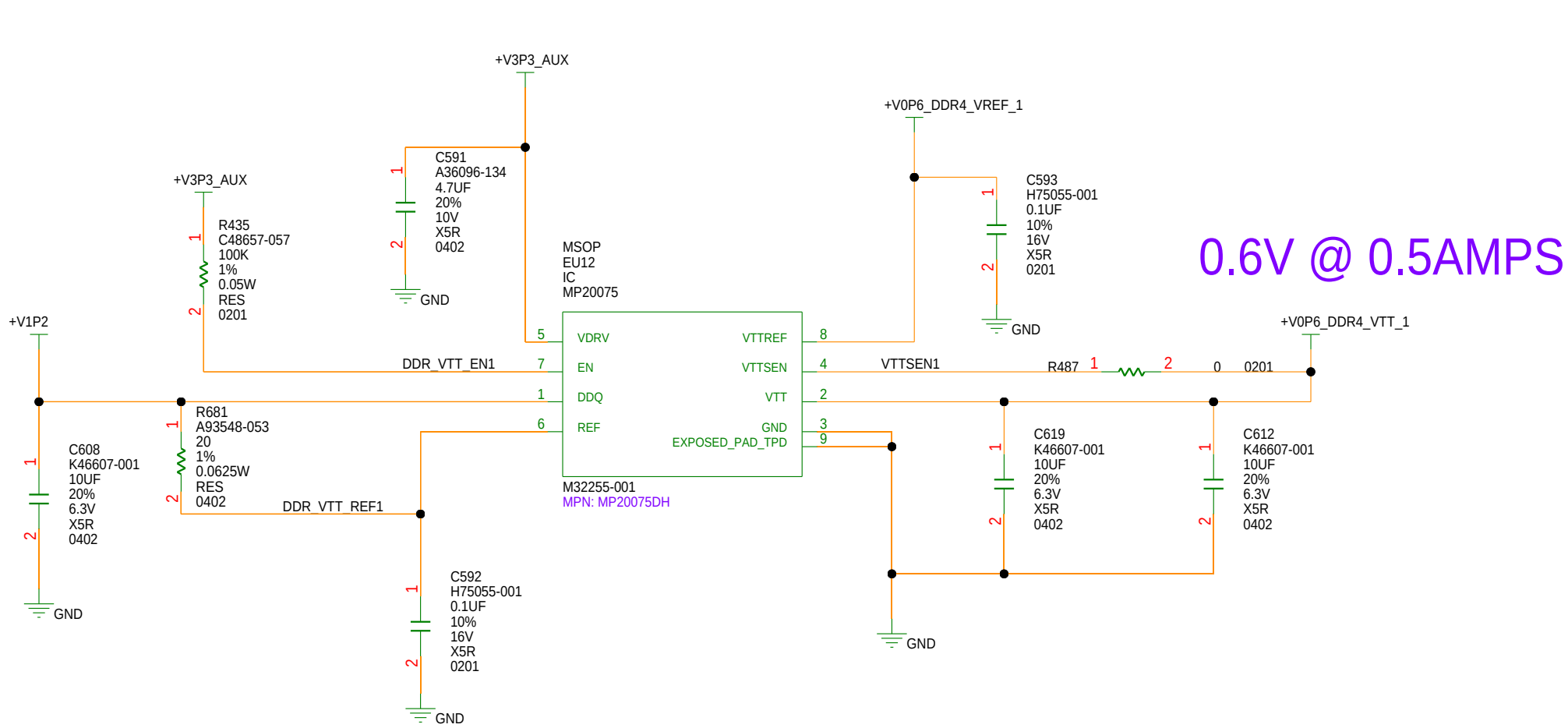
POWER - MP5087 LOAD SWITCH

B

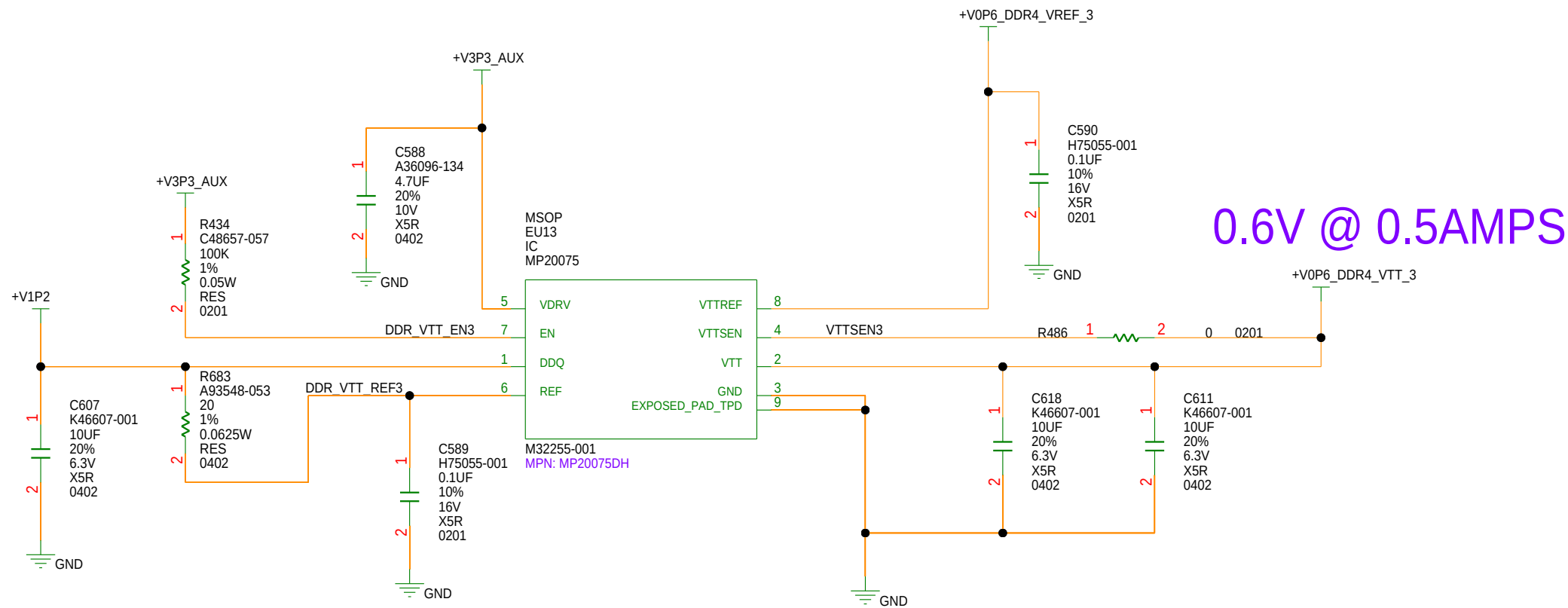
A

B

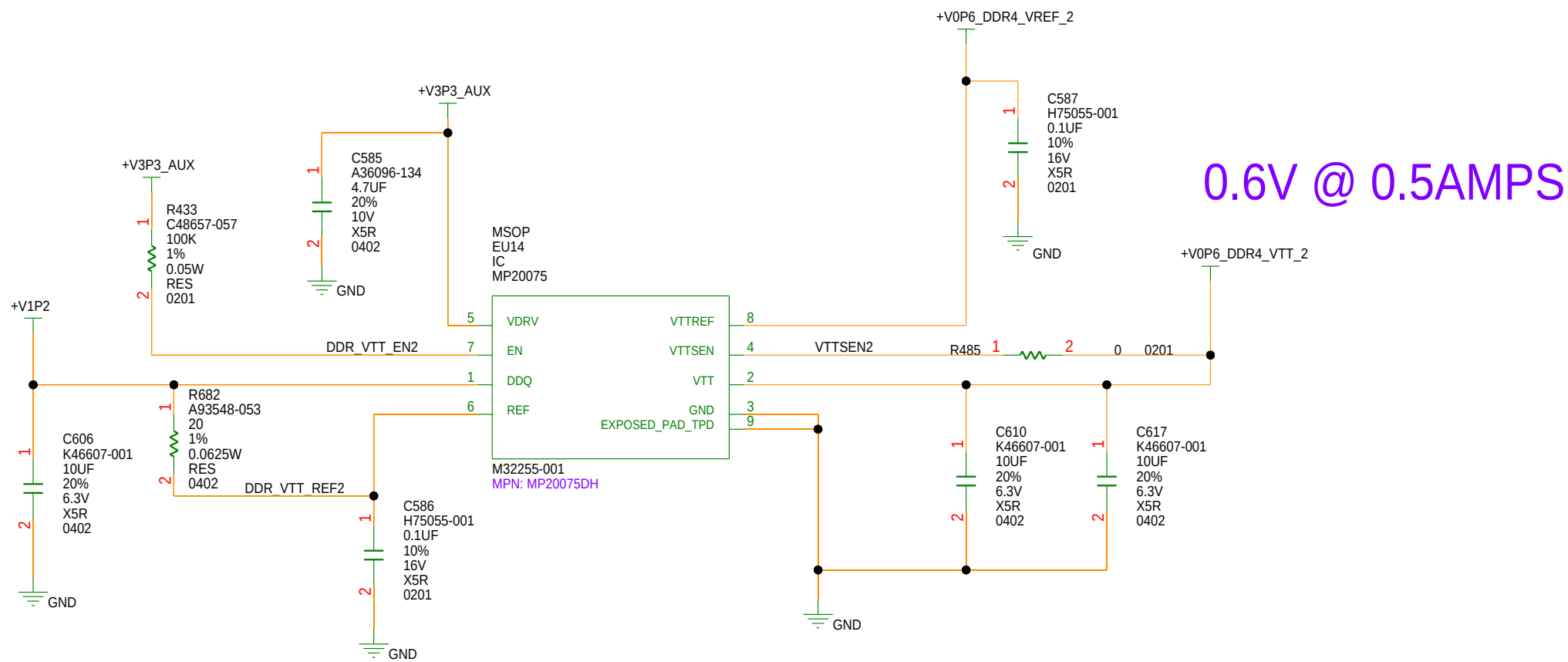
A



0.6V @ 0.5AMPS



0.6V @ 0.5AMPS



0.6V @ 0.5AMPS

POWER - +V0P6_DDR4_VTT

Mon Jul 28 12:47:12 2025

DEPARTMENT	altera	SIZE	CODE	DOCUMENT NUMBER	REV
		C+	34649	150-0330723-B1	2.1
		SCALE:		DO NOT SCALE DRAWING	SHEET 60 OF 62

B

A

B

A

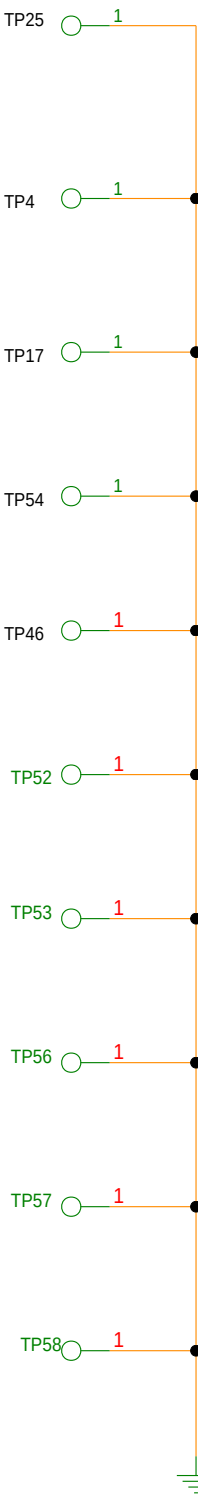
POWER PROBE POINTS

CAD NOTE:
PLACE NEAR PRIMARY LOAD POINTS FOR EACH RAIL
ADD SILK SCREEN LABEL WITH RAIL NAME FOR EACH TP

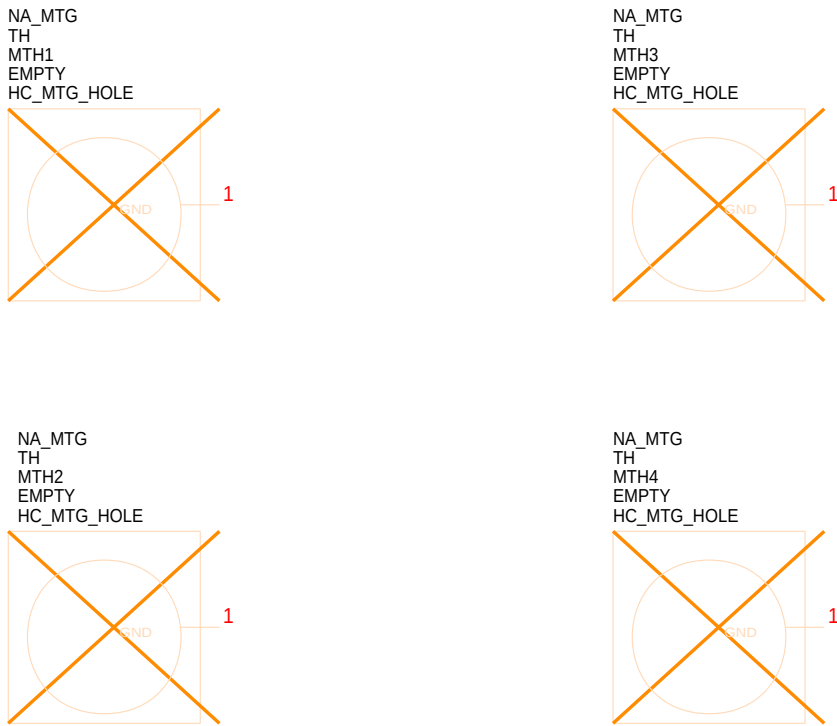


GND PROBE POINTS

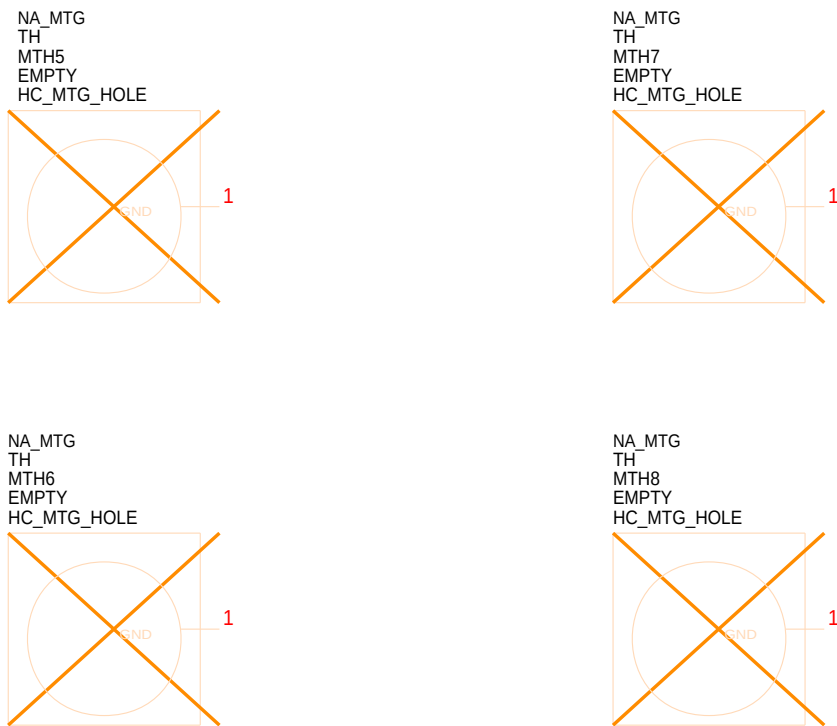
CAD NOTE:
SCATTER AROUND THE BOARD TOP SIDE



MOUNTING HOLE FOR HEAT SINK



MOUNTING HOLES FOR PCB



TEST POINT

Mon Jul 28 12:47:13 2025

DEPARTMENT	altera	SIZE C+	CODE 34649	DOCUMENT NUMBER 150-0330723-B1	REV 2.1
SCALE:		DO NOT SCALE DRAWING		SHEET 61 OF 62	

B

A

B

A